

[阅读中文版本](#)

From technological autonomy to urban autonomy, let Jing-Zhang become a source of original innovation once again.

If the first autonomy refers to the breakthrough achieved a century ago, when the Jing-Zhang Railway was independently designed and built by China despite engineering constraints, then Jing-Zhang's "second autonomy" is oriented toward the city: enabling planning to acquire autonomous sensing, intelligent simulation, human-machine collaborative decision-making, and continuous iteration. Taking the century-old Jing-Zhang Railway as its historical and spatial axis, this proposal uses an AI-native planning operating system to coordinate trusted data, causal diagnosis, an intelligent computing network structured as "primary computational power hub - edge nodes - application scenarios," the spatial framework of "three zones, two wings, and four corridors," and five categories of renewal strategies. It connects planning approval, investment decision-making, urban operations, and public life into one feedback loop, supporting Jing-Zhang's upgrade from a place where AI resources gather into a world-class source of original AI innovation that continuously generates new ideas, new enterprises, new standards, and new scenarios.

Design basis and information list

This chapter fully preserves the team's discussion of the AI-era planning methodology, city history base and project awareness. Static space, dynamic running, institutional projects and social awareness information need to indicate the source, time, process, credibility and use; available regulatory detailed planning, tenure, municipal and engineering conditions remain "pending formal data completion". [Source](#)
[Source](#)

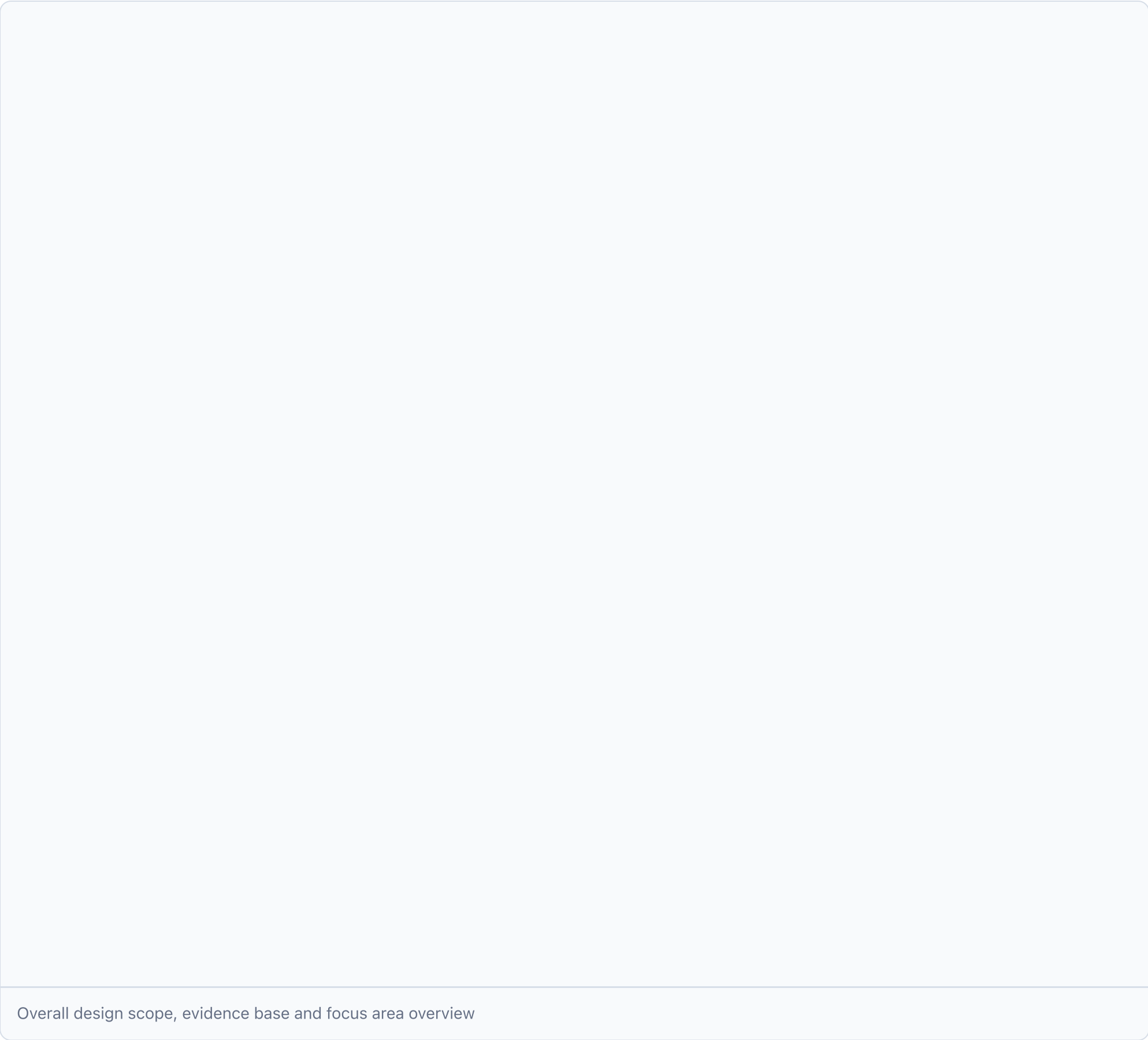


Figure 01 is required for the overall design scope and the overview of the event.

The first chapter of the project understands why it's planned with AI?

1 Project Understanding

The Jing-Zhang Railway innovation belt for the AI era, with the "100-year Jing-Zhang ▪ AI innovation belt" as its flag, and the "world-class source of original innovation" as its core, recreated a China-AI-era science and innovation corridor, cultural corridor, resilience corridor; and the three priority areas of Zhongzhi Park, Beijing AI Origin Community, and Dazhongsi, with a division of functions and resilience.

AI New theme for time planning: eight fundamental changes

These eight changes are not about local upgrading of planning tools, but rather about a systematic shift in urban planning from basic data, cognitive methods, workflows to value targets — AI — not only to improve the efficiency of text, drawings and data processing, but more importantly, how planning recognizes cities, how to form judgment, how to organize proposals, and how to move on into urban construction and governance.

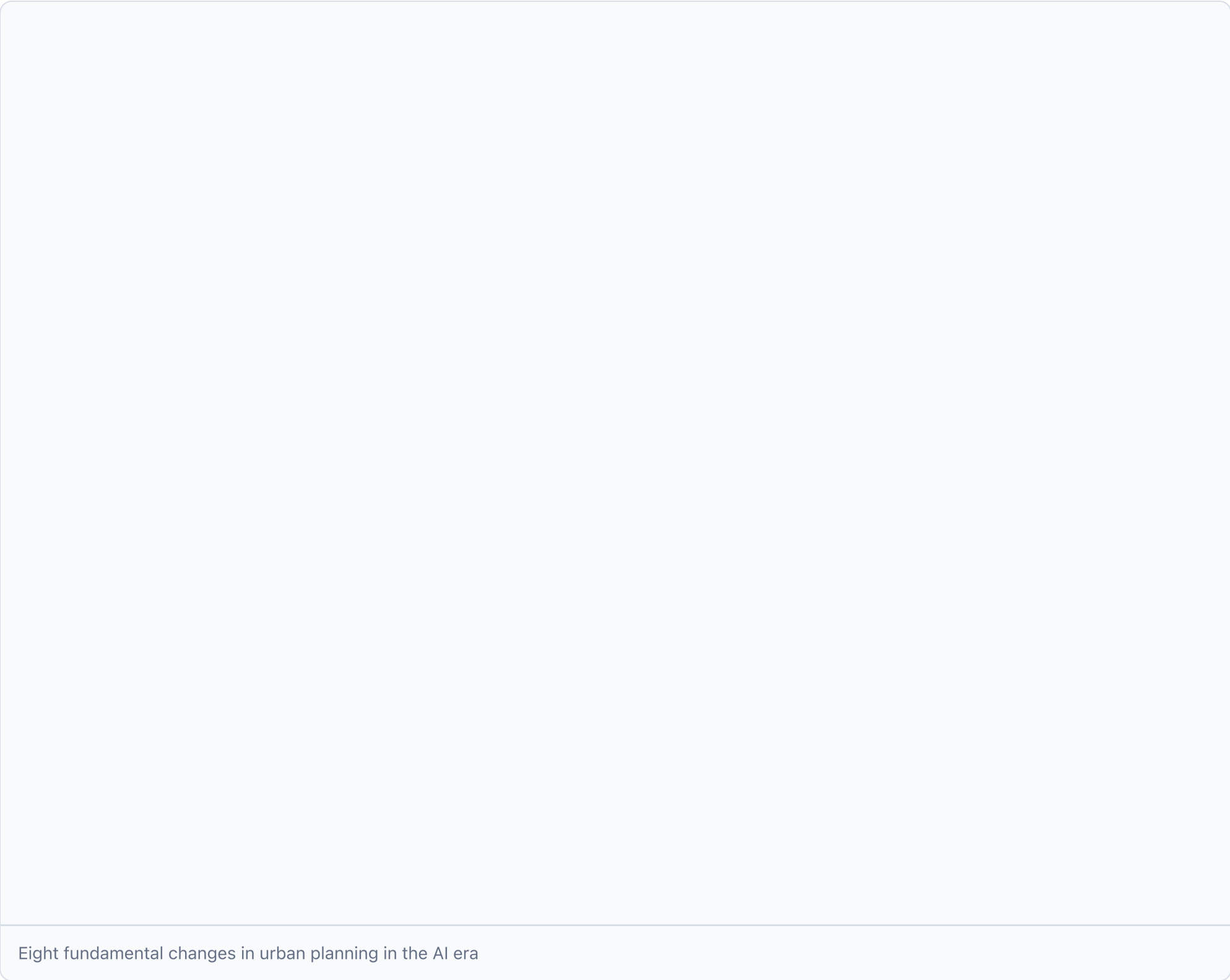


Figure 1

Dimensions	Traditional planning	AI The Changing Age
* Data base*	Small sample static data	real-time full-domain sensing data (remote sensing, IoT, signaling, transportation, energy consumption, feedback)
* Cognitive paradigm*	Experience-driven	Data and knowledge are co-driven, and planners judge which data are credible and which patterns apply locally
* Analytical methodology*	Single analysis	multimodal intelligent analysis, relevant, abnormal and candidate explanations identified and professionally validated
* Planning object*	Physical space	"people-land-city-system", associated property rights, population, industry, energy consumption, operations and social impacts
* Planning process*	Linear Processes	Together, new evidence can fix judgment and optimize proposals
* Planned products*	Static blueprints	Dynamic scenario and strategy library, callable, groupable, iterative
* Decision support*	Auxiliary tools	intelligent decision-making hub, alerting evidence gaps, pre-conditions, conflict of rules and scenario failure
* Value orientation*	Efficiency priority	resilience, equity, sustainability and human dignity

These eight changes are shared: the essence of AI-native planning is based on urban evidence, underpinned by intelligent evolution, centred on human-machine co-judgment, and operated in closed-loop cycles, facilitating planning from static blueprints to dynamic urban decision-making systems.

Question three: How to do it, how to present it, how to apply it.

The overall orientation of "World-class " is based on answers to three core propositions that constitute a whole logic from methodology, expression to application.

How to plan: the logical path of AI-native planning

Instead of adding an AI tool to the end of the traditional workflow, AI-native planning has organized the whole process in a "data readability, evidence traceability, knowledge reasoning, program simulation, decision interpretation, implementation feedback" from the start of the planning process, resulting in six consecutive stages:

Perception and data aggregation evidence construction and governance ► knowledge and reasoning ► program generation and simulation ► decision support and synergy ► monitoring assessment and iterative



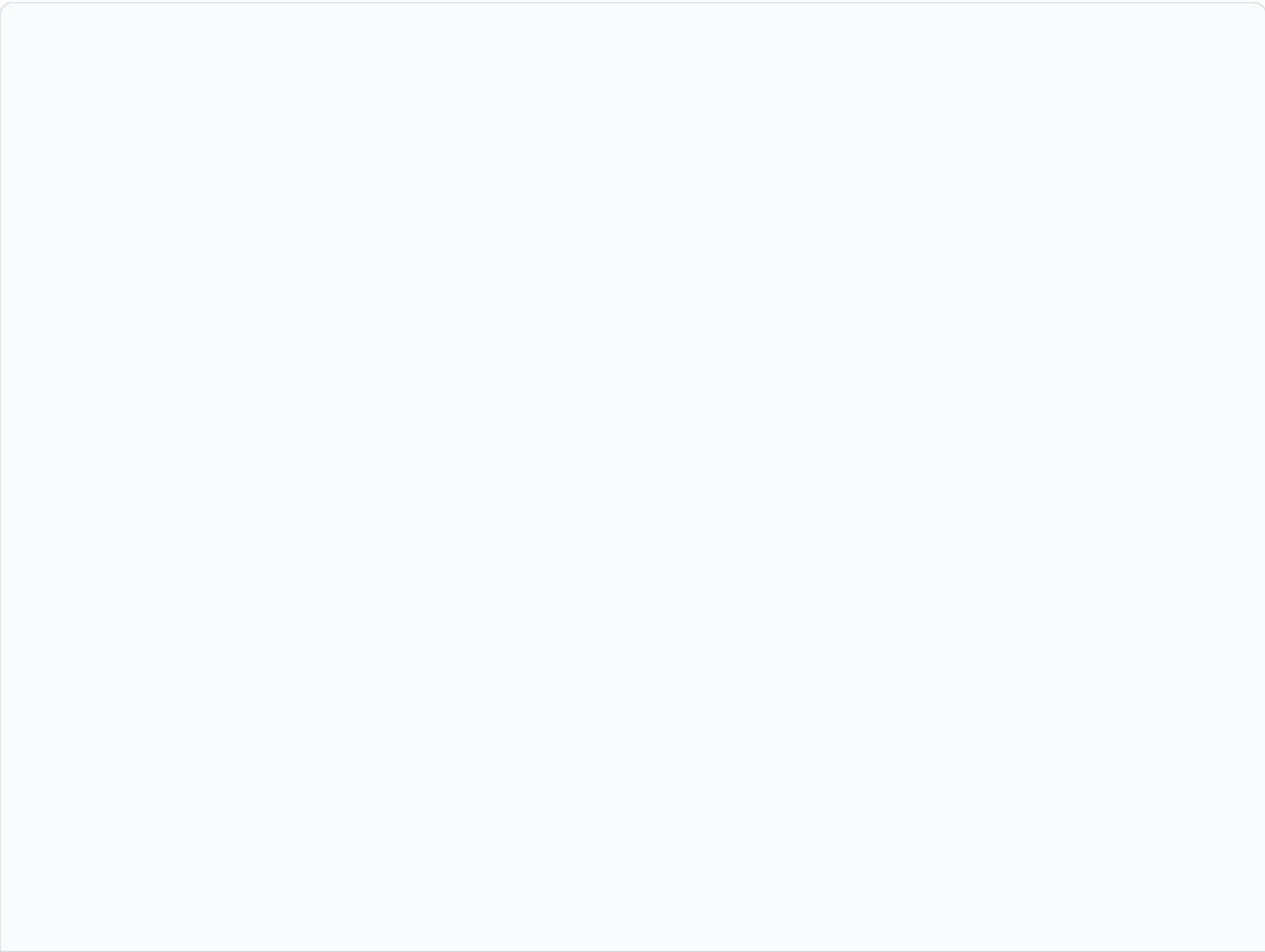
How to plan: the logical path of AI-native planning

Figure 3 How to plan: the logical path of AI-native planning

The entire logical path embodies four core features: ** data continuity, knowledge-driven, AI enabling, closed-loop** — planning no longer ends with outcome reviews, but rather with a continuous process of " perception-evidence-generation-generation-decision-implementation-results".

How to project: a new expression of content interface

AI planning for the era should not be a set of charts or a three-dimensional model, but rather a unified planning matrix, with six types of information interfaces based on the mandates and perceptions of different users: ** decision-making interface** (conclusions, evidence, risk, action), ** professional technical interface** (data, methodology, model, review), ** spatial experience interface** (two-dimensional/three-dimensional/scenario), ** approval rule interface** (indicators, conditions, exemptions), ** project investment interface** (lists, time series, finance, responsibility), ** public and operational interface** (change, progress, feedback, operation).



How to present planning: a new expression of content interface

Figure 4 How planning is presented: a new expression of content interface

Planning should have four features: ** precision expression, layered presentation, visual interaction, traceability** — no longer “programmatic understanding”, but rather, decision-making, review, construction, operation and feedback by different actors around the same planning matrix.

How to apply planning: new scenarios, new applications of the AI era

AI driven planning applications follow a common path — ** problem identification — proposal generation — response evolution — scenario design — implementation delivery — feedback optimized**: transforming real urban issues organizations into a continuous planning system based on credible evidence, multiplanning, integration of space and operations, open collaboration and version iterative mechanisms that transform the Jing-Zhang Railway urban design from a one-time result to a process of urban action capable of continuous learning, validation and updating.



Figure 5 How to apply planning: new scenarios, new applications in the AI era

Chapter two, the establishment of the bottom line...

Establishment of a base: from database to urban time and space data

Build a "all-factor, multi-scale, long-cycle" foundation for the Jing-Zhang Railway innovation belt and raise multi-source data to a calculable, reasonable, traceable knowledge network that supports problem identification and planning.

Rather than centralizing all urban data, the urban evidence system transforms decentralized, heterogeneous, different quality and different permissions into credible evidence that supports planning judgment through governance, validation and interpretation. Its core logic is:** multi-source data access — integrated resource management — data governance — evidence generation — planning application.** Data can enter planning judgment only if they identify sources, time, spatial scope, processing methods, quality levels and boundaries.

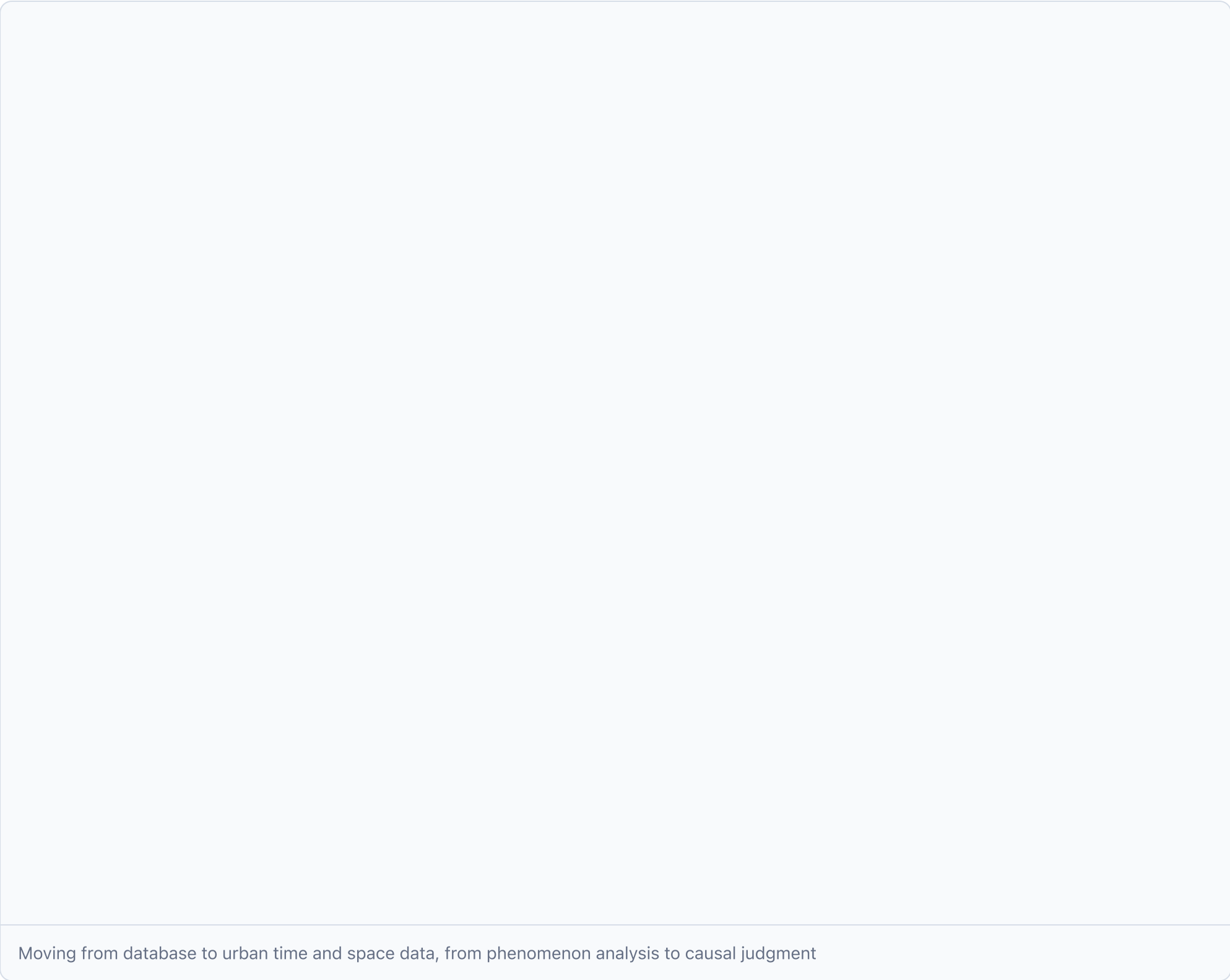


Figure 2 Transition from database to urban time and space data, from phenomenon analysis to causal judgment

2.1 Data Foundation

Static space evidence

- Land, buildings, roads, installations, topography, water systems - Property rights, ecologically sensitive areas, historical and cultural resources - 3D urban form, skyline, public space quality

Dynamic running evidence

- Population activities, traffic, mobile phone messages, consumption, logistics - Energy, water, tourist stopovers, facilities. - Meteorological and environmental monitoring

Institutional and project evidence

- Policy documents, approval conditions, budget, land supply - Construction projects, operating rights, contractual arrangements - Subject and source of funds

Social cognitive evidence

- Resident opinion, business claims, online discussions, complaints recommendations - Local knowledge, expert judgment - Stakeholder interviews

2.2 Data mapping

Instead of drawing urban data into a network, the urban audience organized "where the data came from, what they described, what they supported or refuted, what decisions were made, what results were ultimately produced" into a searchable, verifiable, traceable, up-to-date space-time relationship system.

The bottom of the mark.

- Space units (lots, blocks, corridors, nodes) - Functional facilities (transport, industry, public services, computational power facilities) - Owners of title and subjects of governance

Structural relations

- Space relations (neighbor, inclusive, accessible, visible) - Functional relationships (synergy, conflict, dependence, substitution) - Time series (evolution, update, project sequence)

Professional theme profile

- Transport-land-industry interaction mapping - Historical protection - urban renewal weighs - Science resources - matching of facilities - computational power facility - Security assessment

Other relations

- Policy-project-space mapping relationships - Public demand-planning response

On the "one map" and CIM, add* *semantic links, tracer of evidence, planning reasoning, basis of decision-making and implementation feedback*,* four more critical dimensions than the normal knowledge graph: source and credibility; support and rebuttal; reasoning and decision-making processes; operational results and their feedback**.

System	Main questions answered	Typical content
GIS	Where, how far, how far?	Land parcels, roads, water systems, facilities, administrative districts
CIM/digital twin	What's the status of the city, how it works?	Construction, plumbing, equipment, 3D models, sensors
Data Centre	What data exist, where they exist?	Data directories, interfaces, databases, data services
City	What does the city have to do with it?	Construction belongs to the plot, the enterprise is located in the park, the road link node
* City *	Why form a judgment, whether it is credible and how it affects decision-making	Evidence, sources, assumptions, rebuttals, rules, judgments, decisions, actions, outcomes

2.3 Five core permissions

Image Integration

- Harmonizing information on different sectors, professions and formats to the same urban audience and creating a cross-cutting evidence system

Association Found

- Discover hidden links between space, population, economy, environment and governance and identify common problems in decentralized systems

Path reasoning

- From evidence to judgment, proposal, decision-making and projects, the basis of planning and pre-conditions can be checked in reverse

Impact analysis

- Identification of affected objects and sectors along spectra relationships when policies, data or projects change

Knowledge deposition

- Continuous accumulation of research, judgment, controversy, proposal selection and implementation results as reusable, updated knowledge assets

Three-tier working framework

The proposal is organized on a three-tiered scale, “Integrated Research Scope - Overall Design Scope - Focus Areas”, which is attached to the Industrial and Future Cities Study, to the overall urban design along Jing-Zhang, and to the detailed conceptual design of Zhongzhi Park, Beijing AI Origin Community and Dazhongsi. The temporary binding scope is not used as a statutory red line or as a basis for approval. [data: global/site_business.geojson#SITE-001] [Depth](#)

1.3 Three sectors, two wings: overall division of labour among the three priority sectors

The role assigned to the " two wings of the Three Sectors" under the terms of the mission document, this proposal has identified the three priority areas as three distinct points in the source chain, moving the " source " from abstract concepts to specific areas of division of labour, affordability and synergy:

Focus areas	Positioning	Theme	Role in the supply chain
Zhongzhi Park	AI Autonomous innovation acceleration zone	Open to win, innovative development	All-innovation innovation + central power and governance voice, breaking critical core technology
* Beijing AI Community of Origin*	World-class AI Innovation Ecology	..enhanced advantage.	Element convergence and standard output, producing new ideas, new enterprises, new standards
Dazhongsi	AI Industrial cluster	• Ecological synergy	Intelligent neo-business + Southern low-time response, industrial landing and scene proliferation

Zhongzhi Park.. Breaking through..the community of origin.....

The three are supported and responded by a series of Jing-Zhang gallery links to the orbital network, constituting a complete chain of "*autonomous innovation breakthroughs* * factor sources gather together *industrial ecological convergence**"; two wings (transport, robotics, public space nodes and peripheral training centres) are jointly responsible for the overall positioning of the "World-class Access of Orginal Innovation" phase. This location is the site selection for the urban design phase, the location of informal plots, and the re-election of specific plots within each district after obtaining official boundary and tenure information.

1.7 Functional structure: quadrant space skeleton

Under overall positioning, the Jing-Zhang Railway innovation belt forms the functional structure of "*Four Channels**" as the space skeleton of two wings in Sector 3 (compacting outline 1.2):

Functional Structure	Content	Main location
* Engine core*	University-house-business innovation cluster, source engine of innovation	Beijing AI Community of Origin (core)
* Double iron complex*	Transport-Industry-Public Space Integration Corridor, Space Main Axes of Zone Three	Jing-Zhang corridor (cross zone 3)
resiliencecomputational power district	Safe, low-carbon, resilient infrastructure layouts that underpin autonomous innovation	Zhongzhi Park (core)+ outer training centres
* Biographical belt*	Railway heritage activity and community renewal of integration belts to fulfil the cultural corridor attributes	+ Dazhongsi

The four corridors are not four parallel strips, but are superseding four-fold properties on the same space:*the science engine core** provides the "density" of the source,*the dual iron complex** provides the "connectivity" of the flow,*the resiliencecomputational power area** provides the "security" of operation,*the cultural life belt** provides the "heavy" of identification.

1.8 Division of labour and regional synergies in the three regions

The positioning of "world-class " also requires a clear division of labour between the three sectors so that the supply capacity does not fail by a single-point impact (see also 4.4 for details):

Focus areas	resilience character	Main space carrier	Relation to source of original innovation
Zhongzhi Park	AI Independent Innovation + Computational Power and Governance Speech	Logic and training centre, chip testing workshop, model validation site, safe governance laboratory	Take over the primary power centre and standard governance to backstop the source
Beijing AI Origin Community	World-class innovation ecology	Open Source Publishing Room, Results Conversion Street, Data Elements Lodge, Standard Evaluation, Talent Community, Endside Construction Portal	The scene core, six elements and standard output.
Dazhongsi	Smart original business (situation response + edge response)	Enterprise service, product release, lightweight edge node	Mobilization of a resource and a computational power service for low-time and disaster preparedness in the south

The peripheral training centres carry out large-scale training with high energy consumption over a long period of time; Zhongzhi Park takes responsibility for full-scale reasoning, chip testing and safe governance; the origin community assumes the source of factor, standard-setting and credible data flow; and the three sectors together with transport, robotics, and public space nodes form the complete chain of *"* periphery training — Zhongzhi Park full-barrel reasoning and validation — origin factor source and standard output — the edge response of the three regions*."*

1.9 Design of boundaries

This section is a conceptual orientation oriented towards the urban contest. The specific plots, spatial scope, functional matching and construction scale of the three priority areas are designed to evolve and are not the result of approved site selection, land use or engineering; “World-class Access of regional innovation” is a planning vision and evaluation criterion that does not constitute a commitment to the construction or implementation of any subject.

The formal deepening must be preceded by additional information on the boundaries of the official focus areas, the conditions of regulatory detailed planning, terrain heights, electricity and municipal capacity, tenure and environmental impacts; the information should be obtained by re-calibrating the location, functional structure and division of labour between the three areas and by

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sources.json ' registering external sources of discussion and the unknown conditions of registration in
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assumps.json '.

Main references

1. The 100-year Jing-Zhang AI Innovation Belt Open Job Book for Smart Bodies is a local summary of clean-up rights. 2. The proposal structure outline “1 Project Understanding and Functional Structure”. 3. In April 2023, during a study tour in Guangdong, General Secretary Xi spoke about the importance of "enhanced science, technology and innovation (STI) sources." 4. Keynote address by President Xi at the opening of the World Congress on Artificial Intelligence and the High-level Meeting on Artificial Global Governance. 5. In April 2025, General Secretary Xi Jinping gave an important indication on the occasion of an innovative ecological community visit to the "mode space" in the Shanghai area. 6. The Secretary-General of Xi Jinping has made important presentations on the convergence of artificial intelligence industries, innovative ecological construction and high-quality industrial development. 7. This proposal 4.4, "Soft resilience Smart Program" (two-sided division of labour in Sector 3).

Before being formally incorporated into the submission package, the leader ' s key statements and specific sources of the subject matter case should be registered at

`sources.json` ' and the unknown conditions of location, boundary, tenure, etc. should be registered at

`assumptions.json` '.

Integrated study of industry and future city studies

The focus of AI-native planning: Jing-Zhang's Second Self-Assessment is not on general industrial clustering, but on moving from technological sub-seat autonomy to the autonomy of city recognition, planning, spatial action and resilience, so that institutions of higher learning, business, talent, capital, reliable data, testing the environment and public life together support new ideas, new enterprises, new standards.

1.1 Overall positioning: world class

The competition is aimed at complex urban areas along the Jing-Zhang Railway - a collection of science, technology and innovation resources, double-column collages, and the history of history. Thus, the overall positioning of the proposal is reduced to one sentence:

** Centennial Jing-Zhang ■ AI Innovation — World-class Association.**

Using the 100-year history of the Jing-Zhang Railway as a three-fold attribute, the "AI Generation Science Corridor, Cultural Corridor, Resolienence Channel", the Jing-Zhang Railway innovation belt was built as a source of world-class planning for the Chinese artificial intelligence age.



Figure A: Overall positioning: 100-year Jing-Zhang □ AI Innovative Area

This general orientation is carried out at three levels, corresponding to the Core Three (1.2):

Positioning level	Content	Answering the proposition
* A corridor* (space carrier)	Jing-Zhang corridor as a triple attribute of "Creative + Culture"	How are we planning this?
* One source* (core function)	"World-class access of international innovation" -- producing new ideas, new businesses, new standards, not just aggregate elements.	How does planning work?
* A methodology*	"Planning Operating System" - Human-engineer planning with city-run feedback	Planning how to produce value

“World-class” is not a slogan, but rather a testable positioning standard: source of extraordinary innovation is distinguished from the general AI campus, not by the more intensive elements, but by ** greater adaptive capacity ** — when technology paradigm shifts, conventional power supply shocks, capital enters the lower cycle, and brain drains, the ability of ecology to replace itself, restructure itself, and regenerate itself. This judgment is in line with the dictum of this proposal 4.4 "Soft Reassilient Intelligence":** the core competitiveness of the world-class AI source of original innovation, which is precisely its ability to produce new ideas, new enterprises, new standards in the midst of continued instability.**

1.2 Top-level pull: an important discussion of sources of science, technology and innovation

Secretary-General Xi Jinping.

"To strengthen the science- and innovation-source function and work towards the creation of new scientific discoveries, new technological inventions, new industrial orientations and the development of new ideas from scratch, to become the first discoverer of science patterns, the first inventor of technology inventions, the first pioneer of innovation industries and the first practitioner of innovation ideas, to produce a set of original results of basic research and applied basic research that will break through a critical core of technology in the neck."

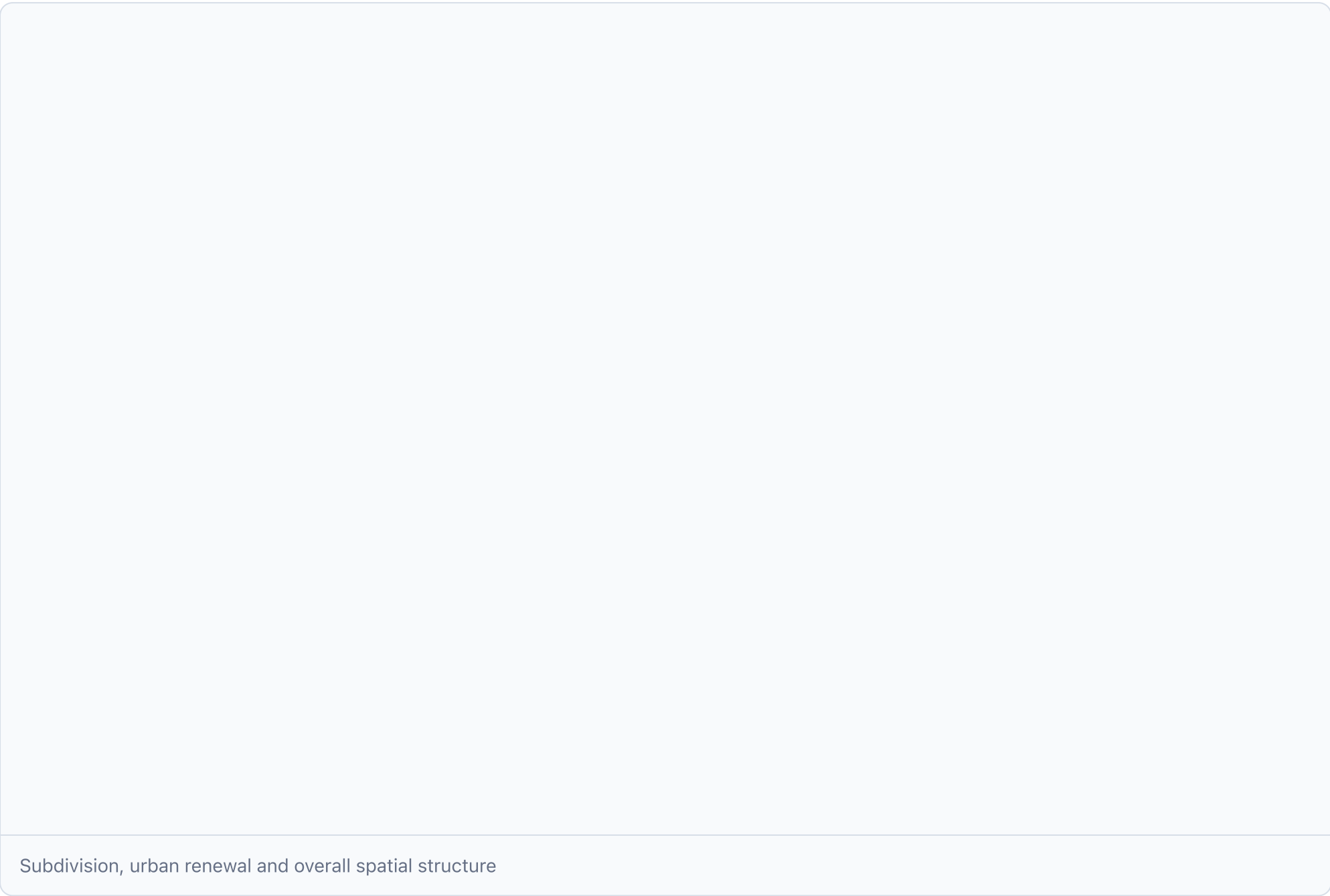
This discussion provides the most direct top-level pull for "world-class access of international innovation" and anchors the four dimensions of this program's positioning - ** "four first"*:

- **First discoverer of scientific law:** based on the collection of institutions of higher education as the source of basic research in AI;
- **First creator of technological invention:** based on Zhongzhi Park ' s autonomous innovation, it has become a source engine for key core technological breakthroughs;
- ** First pioneer of the innovation industry:* ***based on community elements at the origin, becoming the source of new industries, new business;***
- ***First practitioner of the concept of innovation:** based on the "Planning Operating System" approach, it was the first scene of the change in the paradigm of planning in the AI era.

The spatial division of the "Four Ones" and the "Three Twos" mirrors each other and forms a whole chain of ideas, technologies, industry, ideas.

urban renewal and regulatory detailed planning depth urban design

The overall design involves five types of structural problems entering space structures, partitions, public facilities and urban renewal strategies. The conditions of volume, height, density, regression, etc., relying on formal regulatory detailed planning are to be maintained; the submitted map layer is to design models that are not statutory regulatory detailed planning conclusions. [standard: MOHURD-CONTROL-DETAILED-PANNING] [Depth](#) [Spatial data](#)



It is necessary to map 02 to map the area and the overall space structure.

Chapter three, problem identification, planning for what?

Problem identification: from phenomenon analysis to causal diagnosis

To identify "why" and "what's most effective to intervene" rather than "what" and "how relevant" to stay. The floor answers the third question of the six-tiered progressive logic - ** Planning for what to address **: Translating the current state of the Jing-Zhang Railway innovation belt into measurable, monitorable, interventionable causal diagnosis and diagnostic indicators to provide a basis for follow-up "program generation" and "discovery of countermeasures."

Causal diagnosis five core phenomena - cause and effect - diagnostic indicators AI Innovation zone 3.1

Methodological shift: from relevance recognition to causal structure analysis

This level is a key step from the "problem identification" in the evolutionary logic of the six tiers of the overall proposal, and the methodological shift can be summarized in one sentence:

From the identification of relevance to the study of causal relationships, you can identify "why" and "what is most effective to intervene" using causal extrapolations, neurological networks, structural equation models, etc.

Traditional problems are often identified at the level of "situation" and "relevance" and can easily be miscalculated as causes and consequences, thus proposing a response to the symptoms and root causes.

Dimensions	Traditional practices (relevance)	This chapter approach
* Issue description*	"A is related to B presence"	"X leads to Y, aggravated by Z" -- clear causal chain
* Targets of intervention*	Fix the phenomenon itself.	We'll locate the controllable leverage point for what's most effective.
* Result pattern*	Qualitative description + Isolated indicators	Compableable, top-map, monitorable diagnostic indicators

Thus, each core issue is presented in a uniform "** phenomenon – causally – diagnostic indicator**": the phenomenon answers "what it sees", the causal answer "why it happens" and the diagnostic indicator answers "how to quantify and keep tracking", making problem identification a direct input of "program production" and "discovery of countermeasures."

3.2

Overview of the five core issues

The Jing-Zhang Railway innovation is faced with five core issues that are both independent and compatible in the AI era. The following table provides an overview of the problems, with more details of the deeper diagnosis of the problems in the next section.

#	Core issues	phenomena	The chain.	Diagnostic indicators
* /*	AI IRR does not match the facility system	High concentration of institutions, institutions and enterprises, but insufficient space for residence, commerce, culture and interaction	Job separation, long distance travel, break-up of innovation networks, brain drain risk.	Post - Housing balance, density of innovative facilities, time cost of commuting
II	Low-impact reuse along the dual-iron space corridor	Large numbers of inefficient industries, storage, idle land on both sides of the rail corridor	Complex property rights + security controls + lack of power to renew space and value sinking	Land-use efficiency along the routes, availability on both sides of the rail line, updating potential index
III	Conflict between complex transport systems and poor urban traffic management	Subway, peri-urban railways, high-terrestrial and urban roads overlap, with inefficient switching and fragmentation management	Multi-corporate operation + data not reached + insufficient predictive capacity, congestion and experience disproportionately	Switch time, congestion index, multiple mode travel share
IV	The space layout of the hub and its safety	Rapid expansion of new infrastructure, such as the Centre of Intelligence and Accounts, and the Grand Data Centre	Energy-use-land-security constraints gathering, layout risks and lack of resilience	computational power demand density, energy security, disaster risk exposure, networking
V	Contradiction between historical and cultural preservation and spatial renewal	Jing-Zhang Railway heritage is of high value, but conservation and utilization are difficult to balance	Protection of the rigidity of the border + blurring the dynamic path + inadequate public participation	Level of conservation of heritage sites, intensity of active use, public cultural identity

Question one.

AI IRR does not match the facility system

Completed

The phenomenon — causality — is that the complete and deeper diagnosis of the diagnostic indicators is as follows (comprising a conglomeration pattern of creative resources, a mismatch of facilities with six types of gaps, calculators of diagnostics).

Problem identification: AI Generous Resource Pooling, the support system is still in "Securing Basics."

Jing-ZhangAI’s innovation has brought together **535** university faculties and **556** AI enterprises, but the existing public services are still based on the residential “basic” logic. Put talent, business, and basic life together in the same spatial context, the core problem is clear: the intensity of the collection of innovative resources has outpaced the way the current system is carried.

Problem: AI has created a pool of resources, and the system of supporting facilities is still in the "basics"

The "human spot" and the "facility spot" are stacked together in the same picture, and are wrongly matched: where the resources are gathered at extreme speed, but where the combination is still missing.

Resources: 535 university faculties + 556 AI enterprises

The innovation zone has a high concentration of creative resources, with a spatial two-polar pattern of North-end talent, South-east industrial convergence.

- ** 535 places in higher education** Data: using the Qinghua University (400+ department/building) as the source of talent, together with Beijing Forestry University, China Mining University (Beijing), Beijing University Medical Department, Beijing University of Science and Technology, etc. constitute the country ' s most densely populated AI talent supply. - **AI enterprise 556 ** Data: 128 families at five junctions ride dust eradication, several times more than the business area of the second tier, which is the core engine of industrial clustering. - ** In the middle is "the corridor" **: between the North and the South, between the North and the South, between a commuting and collaborative corridor of about a few kilometres long — a combination that should have followed, but not followed.

Where is it?

Existing public services are based on ** basic level of protection**, which is based on a standard set of living standards: the solution is basic life such as "buying food, going to school, going to health care, basic sports" and not "new life" for the AI. The result is:

Core judgment

Where talent density is highest, it is precisely the weakest link in innovation. The current situation is accompanied by the problem of "inhabitation" and failure to solve the problem of "innovation" — an additional support system that is centred on AI,** rather than on the population.

Space overview: tertiary and AI enterprise resource collection Defaultly lays out the original spot for the 535 + 556 AI enterprise; you can switch to the top right panel to see the resource layer.

Current public service provision combing: basic level 298 and six types of gaps

In contrast to the need for innovation, the existing system of public services can be characterized as "one base + six types of gaps".

Level	Current status characteristics	The question of the need for innovation in relation to AI
Strategic layer Official caliber 298	Basic living standards for buying food, attending school, attending medicine, basic exercise, etc.	Targeting "inhabitants" rather than "innovators" and mismatching with high-density spatial distribution; bottom-up rather than empowerment
Innovation package (type 6) Coffee social/fitness exercise / library self-study / conference road show / canteen catering / display community	Scattered, self-emerging, fragmented; Third Space and Innovation Services are generally missing	It's impossible to carry the "unexpected interaction" of AI high-frequency talent -- innovation depends on collisions.

Population caliber further illustrates the size of the mismatch:

- Ph ≈ 4.50 million; AI Innovative Population Pai ≈ 60,000 (horizon-4080,000) - This means that the new package should not be "up to 450,000 inhabitants" but "a continuous spectrum for an innovative population of 60,000 ", taking into account the sharing of the population.

Thus, the core of the proposal is not “supplementation of points”, but ** a new set of logic** — moving from “basic, point supply” to “continuous spectroscopy, four layers”.

Question II

Ineffective reuse along the dual-iron space corridor

Completed

The phenomenon — causality — is the complete and deeper diagnosis of diagnostic indicators (with four sub-problems and measurable diagnostic indicators).

Ineffective reuse along the dual-iron space corridor

Following the entry of the Jing-Zhang Railway, and the transformation of the 13th line “to the sky,” the city that had been cut by the railway was re-sewed into a high-value gallery. But the large number of old factories on both sides of the corridor, the trading market, the negative space under the bridge, the compound on the back to the park was not synchronized, creating a development error that “released potential and the status quo is still inefficient.” This section focuses on the “low-impact redevelopment and utilization problem along the bi-rail space corridor” (core issue two in the overall framework), which follows the logic of the phenomenon-causal diagnosis.

Double-barracked, inefficient redevelopment, land efficiency, updating potential index, AI innovation zone. Overview

Overall judgment: potential release and inefficiencies

At the same time, AI needs industrial land to be a piece, continuum, capable of carrying innovative activities, and there is a clear mismatch between “many, but fragmented, areas of ownership” and “industry needs to be isolated”.

** Cause and effect.** Physical barriers disappear and the value of the land on both sides of the porch rises after the transfer of two irons from the ground to the ground / height; but the complexity of property rights, high security controls, and insufficient power for renewal have been superimposed, resulting in market, squatter, old industry and sub-bridge space not being updated in a timely manner, resulting in ** spatial fragmentation and value sinking.** The causal chain presented in this section is: complex property rights + high safety control requirements + insufficient power for renewal.

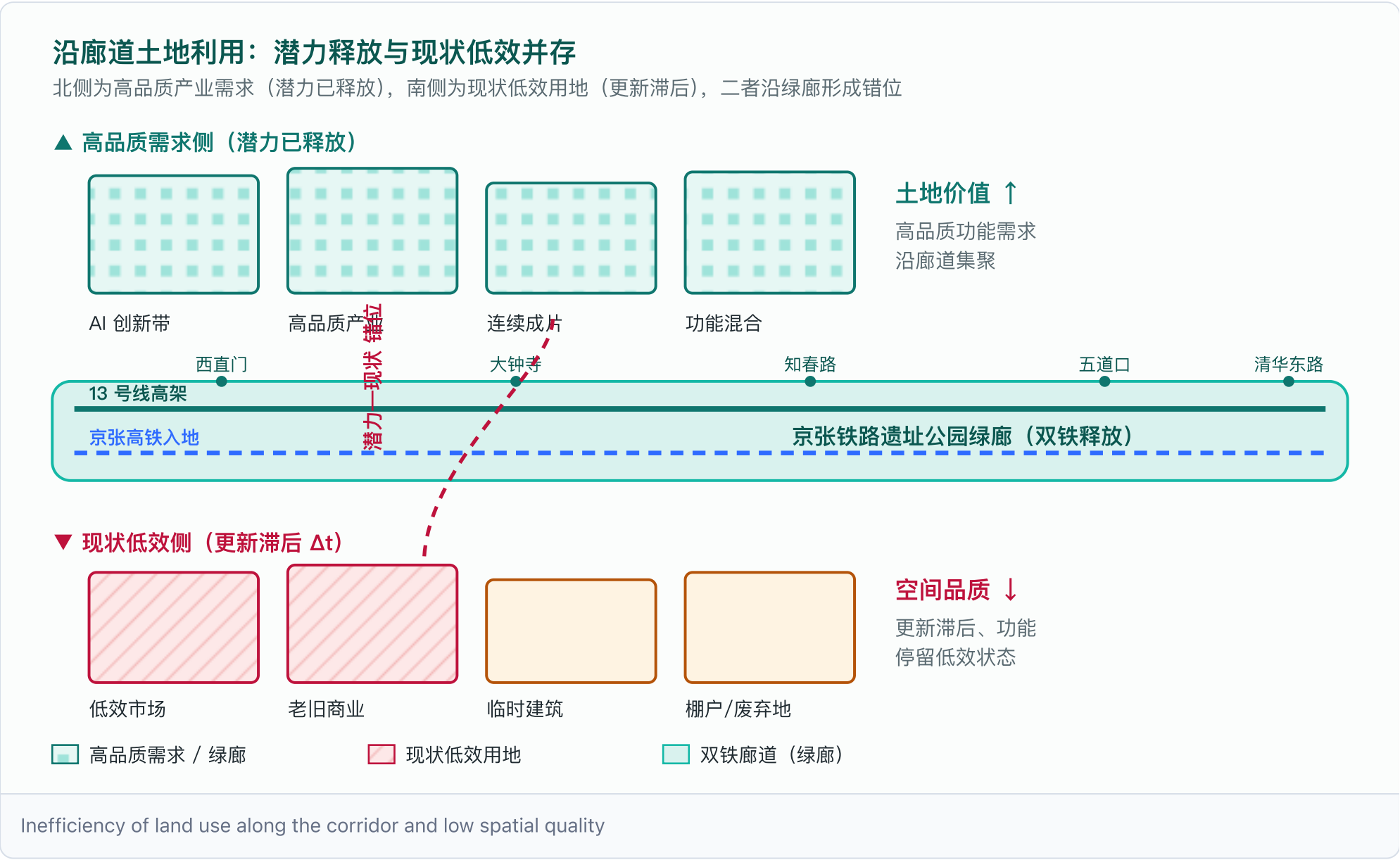
This issue, "Ineffective reuse along the dual-iron space corridor", is at the core of the overall problem framework, with two issues: its base floors, such as land use efficiency along the top-down line, and the availability of both sides of the railway (supported by city-time data foundation), and its fine re-activation and continuum (generation of counterpart proposals and discovery of countermeasures). The results of the problem identification will be directly translated into quantifiable, monitorable diagnostic indicators and digital base construction needs.

Question one.

Label release and ground update "time difference" cause a misalignment

After a two-way landing, the physical cutting of the railway against the city was lifted, and the land on both sides of the corridor was transformed from “the edge of the railway devaluation” to “high-accessible and scarce resources” with a significant increase in the value of the land. But the old markets, squatters, old industries/storages, negative spaces under the bridge were not synchronized into newer orbits – space barriers were lost and development potential was present, while the current quality and function remained inefficient, creating a clear time gap and space error between the “exploitation potential” and the “status quo”.

- 9 km: length of corridor along the Jing-Zhang Railway - * two irons: **entry to the ground (high iron) + simultaneous release (line 13)** - ***T+Δ**: gallery value window period vs updated landing cycle



1 Trading market category (non-capital function defusing)

There is a gap between old construction style, low-quality space, and lack of accommodation.

2 Negative space class (large area, non-functional)

Under the bridge, unoccupied railways, temporary parking lots, the perimeter of the wall — a significant area — the lack of public functions and the persistence of a “visible, useless” negative state.

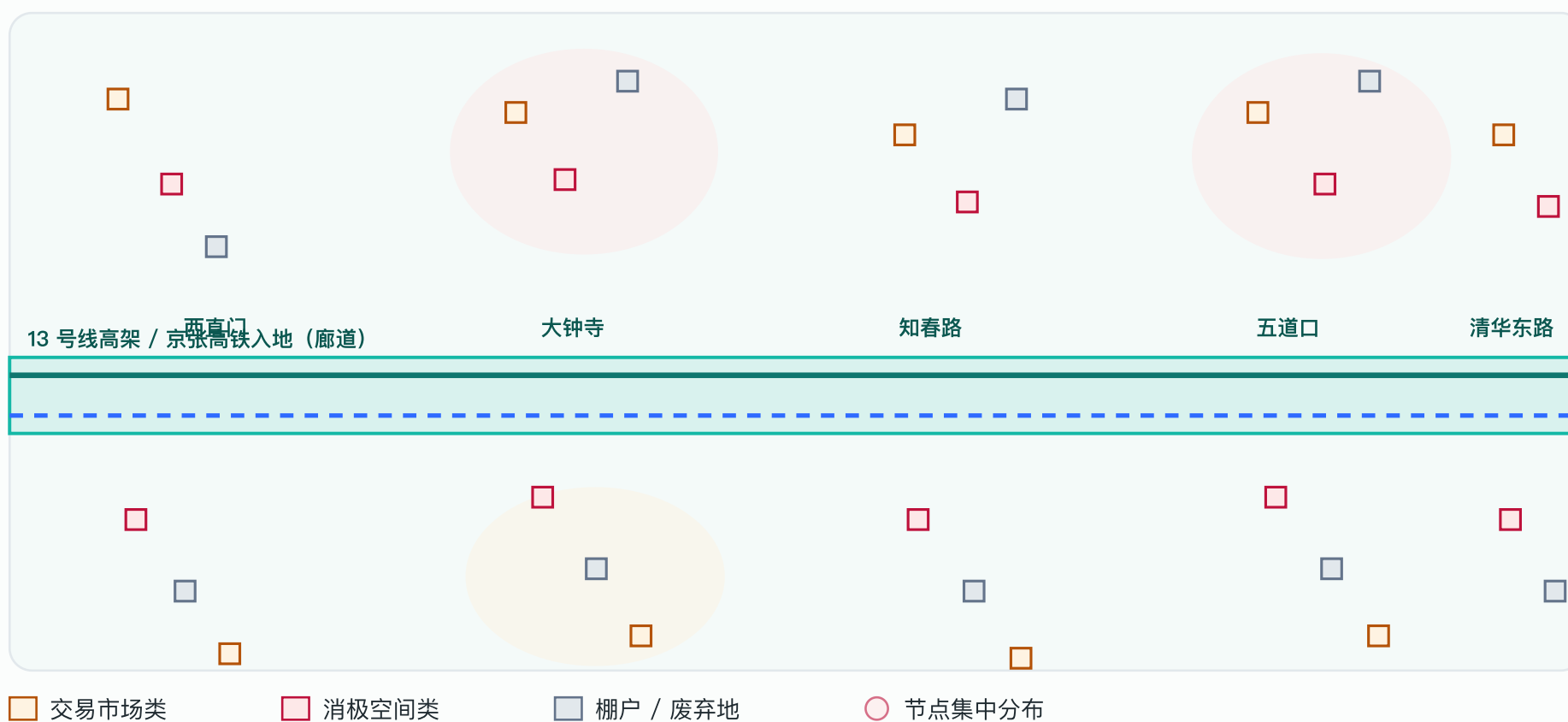
3 shanty towns and abandoned land

There are shantytowns and up-to-date plots around nodes such as Harun Road, the Four Rings West and the Five Crossroads, which are costly and complex to upgrade.

- **under bridge:** about 194 000 m2 negative space (proposal measurement) - * *railway idle:* **about 605 000 m2 (proposal measurement)** - * **temporary parking:** about 489 000 m2 (proposal measurement) - * *wall*:* about 4,977 m closed interface (proposal measurement)



西直门—大钟寺—知春路—五道口—清华东路等节点周边，低效图斑点状散布、难以成片整合



Distribution of old temporary buildings along the corridor

* Chart 2 — *Distribution of old temporary buildings along the corridor with inefficient plots.** The perimeters of the direct West Gate, Dazhongsi, Zhichun Road, five junctions, and East Chonghua Road are concentrated in old temporary buildings, with new spots scattered and difficult to integrate. Question III

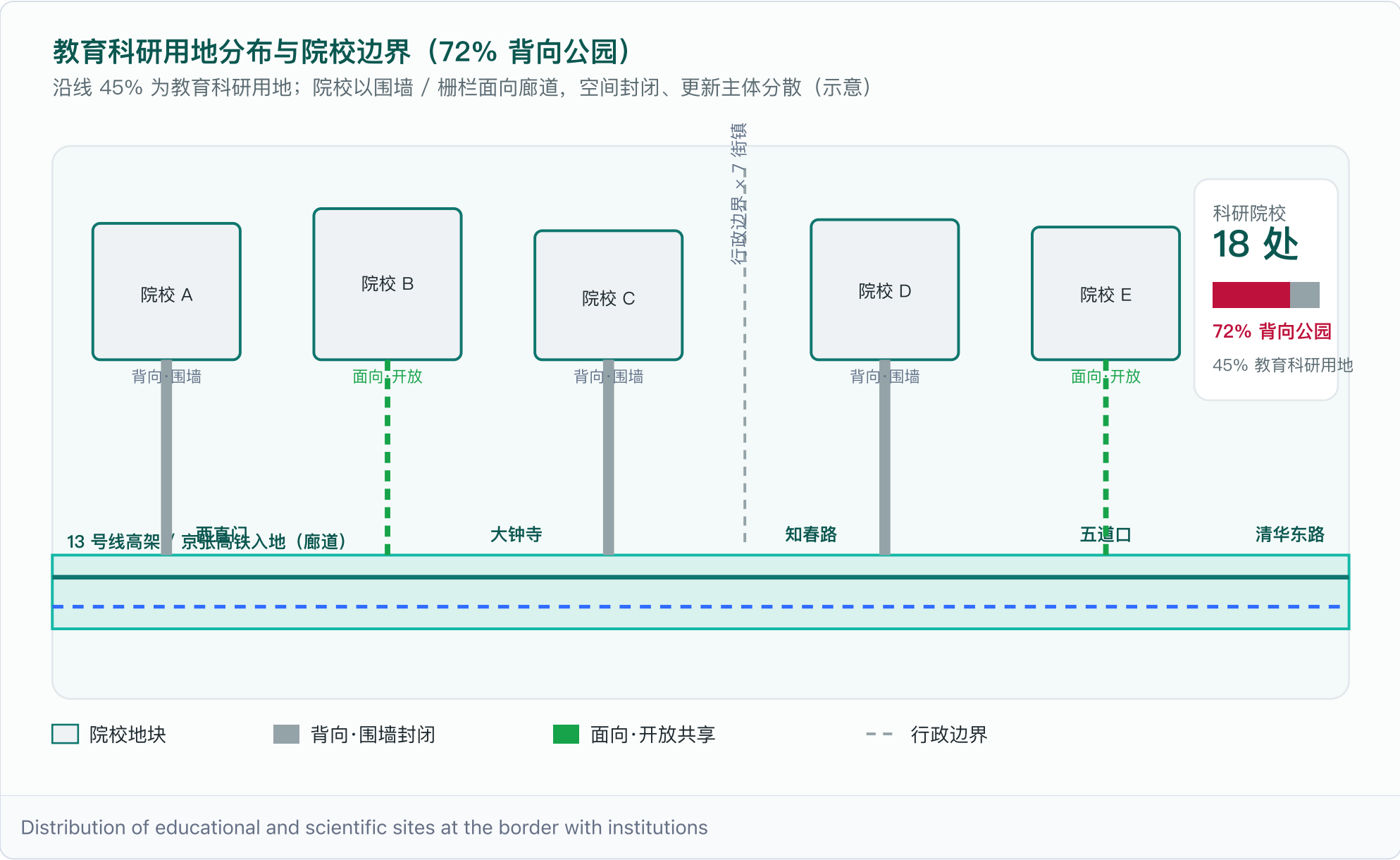
The compound wall is superimposed on the administrative boundary and the main body is updated and dispersed

It is not only a matter of space, but also of governance.

[illegible]

boundary, * administrative boundary, * administrative boundary, * administrative boundary, * administrative boundary, * administrative boundary, * administrative boundary, * administrative boundary, * administrative border, * administrative boundary, * administrative boundary, * administrative boundary, * administrative boundary, * administrative boundary, * administrative boundary, * administrative boundary, * administrative boundary, * administrative border, * administrative, * administrative boundary, * administrative boundary, * administrative, * administrative, * administrative boundary, * administrative, * and **

- **18 places:** scientific institutions, 72 per cent back to parks - **7 **** Wall: closed interfaces blocking accessibility and sharing**



* Chart 3. Distribution of educational and scientific sites along the boundaries of institutions.* 45 per cent of educational and scientific sites are located along the road, and 72 per cent of 18 scientific and research institutes are in the park, with walls and fences, leading to the closure of space and the dispersion of new subjects. Question IV

Lack of public space and ecological networks

Even though the corridor was released as a green belt, its “public” and “ecological” aspects were not fully realized:

- Open space is separated by various interfaces, * 75 per cent negative space*; - Greenness * is scattered *, does not form a continuous green net and the ecological corridor effect is difficult to achieve; - There is a lack of linkages and design synergies between nursery/parks and insufficient release of ecological values and public vitality.

- **75 per cent:** Open space is passive - **22:** Open space paragraph (mostly passive, proposal measure) - * dot*: green land is not a continuous network

Diagnostic indicators

Diagnosis indicator: from phenomenon identification to measurable, monitorable

Following the "from phenomenon analysis to caususal diagnosis" approach, the above-mentioned four questions are translated into quantifiable, top-moderable and continuously monitored * diagnostic indicators* that can be monitored from the digital base. The indicators are both urban spatial data foundation and directly corresponding to the monitor of subsequent digitized products.

Problem	Diagnostic indicators	Status / Threshold (calculated)	Data sources and monitoring tools (digitized base)
<i>* Question I Time difference*</i>	Land efficiency along the routes	Low-utility ratio, unit output intensity; land value index after release of corridor, but current quality not synchronized	A map of land + land price monitoring + remote sensing image recognition (AI one graph)
Update potential index (time dimension)	Local block "Identification Planner Update Operations" phase and window loss risk warning	Update the special planning library + the progress board (camp town applet)	
<i>* Issue II Type fragmentation*</i>	Land use efficiency (structural dimensions) along the routes	Trading market category / Negative space category / Area of abandoned squatters: ≈ 194,000 m2 under bridge, ≈ 605 000 m2 for railways, ≈ 489 000 m2 for temporary parking, / 4,977 m for walls	Low Use Current Status Bank (Class-Package) + Program calibration
Update of the Potential Index (one dimension)	Number of composing units, fragmentation (platform connectivity)	Spatial unit spectrometry + image connectivity calculations	
** Question No. 3	Accessibility on both sides of the railway (shared dimensions)	72 per cent back to park colleges (18 places); number of cross- Street town coordination units (7 blocks); length of barrier interface	Open School Data + Street and Town Border + Multi-Purpose Access
<i>* Issue IV Public ecological deficiencies*</i>	Update Potential Index (Quality Dimensions)	Negative spatial ratio 75 per cent (22 open spaces); Greenfield network connectivity (points vs continuous green network gap)	Open Space / Greenlands Vector + Econet Gaps Visualization

The three core indicators mentioned above - land-use efficiency along the routes, rail availability on both sides, and an index of renewed potential - are the diagnostic core of the "low-impact reuse problems along the dual-iron space corridor"; this identification further refines them in this section into measurable sub-indicators and data support, as inputs for the successor "programs to generate" and "digitized products (campaigns)".

Question III

Conflict between complex transport systems and poor urban traffic management

Completed

The phenomenon — causality — is the complete and deeper diagnosis of diagnostic indicators (with five sub-problems and measurable diagnostic indicators).

Conflict between complex transport systems and poor urban traffic management

Jing-Zhang high-terrestrial, 13th high-line, superstitious urban road network re-construction, central eco-paste belt accompaniment, multiple TOD hubs and stereo-static traffic have evolved into a three-tiered complex transport system. But integrated traffic management is still at the level of static planning and manual experience, and it is difficult to cope with sharp contradictions between the complexity of multi-layered, multi-corporate, peak-dispersional transport systems and the lag in management levels.

3-storey composite traffic, switching efficiency, managing debrisification, congestion index, AI innovation zone Overview

Overall judgment: the system is complex and management is lagging behind

** The phenomenon.** Subway, peri-urban railways (Jing-Zhang high iron), line 13, urban roads are overlapping, creating a “three-storey complex” system of underground high iron, ground park green halls, high-mounted orbits; along the TOD hub, P+R, multi-public transport, and static traffic. However, the experiences of residents and visitors are not improving simultaneously with hardware upgrades.

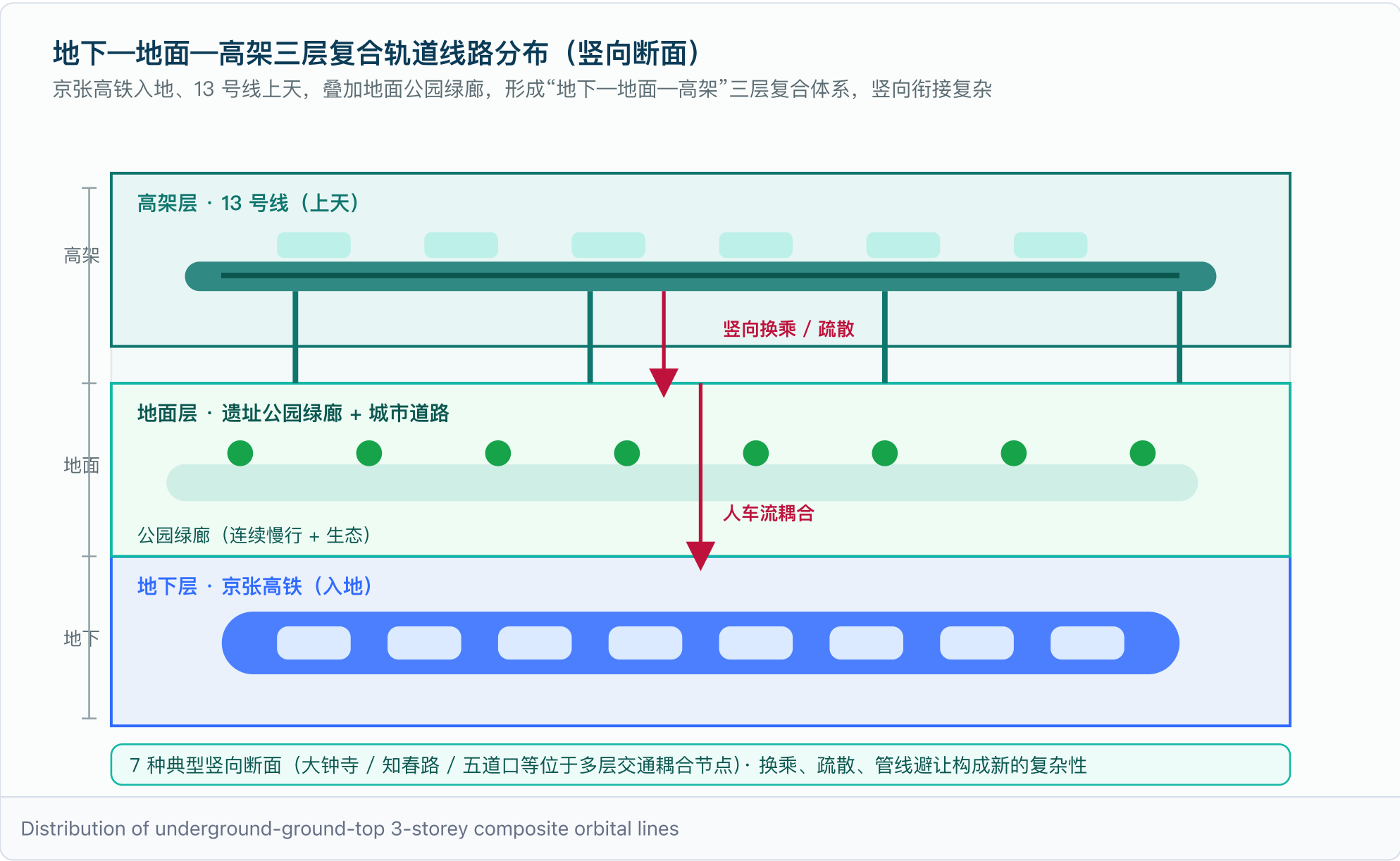
** Cause and effect.** Multi-subject operation (rail / subway / municipality / university / street town) + data not accessible + insufficient predictive capacity, leading to “hardware complexity” coexisting with “software flaring” and eventually ** both congestion and experience**. The causal chain that emerges in this section is: multi-subject operation + data not broken + insufficient predictive capacity.

This issue, "The contradiction between complex transport systems and poor urban traffic management", is at the core of the overall problem framework: its dynamic operation of traffic flows, traffic flows, public transport, etc., points down to menu-oriented responses such as smart-line strategies, rail corridor integration and active migration. Its diagnosis will be translated into measurable, monitorable diagnostic indicators and form the core of monitoring of intelligent transport platforms.

Question one.

Underground-ground-up three-storey complex, vertically complex.

When Jing-Zhang's high-terrains entered the ground with the 13th line, the zone formed **7 typical ground-to-ground space faults**. Underground heavy iron, ground parks, high-mounted orbits, traffic, piped connections are at a high level, and the traditional organization based on “two-dimensional plane” is no longer adequate — vertical dimension switching, evacuation, pipe avoidance as a new source of complexity.

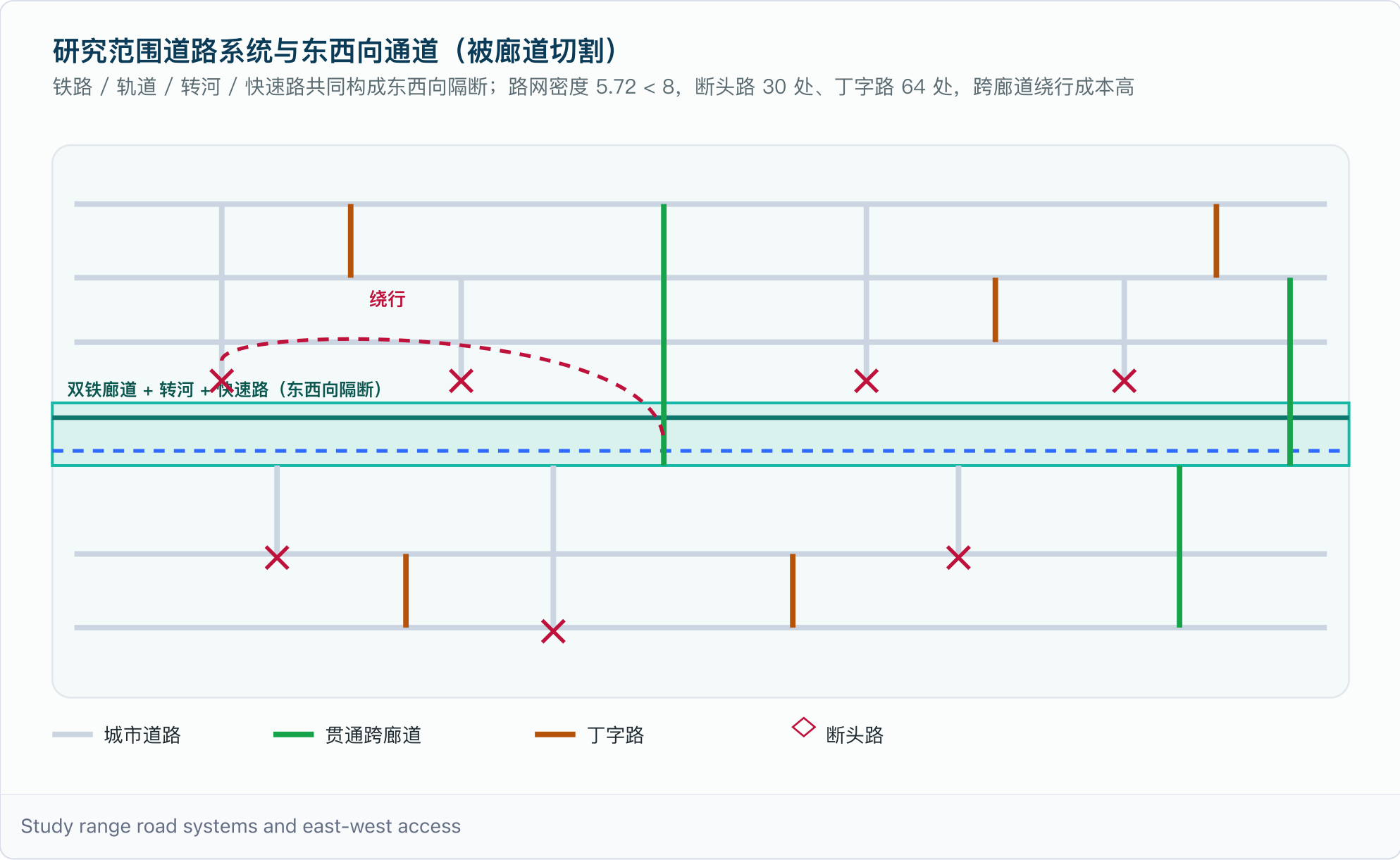


** Underground 1 — Ground - Three-storey composite orbital line distribution.** Jing-Zhang high-terrain, 13-line elevation/ground/land-entry alternation, forming seven typical vertical faults, Dazhongsi, Sichuan Road, five junctions, etc. are located at multi-storey traffic nodes. Question II

East-West connections have been cut for a long time, and traffic density is still low.

The combination of railways, orbits, trans-rivers, and fast roads constitutes an east-west cut-off, with residents having to travel further across the corridor. The study's comprehensive network density * 5.72 km, which is below the requirements of the new general rule 8 km/km2; **the current situation is * 30 break-ups, 64 tids** and inadequate network connectivity further exacerbates the inconvenience of crossing the corridor.

- **5.72:** Network density km/km2 (new general rule 8) - *30 *: * * * * 64 * * * * typography



* Chart 2 Study range road systems and east-west access.* Superhighways, fast roads, main roads, secondary roads, secondary roads, and secondary roads are well graded, but cut by rail / orbit / trans-river, there is insufficient link between east and west to cross-council, and the cost of bypassing is further amplified by the break-head and thong roads. Question III

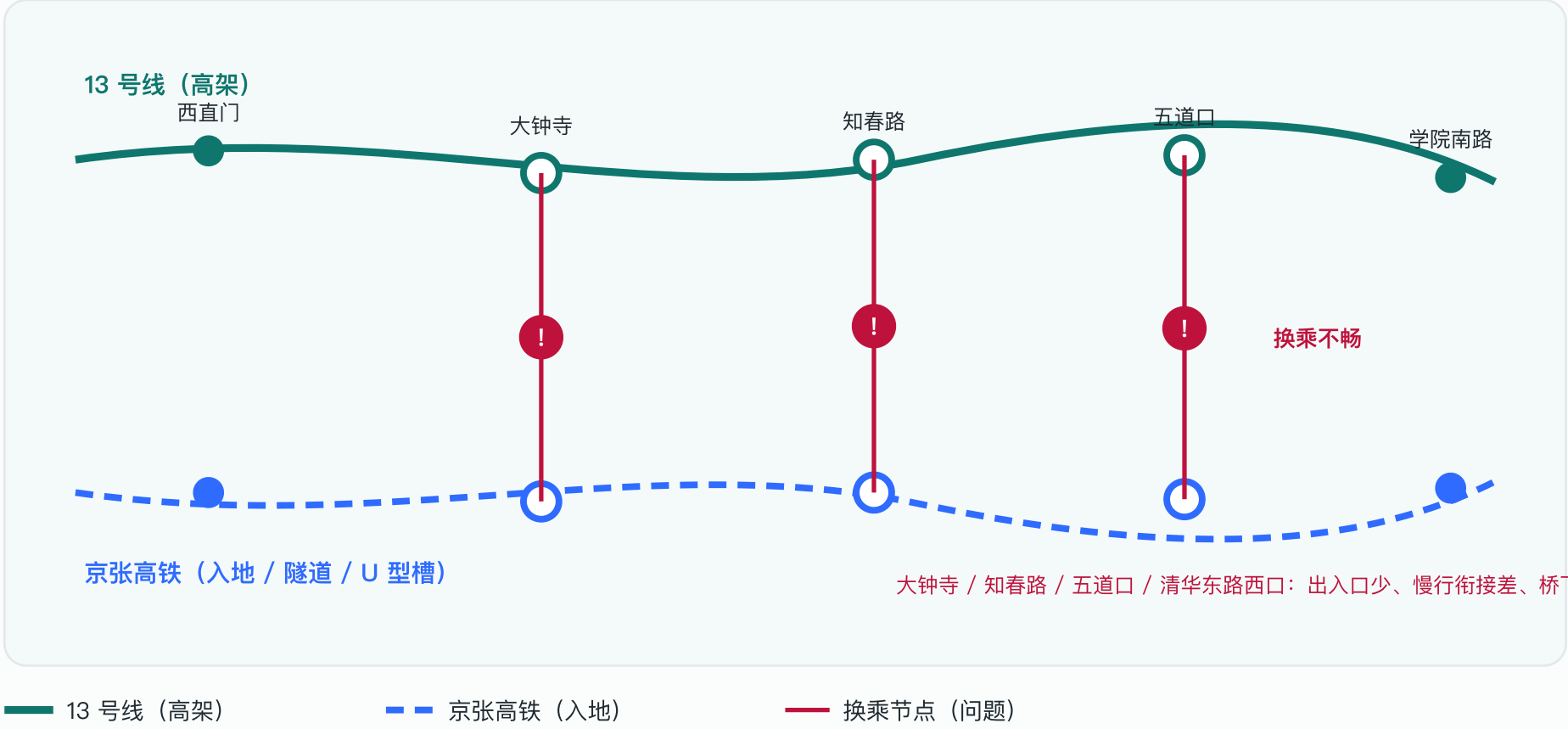
Site switching and access is inefficient

Line 13 is parallel to the Jing-Zhang culture, but traffic is not smooth, and there is no organic link to park land and traffic on both sides.* TOD facilitation area is only 70%*, and there is still *30% regional accessibility *; most site parks need to walk for more than 10 minutes from the station. There are problems such as low access to Dazhongsi, Zhichun Road, five exits, Tsinghua East Pass, lack of connection to the active mobility system, and interference with parking under the bridge.

- **70%: TOD convenience area (30% insufficient)** - **10 min+: majority of site to park walk hours - **4 stop:** Dazhongsi/Sichuan Road/5 junctions/Cinghua East Lush

城市轨道交通与站点换乘关系

13 号线（高架）与京张高铁（入地 / 隧道 / U 型槽）交织，西直门—大钟寺—知春路—五道口—清华东路西口—学院南路等站点换乘关系复杂



Urban orbit and site migration

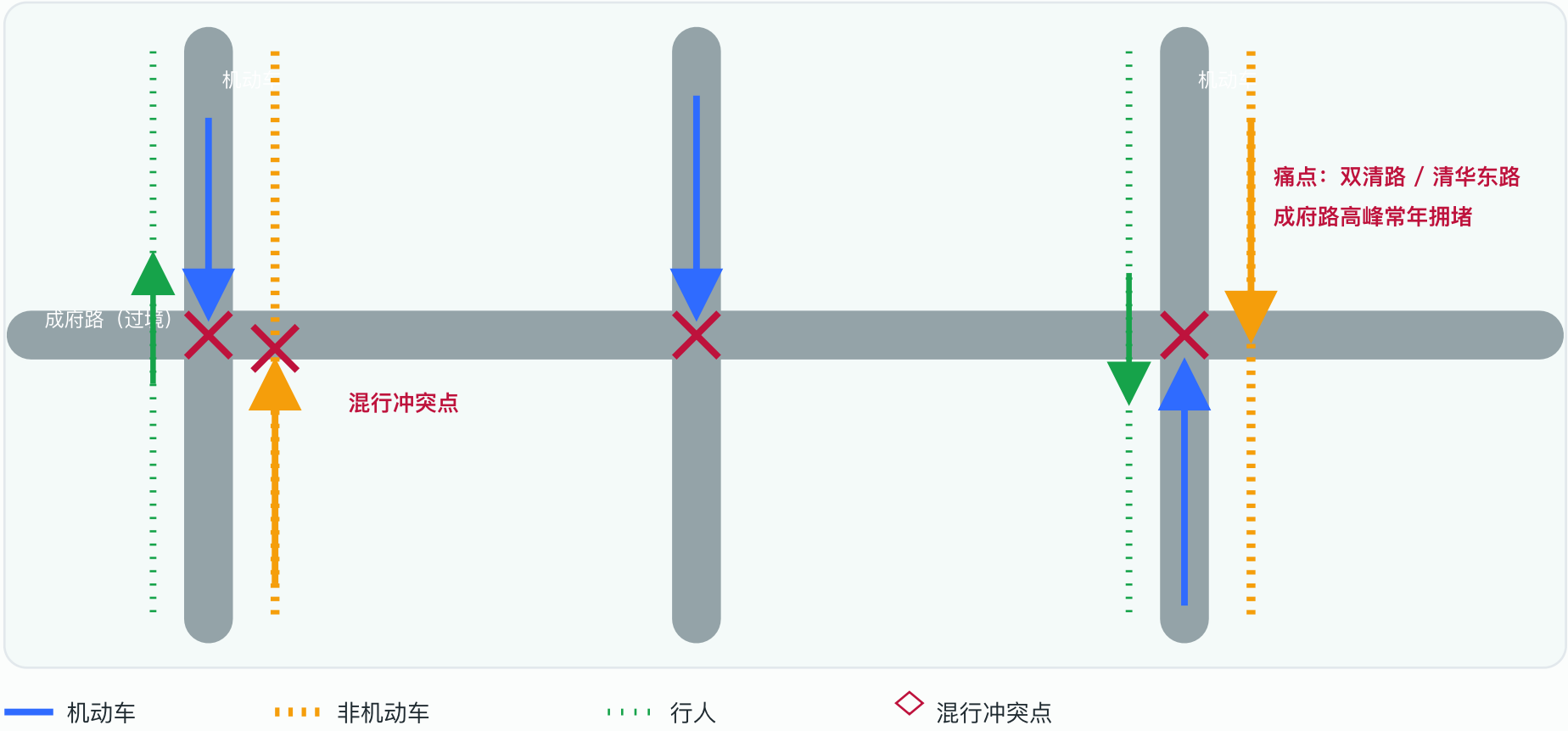
Chart 3 * Urban orbit and site exchange. Jing-Zhang high iron surface segment/ Tunnel segment/ U-train section is interwoven with line 13 above/underground segment / entry segment, with complex transject relationships between Xinjiang Gate, Dazhongsi, Sichuan Road, five junctions, Xinhua East Pass, South Road, etc. Question IV

Multiple traffic streams, severe nodes

Transit traffic accounts for a high proportion of traffic,* *t-roads, tilts towards two clear roads,** proximity to intersections* *leading to serious clashes between pedestrians / non-motorized vehicles / motorized vehicles, and disorderly traffic. The peaks of international contacts are constantly congested,** the twin roads and the East Qinghua road / the East Qing Dynasty road** form a traffic “point” where people, non-motor vehicles and motor vehicles interfere with each other in limited space.

城市道路交通分析与跨廊道联系（多种流混行冲突）

过境交通占比高，T 型路口、斜向双清路、交叉口间距过近，行人 / 非机动车 / 机动车混行冲突严重



Urban road traffic analysis and intersectional linkages

* Figure 4. Urban road traffic analysis and cross-corridor links.* Urban rapid roads, main roads and planning are not implemented, transit traffic is intertwined with transport to and from a limited corridor, and high-intensity mixed pressure is placed on the route to and from East Chonghua East Road, Chengzhou Road and South Road of the College. Question 5

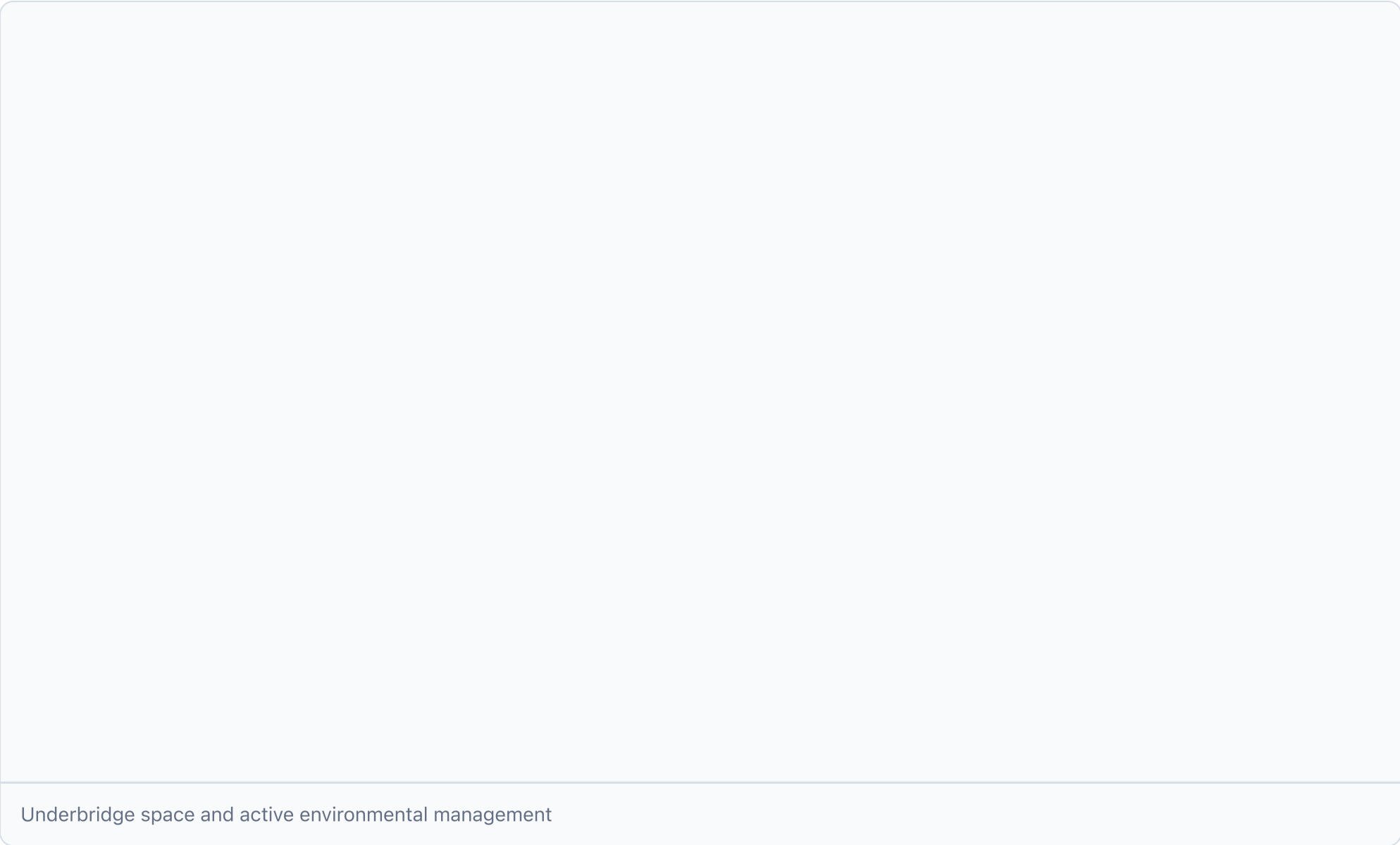
Traffic management tools in broad distribution: A lack of spatial order and a lack of physical mobility under the bridge

1 Under Bridge Space Management

Under-bridge space is occupied by a large number of illegal parking, circular parking and temporary construction, which has not been converted into orderly public space and has instead become a source of interference with access to an active level (which has also exacerbated the lack of access to problem III sites).

2 active system administration missing

Bicycles are parked in an disorderly manner, encroaching on bus stops; the narrowness of the sidewalks, their disrepute, their darkness and poor access experience have weakened the continuous activation of the green porch that should have been provided.



** Chart 5 — Underbridge space and physical environment management is widespread.** Underbridge and street space are too wide, functionally mixed, pedestrian access is usurped, bicycles and bus spots interfere with each other, and experience is poor and poorly managed. Diagnostic indicators

Diagnosis indicator: from phenomenon identification to measurable, monitorable

Following the "from phenomenon analysis to causal diagnosis" approach, the five above-mentioned issues were converted into quantifiable, top-map, ** diagnostic indicators that could be continuously monitored at digitized bottoms**, directly corresponding to the targets of the monitoring of the Smart Transport Platform.

Problem	Diagnostic indicators	Status / Threshold (calculated)	Data sources and monitoring tools (digitized base)
<i>* Question one, triple compound*</i>	Switch time (stereolink)	7 vertical switching of faults / length of evacuation; points of multilayered traffic conflict	+ BIM/IOT twin platform
<i>* Question two, east-west cutting*</i>	Concussion index (accessible)	Network density 5.72 < 8; break-up 30, typography 64; average bypass distance across the corridor	Net vector + real-time traffic + bypass path calculation
** Question three.	Swap Time / Multi-Model Travel Sharing Rate	TOD convenience zone 70% vs 30% inconvenient; station to park walk 10min; access gap	Public transport/orbit/active migration + site access monitoring
<i>* Question four, mixed conflict*</i>	Congestion index (order)	Intensities of conflict between nodal groups; periods of peak congestion in painful sections such as Chengdu Roads/Two Clear Roads	Multisource Perception Integration (Video/ Radar)+Auto-Detection of Events
<i>* Question V Managing roughness*</i>	Multi-modal travel sharing (quality)	Ground occupancy rate below bridge violation; rate of damage to active mobility lanes, frequency of bicycle invasions	Subbridge Portal AI Test + action quality rating + hand-shot closed loop

The core of the above-mentioned core indicators — time switching, congestion index, multi-mode travel sharing rates — is the diagnostic core of “complex transport systems and poor urban traffic management”; this recognition has been refined into measurable sub-indicators and data underpinnings in this sector, as a follow-up “proposal generation” and “digitized products (smart transport platforms)”.

Question IV

The space layout of the hub and its safety

Completed

Symptoms - Cause and consequences - The complete and deeper diagnosis of diagnostic indicators is as follows (including general diagnosis, four phenomena, causal chain and calculable diagnostic indicators).

The space layout of the hub and its safety

When smart clusters are built in high-density areas with “city-level infrastructure” volumes, site constraints are dominated by cost logic, resilience designs are simplified by “one-stop” and running governance is set aside by “building-light” – the “hot” of layout and “cool” of resilience constitute a systemic mismatch. This section deepens diagnosis by the three-part pattern of “results-diagnostic indicators.”

3.4.1 General judgment: “hot” of the general power layout and “cold” of security

** Large model training and intelligent reasoning have led to an explosive growth in demand for computational power, 100,000-calories calibrated smart computational power infrastructure, upgrading the computational power facility from an “enterprise room” to an “city-level infrastructure” that determines regional competitiveness. In this context, the centre of intelligence and extrapolation has rapidly expanded and competed along innovative corridors, presenting a clear tendency to “computal power up the street” – gathering in high-density built-up areas, in the vicinity of high schools, in the vicinity of orbital stations, to gain access to near-computation power and talent facilities.

computational power demand density continues to rise, with energy consumption, land and security constraints coming together on the same spatial scale, making the “gap” and “resilience” a structural contradiction: the layout seeks close to the user, close to the grid, close to the talent, demand for disaster avoidance, retention of redundancy, support for re-engineering — both difficult to meet in the context of an innovative corridor of indigence, and the prevailing practice of priority setting, compromise resilience, ultimately ** build up the risk of the layout together with the deficit of resilience. The causal chain presented by this problem is: **computational power demand outbreak + location constraint + simplification of site design + lack of governance risk and lack of resilience.**

This question is “the spatial layout of the Centre and its security response”, which is at the core of the overall problem framework: it is going to take up basic floors such as computational power demand density, energy security, disaster risk exposure, cyberresilience (which needs to be supported by city-time data foundation), and it is going down to ‘AI+softresilience intelligence’, factor redundancy and system self-resilient strategies (generation and discovery of countermeasures). The results of problem identification will be directly translated into calculable, monitorable diagnostic indicators and digital base construction needs.

3.4.2 Symptom I: Positive conflict between demand outbreak and location constraints

The communicative power centre, which is not an ordinary industrial building, has strict rigid requirements for site selection, and there is a systemic conflict between the spatial realities of high-density built areas:

- * *Energy constraints: **electricity supply must be sufficient and reliable, communications must be fast and accessible - but grid capacity in the core of innovative corridors is limited and transformer facilities are limited, and large-scale power power uses the same grid as electricity for urban life;*** - * **Environmental constraints:** the natural environment needs to be clean, ambient temperatures need to be conducive to energy conservation - but high density built-up areas have significant thermal effects, dust and

dust sources are complex, natural cooling is compressed for long periods of time throughout the year and PUE passive lifting; - * *Hazardous sources : **sites that must be far from dust, oil and smoke, harmful gases and corrosive, flammable and explosive objects — but where gas stations, storage logistics, catering and so forth in the built-up areas are scattered, the safety distance is difficult to meet;*** - * **Disaster avoidance:** Areas at risk of natural disasters, such as floods, earthquakes, need to be removed from areas where there is a risk of site-based pressure lines in areas along the corridors, in areas where water tables rise, and in areas affected by fracture zones; - **Function Compatibility:** The medium-sized and medium-sized Computal Power Center is not appropriate to be built in residential and commercial areas — but “computational power up the street” is precisely pushing the Commutational Power node towards high-density residential and commercial mixed areas with noise, heat dispersion, transport and direct proximity to the lives of the population.

* *Figure 1* * Site-based conflict illustration.** Rigorous location requirements (green) from the Commutational Power facility are in conflict with the spatial reality (red) of a high-density built-up area, with four dimensions of energy, environment, disaster, function, and available secure site space being squeezed in two directions.

3.4.3 Scenario II: concentrated layout and disaster risk exposure

In the context of frequent disasters and extreme weather regularization, centralized layouts are compressing systemic risks to single points:

- * *Flood risk: **low-lying, flood-prone areas such as riverine, ditch, low-dry bridge areas and upwelling areas, with a risk of pouring, leaching and general disengagement in heavy rains and sub-standard floods; catastrophic damage can be caused by an extreme rainfall when critical equipment in the first layer is below the flood level in 100 years;*** - * **Seismic risk:** generic power centre earthquake resistance category. If structural systems are implemented according to general type C construction, the probability of a strong-shocking cabinet being overturned, pipe fractured and cluster paralysis increases significantly; - **Environmental risk:** strong vibrating sources (orbit traffic, large machinery), powerful electromagnetic field interference sources (substations, high-voltage corridors) with continuous pressure on equipment stability and data integrity; - * *Single-point risk**: communal power cluster “on-island” is concentrated and will face overall failure in the event of power outages, fires, internal flooding, lack of cross-point takeovers and off-site disaster preparedness, and direct transmission of power to urban operations and industrial innovation dependent on it.

**Fig. 2* * Centralized single-point risk profile. **Communicational Power group shares the same grid and route, does not physically disperse the disaster preparedness node, and the disaster impact surface cover is completely ineffective.

3.4.4 Symptom III: Space resilience design gaps

Even when the site is selected, there is still a widespread four-fold gap in the spatial resilience design of the computational power facility, leaving “resilient” and “reconstructable” on paper:

1 Flood resistance position. Key data, governance control and core government power are placed in flood-prone underground space or low-lying areas; there is a lack of inflow control at building entrances, ramps, vents, pipes; underground garage ramps are isolated from the pipe entrances — rain pouring paths in extreme rainfall point directly to core equipment.

Fire-resistant position. There is no physical separation of rooms, power and energy areas, and a single fire can spread to the entire commutation power cluster; park ring fire lanes, separate fire zones, inadequate protection of fire and emergency flow lines; critical systems such as power supply, refrigeration, etc. lack a sub-class design for “a failure without transmission”.

**The total plane is unplanned and not reserved for long-term expansion land; the main network for the room, distribution and refrigeration is not built once; and the commutational power unit is not organized in a layout, resulting in a “step-in” full complement that is out of step with the real growth of the business, boosting investment risk and forcing subsequent build-up to disrupting already-on-line operations.

* *Peaks can be re-engineered.* * Civil construction is insulated with hardware depths, which cannot accommodate the evolution of aircraft containers from low-capacity servers to ultra-high-intensity liquid cooling cabinets; distribution, refrigeration modules are not superstitious; pipelines, bridges, interfaces are not standardized, hardware replacements or structural adjustments require large-scale civilization; individual rooms, groups fail in isolation and malfunctions are transmitted along shared systems.

Figure 3. Space availability compares to ideal framework. The status quo is four-dimensional gap (red) and there is a systemic gap between the ideal resilience profile (green) of "Key Layer Lifting + Separatization + Growth + Module Decomposition".

3.4.5 Symptom IV: Inadequate Elements and Systems

The central power centre depends not only on the room itself, but also on the system of supply and operation of the elements. There is now a clear “softreliance” shortboard at both the factor and system levels:

Element resilience

- * *One-point reliance on manpower: **over-reliance on a small number of key local personnel, lack of off-site specialist support routes, AB round value and the sinking of knowledge base, and loss of critical personnel as a systemic risk;*** - * **Vendor lock-in risk:** complete power supply, hardware, software, movement control platform tied to a single supplier, single chip, single cloud platform, single movement control system failure or break-out for direct transmission to the entire cluster; - * *Open-source supply chain risk: **no upstream stoppage or loophole explosion would be possible if the basic movement control framework, cluster management, surveillance components were directly and nakedly used in an open-source version from overseas and were not reinforced by nationalization or internal maintenance branches;*** - * **Data tolerance disaster gap:** ungraded multi-copy storage of heat/temperature/cold data, physical fragmentation of hazard and disaster preparedness nodes not achieved across risk zones, uninterrupted running mechanisms for training missions and loss of hardware or zero; - **Commutational power resources full of hard binding:** generic power, intelligent power, over-calculated power, no downgrade mode of operation, inability to retain core power, shut down non-core operations under extreme malfunctions, and lack of cross-regional movement channels for local power.

System resilience

- * *Construction light runs: **resilience indicator (acceptable downscaling rate, RTO/RPO, failure disposal process, confusion exercise requirements) does not embed the full project research, design, acceptance and operational appraisal process, only with PUE and general power peaks;*** - * **Confusion of standard systems:** confusion of infrastructure levels (standards for building rooms) and of operational levels (levels of intellectual ability) of companies, with private closed standards for interfaces, hardware, movement control agreements, leaving insufficient for future new standards and new hardware iterative; - **Governance synergy deficit:** computational power operators, grid companies, communications operators, tenants, regulators have unclear emergency connection and response processes, smart body alerts are isolated, and there is a lack of unified incident bus and global visualization.

3.4.6 Correlation chain: from demand outbreak to deficit

Linking these phenomena together, the causal structure of the problem presents a clear chain of progression:

- Construction surges towards high-density built-up areas (for near-computational power, helper, helper)

The key leverage points are two interventional links: * *the site end's “safe border front”** (which makes disaster avoidance, hazard-source distance, functional compatibility a non-compromising rigid constraint, with a negative vote at the downside)* *and* * *the design operating end's “resilience indicator embedded”** (which incorporates redundancy, downgrading, re-engineering, self-healing into research, design, acceptance and operational appraisals). The former determines whether the risk enters the field and the latter determines whether the risk is absorbed.

3.4.7 Diagnostic indicators: calculable, mapable, monitorable

Following the “from phenomenon analysis to causal diagnosis” approach, the above-mentioned phenomena are translated into quantifiable, top-moderable diagnostic indicators * *that can be continuously monitored at the digitized bottom.* * The indicators are both city-time data foundation and directly related to the monitoring of subsequent digitized products (soft-resilience intellectual platform, computational power-energy-security coordinated movement).

Diagnosis dimensions	Diagnostic indicators	Status / Threshold (calculated)	Monitoring instruments (digitized base)
<i>* Layout constraints (site security)*</i>	Site Compliance	Computational power facility and flood zone, flood holding area, seismic fracture zone, hazardous chemical sites, strong vibrancy/magnetic sources; whether level A facilities avoid residential and commercial areas	National Space "One Chart" superimposed site compliance automatic verification
COMPutal power demand density	Area grid capacity available in a unit area COMPutal power deployment (FLOPS/km2) vs; intellectual nodes density match load load capacity	+ grid load monitoring	
<i>* Catastrophe exposure (space resistance)*</i>	Exposure to floods	High ratio of critical equipment to flood level per 100 year (1.0 1.0 m compliance rate); ratio of sub-dry bridge area to low-lying river area commutational power node; groundwater level distance	Digital twin Water Safety "four-prevention system" + simulation of inundation
Structural earthquake resistance and fire safety compliance	Seismic resistance class compliance rate (A level B); machine/electricity/storage module ratio; fire lanes and fire ward independence rate	Construction BIM billboard + network monitoring of fire protection items	
<i>* Energy and Networks (Elemental Redundancy)*</i>	Energy security	Electricity reliability rate (target 99.999); two-way independent municipal electricity + diesel generators + storage redundancy rate; PUE versus green electricity ratio	Electricity SCADA + Energy Efficiency Monitoring Platform
Network	Multi-operator, multi-physic route cable access; cross-regional command movement success; physical dispersion of off-site disaster preparedness nodes (unsimilar risk zone)	Network Topup Monitoring + Cross-Drive Movement Control Platform	
Space and operation (reformed self-healing)	Space Reconstructability	Percentage of modularized rooms; percentage of reserved expanded land use (target 20-30 per cent); civil-hardware decomposition; number of malfunction isolation units	Assets digitized desk account + phased board
Institutional compliance rate	RRO/RPO compliance rate; downgraded operational capability (core data power retention ratio); mixed exercises frequency and failback loop closure rate	resilience Indicator Examination Platform + Exercise Event Library	

These four core categories of indicators — site compliance, flood risk exposure, energy security, spatial reconstructability — are the diagnostic core of “the spatial layout of the extrapolation centre and its safety; this identification further refines them into measurable sub-indicators and data support, as inputs for subsequent “proposal generation” and “digitized products (soft resilience platform)”.

Contradiction between historical and cultural preservation and spatial renewal

Completed

Symptoms - Cause and consequences - The complete and deeper diagnosis of diagnostic indicators is as follows (including general diagnosis, four phenomena, causal chain and calculable diagnostic indicators).

Contradiction between historical and cultural preservation and spatial renewal

When a hundred-year-old railway line carrying the national autonomous and innovative memory is asked to “reserve the original truth” and “high-intensity renewal” to protect the authenticity, integrity and functionality of the renewed, and humanity claims meet directly on a nine-kilometre linear corridor. This section deepens the diagnosis along the three lines of “generic – diagnostic indicators.”

3.5.1 General judgment: the “high” value of heritage and the “mixed” path of activation

****** Jing-Zhang Railway, China's autonomous first-line railway, has reached the level of preservation beyond the single-body construction level to the level of preservation of the industrial heritage as a whole and its patrimony: the path of the primary industry, such as tracks, roads, stations, bridges, tunnels, welding units, etc., constitutes a linear heritage gallery of historical information authenticity and readability; the site park has been opened from the “sleeping abandoned railway” to a public space that citizens can sense. However, the balance between conservation and revitalization has been highlighted – the value of the heritage is widely recognized, but the path of “what to protect, how to protect, what to change, who to use, how to operate” remains blurred, protecting the porousness of the frontier and the functional path, and the fact that both sides of the corridor are still in a state of “preserving an unstructured, updated climate”.

** The recognition of the value of the heritage tends to be rigidly regulated; but there is a blurring of the dynamic path (lack of intermediate pathways and regulations for hierarchical control to promote protection) and a lack of power to update and space continues to break; the absence of mechanisms for public participation is compounded by the lack of a cultural identity that is difficult to translate into co-regulating action, which ultimately manifests itself in * a broken * *. The causal chain that emerges from this problem is: * protection of the rigidity of the boundary + a blurred dynamic path + insufficient public participation *.*

This issue, which is “a contradiction between historical cultural preservation and the use of space for renewal”, is at the core of the overall problem framework: it provides basic data on the uplinking of heritage resource floors, the status of corridor spaces, and public cultural needs (supported by city-time data foundation), and goes down to a series of responses to heritage activations, seams and cultural empowerments in the context of the “articular continuity development strategy” and the “precision-activated camp strategy”. The results of the problem identification will be directly translated into quantifiable, monitorable diagnostic indicators and the construction needs of digital bottoms.

3.5.2 Symptom I: Space competition for protection and renewal

The protection and renewal of space demands on the same corridor, both of which are considered “non-negotiable”:

- *** Protector end: requirement for authenticity protection — complete retention of primary industrial remnant of railway tracks, road bridges, welding components to ensure the authenticity and readability of historical information; requirement for holistic protection — extension of protection from monogonal relics to linear porches and historical nodes along the routes, covering all elements of road foundations, stations, factories, bridges, etc.; requirement for active protection — transformation of static remnant into living landscapes in public space, allowing historical elements to participate and experience;** - ***Update end:** call for the seaming of cities — breaking the long-term break between railways and towering facilities on urban spaces, breaking the break-up of head roads and reshaping the fractured urban muscles; demand for priority for people — precision in responding to the daily needs of diverse groups, such as the surrounding population, the commuting population, and the students and students of higher education; request for ecological convergence — to convert idle “grey infrastructure” into green public space.

On the face of complementarity, there is a competition for the same corridor:** protection of the space that requires “main” retention, and upgrading of the space that requires “functional implantation” — often the same road-base, the same station, the same bridge —** when

“protection first” is understood to be “freezing as it is” and new corridors, facilities and industrial functions are not installed; and when “cracking cities” are given priority, the original nature of the original remains is at risk of being cut off, wrapped up and concealed.

** Chart 1 Space competition is indicative.** Protection (green) requires “remaining as it is” and updating (red) requires “functional implantation”, both of which compete positively in the same corridor space and lack a hierarchy, zoning and time-series reconciliation framework.

3.5.3 Symptom 2: Point protection and structural contradiction of linear corridors

The legacy properties of the Jing-Zhang Railway are “linear”, but the protection and renewal in reality are spread out in “points”, creating structural and scale errors:

- ** Legacy sites are separate: **heritage nodes, such as stations, bridges, tunnels, forklifts, are often single-point identified, single-point protected, lack a unified framework for heritage value spectrometry and presentation along the routes, and historical narratives are cut to pieces;*** - **Consequencing arteries fractured *:9 km of linear porches are in a completely different state in different sectors — some paragraphs have been built up in parks, others remain enclosed, others are occupied by temporary parking and inefficient functions, and the arteries are intermittent in space and cannot form an overall pattern of “a series of multiple activations”;* - **Systems of protection and up-to-date scales:* protection is based on “points”, updating is based on “blocks” and operation is based on “blocks” — the borders of the three are inconsistent, making it difficult to align protection planning, the urban renewal proposal with operational management in space; - ** Value interpretation gap*:* the value of heritage is interpreted in terms of “what is retained”, lack of a systematic expression of “why is it important and how it is experienced”, and lack of a perceived narrative link between industrial memory and contemporary urban life.

3.5.4 Symptom III: The bipolar risk of static freezing and renewed destruction

In conservation and renewal chainsaws, practice tends to slide to two extremes, while the genuinely needed midway “to promote” is chronically missing:

Extreme I: Static freeze. The term “protection first” has been simplified to “encapsulation” and the heritage has been fenced, put out of use and turned into a distant and improbable “dead specimen”. The consequences are: the isolation of the legacy from everyday life, the loss of cultural values, and the loss of protection into a financial burden; the rapid deterioration of functionally supported spaces and the emergence of new security and environmental problems; and the difficulty of protecting inputs to form sustainable operational closed loops.

Extreme II: Renewed damage. Simplify the city to “pull back” and remove, cut, package the original remnant in the name of the update, or implant large-scale new functions without assessment of the heritage. The consequences are: damage to the authenticity and readability of historical information and the disappearance of railway memory carriers; irreversibly if the damage occurs, the arterial fracture becomes a fait accompli; public trust is compromised and social consensus for subsequent protection and renewal becomes more difficult to establish.

What is missing between the two poles is ** the intermediate path of “promoting protection”**: allowing the heritage to continue in use through rational functional implantation, and acquiring modern life through functional replacement rather than functional freezing of industrial relics – the transformation of old-style waiting facilities into stools, industrial dragon-gate hangings into sign points, abandoned cars into cultural displays, or light-food spaces. The absence of this path has left protection and renewal in a “no-no-no-no-no-no-no-no-no-no-no” paradox.

Figure 2 * Two-polar risk indicator. Static freeze (left) and renewed destruction (right) are protective-update two failed exports of chainsaws, with the middle “for safety” route (fiction line box) missing.

3.5.5 Pattern IV: Decentralization of diverse subjects and inadequate public participation

The Jing-Zhang corridor cuts across a number of street towns and property owners, with a high degree of fragmentation of responsibilities and interests for protection and renewal, while synergy mechanisms and public participation are chronically lacking:

- ** Decentralized: **railway title holders, units along the road, street communities, merchants, each part of the household, lack of a unified responsibility boundary and coordination platform, and a situation of “protecting unled, updating and operating***

“unaccountable”; - * **Unequal interests**: the responsibility to protect (limitation of construction, preservation of remnant) tends to fall on local subjects, while the renewed gains (valuation of land prices, human import) are shared more widely, and unequal rights weaken the incentives for protecting subjects; - * **Inadequate public participation**: **residents, teachers and students, local enterprises as day-to-day users of heritage, under-involved in key aspects such as the identification of protection lists, the updating of proposals, the planning of operational content, etc., and the difficulty of translating cultural identity into common governance**; - * **Lack of sustainable operations**: lack of a regularized cultural content operation when public spaces are built — lack of railways thematic exhibitions, creative markets, historical cultural guides, etc., space is re-deplete by the lack of management, creating a cycle of “building or idle”.

3.5.6 Correlation chain: from protective rigidity to arterial fractures

Linking these phenomena together, the causal structure of the problem presents two mutually reinforcing paths:

High recognition of heritage value (first autonomous trunk rail) Protection of borderlines for rigid control — renewal of space limits,
fuzzy (i.e., how) activation paths — — — — — lack of energy — — — — — continued fragmentation, isolation of the gallery — — — —
— — insufficient public participation — — — — — decline in cultural identity — — — — — protection and renewal in a “either or not”
impasse — — end result: — — — — history of memory, — continuity of space muscles, — — — continuity of social identity — — —
tectonic — — — — — three — — — — and — — — — — and — — — — and — — — — — and — — — — — and — — — — — and — — — — —

The key leverage points are two interventional links: * *the protection end's "segregated regulation rules"** (restricting rigid controls to the protection core area, leaving flexibility in the buffer coordination area, achieving a "fresh-fired, one-size-fits-all approach") and * *the active end's "mobilization mechanisms"** (functional implantation, cultural operation, public construction to sustain heritage in use). The former resolves "suspended death" and the latter resolves "overhanging" and the two combine to reduce the two polar risks to the intermediate path.

3.5.7 Diagnostic indicators: calculable, mapable, monitorable

In line with the “from phenomenon analysis to causal diagnosis” approach, the above-mentioned phenomena are converted into quantifiable, top-moderable diagnostic indicators ** that can be continuously monitored at the digital bottom*. * The indicators are both urban-time data foundation and directly corresponding to the monitoring of subsequent digitized products (cultural digital archives, public participation platforms, cultural operations monitoring).

Diagnosis dimensions	Diagnostic indicators	Status / Threshold (calculated)	Monitoring instruments (digitized base)
<i>* Heritage protection (realty and integrity)*</i>	Level and completeness of heritage protection	Leveled coverage of heritage sites along the routes; total retention rate of primary residues, such as tracks/roads/stations/bridges; enforcement rate of rigid controls to protect core areas	Heritage resource "One Map" + Heritage Digital Archives + Remote Sensing Comparison Monitoring
Pulse continuity index	Proportion of series of narratives of heritage along the porch (points vs continuous); artery identification/direction of miles covered; historical scene restoration and display point density	A narrative of the gallery heritage + show a little vector superdition	
<i>* Space renewal (suture and activation)*</i>	Accessibility on both sides of the railway (suture convergence)	Breakway pass rate; number of additional pedestrian/carwalks; average east-west bypass distance across the corridor; release length of closed wall	Web Vector + Action Mobile Path Calculation + Wall Demolition Progress Watch
Active utilization intensity	Injection rate of heritage functions (injection function/enabled space); number of active carriers (museums/exhibits/cultural spaces/initiatives); frequency of cultural operations	Functional implant account + space heating + operating calendar	
Social identity (co-governance)	Public cultural identity	Rate of awareness of heritage values among residents/teachers/tourists; number and satisfaction of cultural participation; conversion rate of creative consumption	Public participation in the Platform questionnaire + activity sign-up data + consumption monitoring
Multiple Synergy	Number of cross-town coordination units; protection-update-operating subject-matter agreement coverage; rate of acceptance of public opinion and feedback closed loop	Joint platform to work with desk accounts + voice-processing stream tracking	

The three core indicators mentioned above - the Pulse Consistency Index, the availability of both sides of the railway, public cultural identity - are at the core of the diagnosis of the “continuation between historical cultural preservation and the use of space for renewal”; this identification further refines them into quantifiable sub-indicators and data support, as inputs for the successor “proposal generation” and “digitalized products (digital archives and public participation platforms)”.

The question-identification section of AI urban design | this page is the main page of the `issue-identification' chapter, and the completed questions are linked to the corresponding sub-chapter file, to be added. All data and indicators must be cited after the source of the source.json/Metrics.json.json.

Chapter four, the discovery of the response, the strategy planned?

Discovery of responses: from emerging issues, new tools to new responses, new scenarios

For the five core issues identified in Chapter III, a five-pronged strategy is "continuous spectroscopy matching, fine activation of the camp city, menu-choice thinking, soft resilience calculations, and successional continuity."

4.1 Consisting spectroscopy policy *Public service packages and AI infrastructure (continuous spectroscopy public facility network S/M/H/L + general power hub and edge node)*

Continuous spectral policy S / M / H / L network and AI infrastructure

On the basis of problem identification, the proposal has recast public service facilities from "point supply" to a continuous spectrum covering solo, occasional encounters, coexistence, communication, collaboration, demonstration, community and urban public life, and has completed the AI infrastructure with **3** general power hub + **12** edge node.

Space overview: continuous spectrum support and AI infrastructure Defaultly lays out the original location of 535 tertiary + 556 AI enterprises; the interface (S/M/H/L) and infrastructure (junction/edge node) layers can be viewed on the upper right panel.

Continuous spectral support policy: S / M / H / L 4-level network

Reconstructing the facility from the "point supply" to an **continuous spectrum covering eight states of "single urban public life" **, allowing for a drop in the "present" of various intensity.

spectral axis: eight states, none of which should be missing

The public life of the human person is a continuous axis of intensity, from the most private isolation to the most public city celebrations:

spectral 8-state

Solo, occasional, co-existence, communication, collaboration, display, community, urban public life.

Case experience values broken down by area per person (total 3.0 m2/person):

Status	Alone.	I met him once.	Together.	Communications	Collaboration	Presentation	Community	Urban public life
<i>* Per capita (m2)*</i>	0.3	0.2	0.4	0.5	0.8	0.3	0.3	0.2

This gives rise to the addition of a total of D = Pai x 3.0 ≈ 180,000 m2 (15–210,000 m2) extrapolation.

Level 4: S / M / H / L (set L ⊃ H ⊃ M ⊃ S)

Allocating 180,000 m2 to four levels, which constitute the "embedded amplification" of the skeletal skeletons - each level is a first level of functional amplification and public upgrading:

Level	Number	Area share	Roles and Typicals
S Focus and encounter	150	26% (4.47,000 m2)	Minimal contact point, hourly rent: Focused position / Occasional encounter at bar, scattered between university and enterprise
M Innovation / Life / Community / Integration / Body	50	≈ 41% (7.74,000 m2)	Day-to-day dominance of corridors and neighbourhoods: innovation living room, life node, third space, complex node, physical health
H Industrial flagship engine	3	23% (41 000 m2)	Zhongzhi Park / Beijing AI Origin Community / Dazhongsi, carrying industry flagship and test
L Urban pilgrimage destination	3	≈ 9% (≈16,000 m2)	Chile/ Talent / Industrial Zone, urban destination and public interface

Why "spectrum" instead of "compensation"?

A single dot can only solve the problem of "no" and not "if there can be a collision." The essence of the innovation is that it interacts unexpectedly, requiring a continuous combination of spatial gradients: an AI researcher goes from "one-on-one" to "sometimes colleagues" to "exhibiting results" in a "collaborating road" and each strength should be matched by a corresponding facility. Spectra strategies allow facilities to work with AI talent / the rhythm of life**, rather than simply "sufficient."

Add AI infrastructure: general power hub + Edge node

An AI city needs a "computational power facility" in addition to its human facilities. This program adds **3 central power hubs + 12 Edge node**, which constitutes a double cycle of "Data Up, computational power Down".

Why: Double-track demand of time delay x size

- AI * "Size" * — handed over to the 100,000-calorie cluster in Kyznu; AI * "Time delay" * — local millisecond response required. - Therefore, the strategy is the "** Local Low Cards*" **extrapolation: only about 2000 cards (0.0.6 EFLOPs, approximately 2% of the foreign power) are available locally, taking on the "low-time delay reasoning + data top-line interface" and training is given to Jing-Zhang. - The central power hub is not just a room, but an inference of * city-level interface** for the two elements of "computal power + data".

computational power hub: 3 (1 main port + 2 sub-port)

Hub	Level	Local cards	It's not like I'm going to have to do this.	Area	Hostroom down.
* Zhongzhi Park Port*	Main port	1000	~0.30 EFLOPs	~3800 m²	Renovation plant / New Heavy Building
* Community ports of origin*	Port	600	~0.18 EFLOPs	~2500 m²	Stock implant/ new heavyweight building
Dazhongsi port	Port	400	~0.12 EFLOPs	~2000 m²	Stock implant/ new heavyweight building

The three hubs use "** stock implantation + light separation*" **user decision: each hub = *1 light landmark** (research and development building, accessible sanctuaries) + **1 heavy bottom* * (aircraft room, heavy board / liquid cooling) — instead of being an island-based data centre, they make the computational power a visible and accessible urban element.

edge node: 12 (16–64 cards)

12 Edge Node totals approximately 384 cards (0. 0.1 EFLOPs), downgraded in the AI Enterprise POI Group (e.g. Zhongzhi Park Core 12, Five Crossings, etc.), service** low-time delay reasoning** and data acquisition**.

Double cycle: Data topline. data power down

Double Cycle

* *Uplined data: **Edge collection desensitized hub → Jing-Zhang training***; ***computational power down**: Jing-Zhang wants hub movement control ** Marginal reasoning scenario execution.

Site selection follows the norm drop-off values: 100 m from railway category A ≥/ B_C ≥ 100m from subway category A ≥ 100m/ B_C ≥ 80m from residential area 100m; large- and medium-sized aircraft rooms do not enter residential/commercial areas; flood restricted areas.

Main references and data calibres

Category	Contents	Source/ caliber
University places 535	Qinghua 400+System / Building + Northwood + Mining (Beijing) + North Great Medical Department + North Coda, etc.	Map POI, 2026-08 grab, WGS84
AI Enterprise POI 556	5 mouths 128 families on one horse; only 58 AI high related entities	Map POI, 2026-08 grab, high faith
Basic maintenance 298	Official-calibre public service facilities	Standardized Bottom
Pai ≈ 60,000	AI Innovative population (4-80,000)	Push Value
Per capita 3.0 m2 / VIII	0.3 / Occasional 0.2 / Co-existence 0.4 / Exchange 0.5 / Collaboration 0.8 / Demonstration 0.3 / Community 0.3 / Urban public life 0.2	Case experience
S/M/H/L ratio	≈ 26% / 41% / 23% / 9%; S=150 / M=50 / H=3 / L=3	Push Value
COMPUTATION POWER CENTRE 3	Zhongzhi Park Main Port 1000 Kcal/ Origin Community Port 600 Kcal/ Dazhongsi 400 Kcal (local 2000 Ccal 0.6 EFLOPS 2%)	extrapolation value; host room down to proposal decision
Edge Node 12	≈384 ≈0.1 EFLOPs	Data (POI group) + evolution value
Location Back	Railway A 800m / B-C 100m; Subway A 100m / B-C 80m; Housing 100m; Flood Ban	Normal value

The values in this report are divided into three categories and are indicated on a case-by-case basis:* *[norms]** (industry/norms),* *[cases experience]** (domestic and international case experience), [extractable values]* (*retroactivity*)* (retroactive);* *[data]** (original POI/statistical calibre),* *[in extrapolation]** for analytical judgment based on data,* *[user decision]** for manual decision-making at the proposal level. All point coordinates are presented directly by WGS84, consistent with the bottom map, without any deviation.

Detailed design of focus areas

Zhongzhi Park, Beijing AI Origin Community and Dazhongsi, respectively, take responsibility for innovation and certification, young talent and open-source living, business growth and results. The division of labour between team members, grouping, public space, scenery, operation and resilience is fully maintained; all three borders are subject to formal review. [data: geometry/key_areas.geojson#PROV-KEY-001]

The other two provisional ranges are independent elements of the same data concentration. [data: geometry/key_areas.geojson#PROV-KEY-002] [data: geometry/key_areas.geojson#PROV-KEY-003]



Zhongzhi Park, Beijing AI Origin Community and Dazhongsi Focus Design

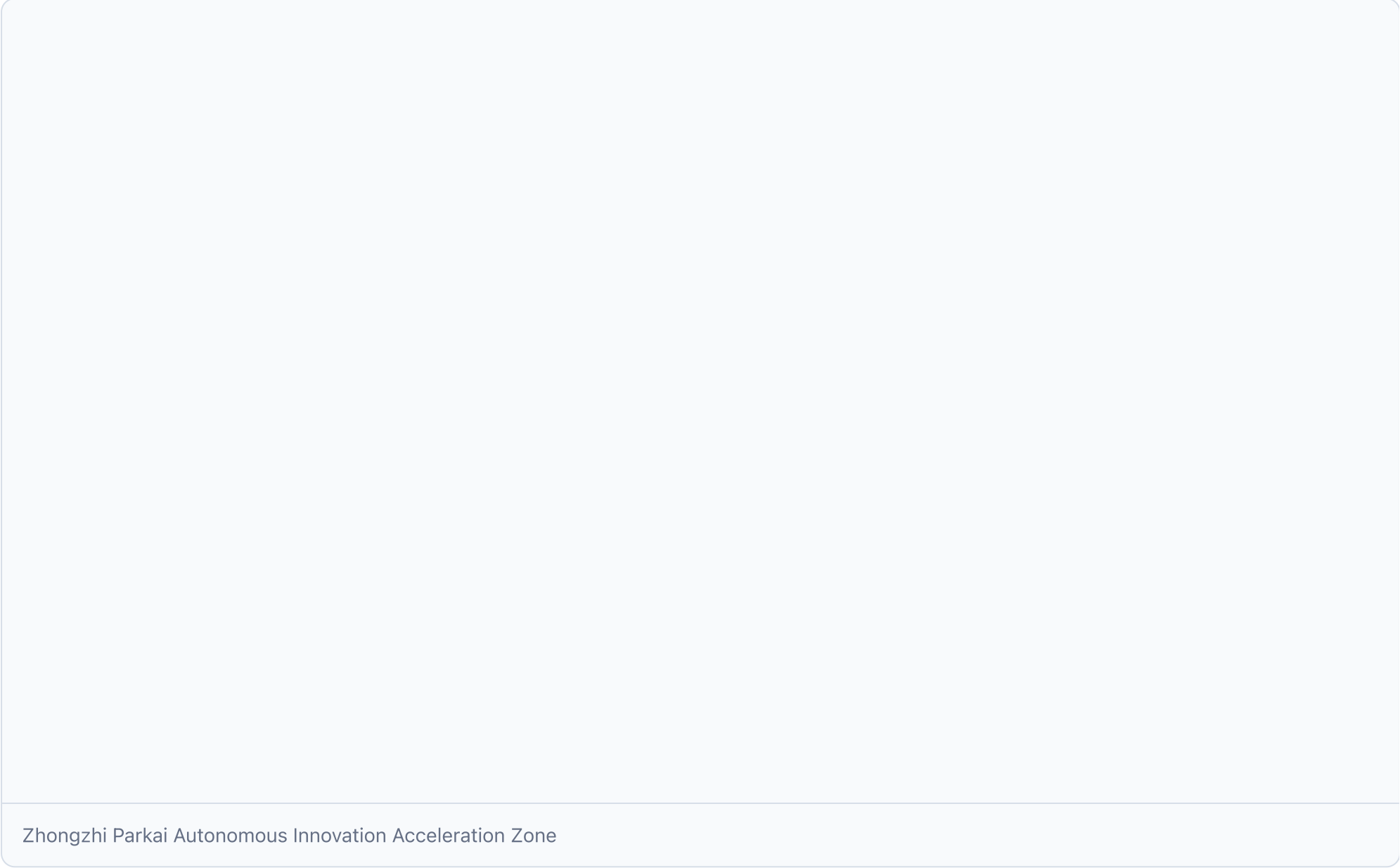
Three priority areas and pending conditions must be shown in figure 03.

1.4 Zhongzhi ParkAI Autonomous Innovation Acceleration Area

Focus area I

Zhongzhi Parkai Autonomous Innovation Acceleration Zone

Open win-win, innovation development



Zhongzhi Parkai Autonomous Innovation Acceleration Zone

B/Zhongzhi ParkAI Autonomous Innovation Acceleration Area President Xi Jinping 2026 World Congress on Artificial Intelligence and High-level Meeting on Artificial Intelligence Global Governance

"As a new engine of world economic growth and as an accelerator for the transformation of new and old dynamic energy, we are moving from a "digital world" to a "physical world." We need to seize the historic opportunity to encourage open-source openness, cooperative sharing, comprehensive promotion of artificial intellectual technology and innovation, industrial development, landscape applications, concerted upgrading of traditional industries, growth of new industries, and forward-looking arrangements for future industries, so that artificial intelligence can be made to do everything."

Zhongzhi Park assumed the link "** technology breakthrough**" in the source chain, located in **AI Autonomous Innovation Acceleration **:

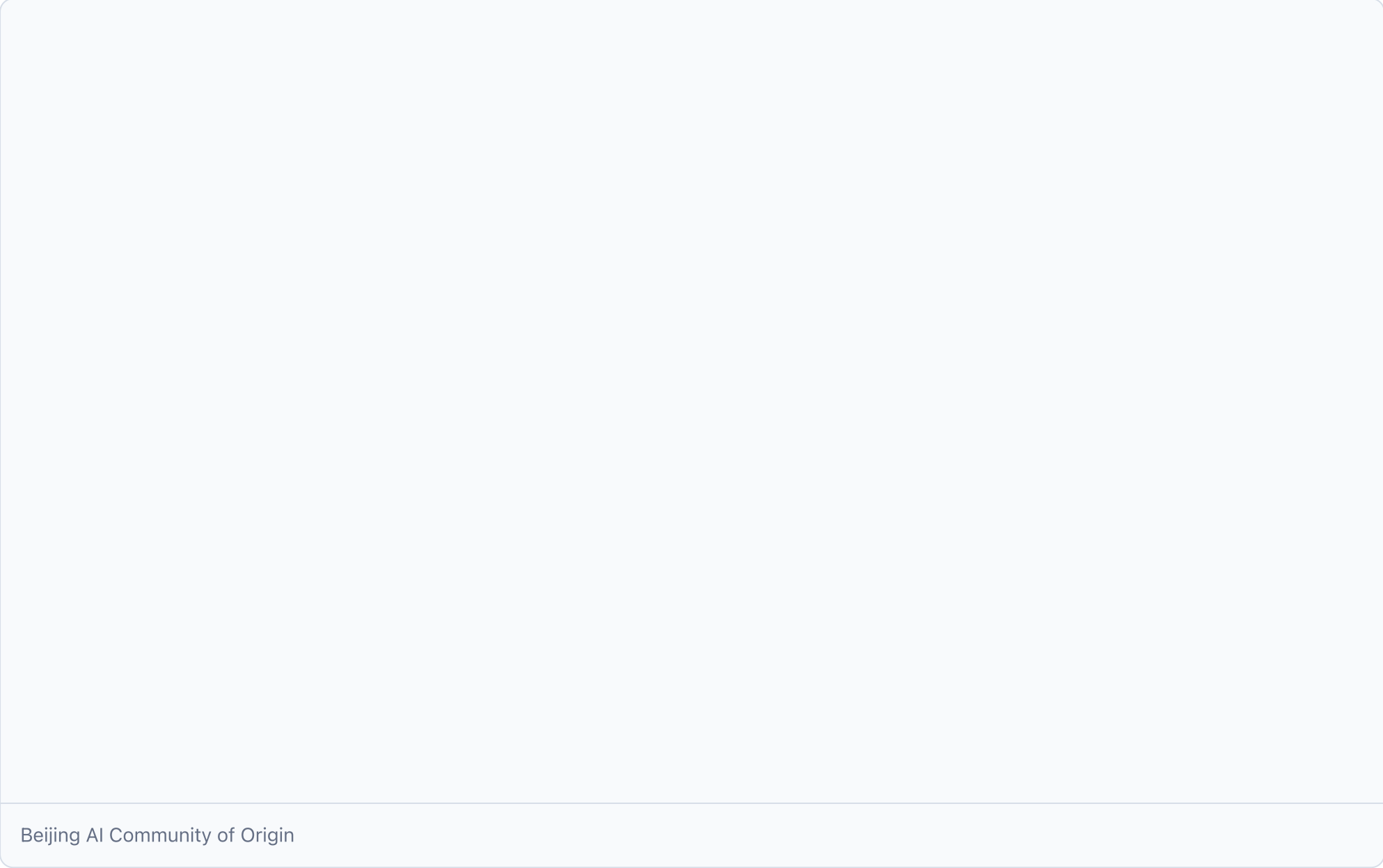
- ** All-in-house autonomous innovation: **All-in-house autonomous innovation system covering reasoning and training centres, chip testing workshops, model validation sites and safety governance laboratories to underpin key core technologies (the "first creator of technological inventions")***; - **Community power and governance voice*: as a regional central power centre and as a provider of standard governance, the key source of support for communities of origin in reverse; - ** Open win-win**: Open-win, innovative development is the theme, encouraging open-source, cooperative and shared sharing, and placing autonomous innovation in an open pattern of global collaboration.

1.5 Beijing AI Community of Origin

Focus area II

Beijing AI Community of Origin

[Empowerment, Young Power]



Beijing AI Community of Origin

Figure C: Beijing AI Community of Origin General Secretary Xi Jinping, an important directive during a study of innovative eco-communitys in Shanghai in April 2025, when the big model of "mode space" in the Shanghai area was created. "Advanced intelligence technology accelerates over time. Artificial intelligence is both a young enterprise and a young one. Fulfilling the great dreams of the Chinese nation, it is in the hope of young people. All of you must be patriotic, assert the will of the nation, strengthen its capabilities, link individual struggles to the destiny of the nation, run the batons of history, and shine on the broad stage of China's modernization.

The Beijing AI community of origin is the "World class of original innovation" ** scene core, **assuming part of the "** Thoughts and Standards" in the source chain:

- * *Elemental source and standard output:* **at the core of talent, enterprise, open source, data, convergence of standard five elements and standard output, load open source publishing hall, results transformation street, data element living room, standard assessment and talent community;** - * **Reinforcement of the advantage:** transforming "brain density" into "source concentration" based on the established advantage of the convergence of institutions of higher education with young people; - * *Young Power* *: Using the theme of "enhanced advantage, youth strength" as the first resource for a source of ideas, AI became the setting stage of "young cause, youth career" (for detailed spatial design, see 4.4 "source of original community").

1.6 DazhongsiAI Industrial Cluster

Focus area III

DazhongsiAI Industrial Cluster

Elemental convergence, ecological synergy



DazhongsiAI Industrial Cluster

D/DazhongsiAI Industrial Cluster General Secretary of Xi Jinping. "The country is rich in data resources, industrial systems, market space and prospects for developing artificial intelligence. It is important to strengthen policy support and talent development, and to strive for more high-quality products that are safe and reliable.

Dazhongsi is responsible for the "** industrial landslide and ecological clustering**" section of the source chain.

- ** Intelligent neo-business: **carrier enterprise service, product release and lightweight edge node to transform source results into a landable, diffuse industrial landscape;*** - ** Southern low-temporal response:* mobilization of source and community power service to cover low-temporal response and disaster preparedness in the South to complement regional synergetic end-response capacity; - **Element convergence, ecological synergy **: The theme of "Element convergence, ecological synergy" is to develop healthy ecosystems in which all subjects live together and live together, enabling AI to do everything.

4.4.4 Space evolution of strategy: Two Wings Down in Sector 3 and resilience Space Organization

The three core strategies are not on the principle level, but on the spatial division of the "two wings of the Three Sectors" and the specific spatial organization of the community of origin in AI. This section is a layering exercise of three strategies: the location of the scene, the total plane, the three layers of resilience, the water park relationship, the flow lines, the structure of the building, the ratio and the drawings.

Site location: AIS-based community division of labour between two wings of sector 3

The proposal assigns roles to the "two wings of the Three Sectors" in accordance with the mission document. **AI Origin Community as "World Class AI Innovation Ecology Source of Innovation"**, with six elements of convergence and standard output; ** Zhongzhi Park** with AI Self-Innovation and Computation Power/Governance Speech; ** Dazhongsi** with intellectual novelty and low-time response in the South. This division of labour directly corrects the misplacement of "the Centre of reasoning as the primary site": source of original innovation as a place of ideas, business and standards.

Once you have access to information on official boundaries, terrain, electricity, municipalities, tenure and fire protection, you should re-elect specific plots within the community of origin of AI.

Focus areas	resilience character	Main space carrier	Relation to source of original innovation
Zhongzhi Park	AI Independent Innovation + Computational Power and Governance Speech	Logic and training centre, chip testing workshop, model validation site, safe governance laboratory	Take over the primary power centre and standard governance to backstop the source
AI Community of Origin	World-class innovation ecology	Open Source Publishing Room, Results Conversion Street, Data Elements Lodge, Standard Evaluation, Talent Community, Endside Construction Portal	The scene core, six elements and standard output.
Dazhongsi	Smart original business (situation response + edge response)	Enterprise service, product release, lightweight edge node	Mobilization of a resource and a computational power service for low-time and disaster preparedness in the south

Zhongzhi Park is responsible for full-scale energy consumption, long-term large-scale training; Zhongzhi Park is responsible for ink reasoning, chip testing and safe governance; AI Community of Origin is responsible for factor source, standard-setting and credible data flow; Sector Three, together with Transport, Robots, Public Space Nodes, forms a complete chain of *"* periphery training - Zhongzhi Park full-barrel reasoning and validation — source of origin element and standard output — Zone Three Edge Response*"*.

The roads, orbits, water systems and green areas in the map are mapped by the local OSM cache unified projection of `AI maap/osm-data.js` '; the overall design scope and the three focus areas follow the provisional presumed coordinates of `map-three-layers.html` ', which are used only for the identification of space relations during the competition phase. Zhongzhi Park is responsible for the whole house of innovation and general power management; the community of origin is responsible for ecological source of original innovations and standard output; Dazhongsi is responsible for the transformation and marginal response of the new industry. The three are linked by Jing-Zhang corridor and constitute a "two wing" system of collaboration with peripheral training resources, the network of the urban landscape.

General plane design: one session, two rings, three interfaces, six groups

The AI community of origin uses a space structure ** "One Court, Two Rings, Three Interfaces, Six Clusters"**, corresponding to a six-factor space carrier.

Division I: Element sharing courtyard

As a public living room for the public:

- Ground-based embedding of real-time eco-resilience information belts showing the aggregate health of talent flows, enterprise growth, open-source dynamics, data compliance flows, standard iteratives and usage of the computational power;
- Set up an appointmentable outcome press, open source demonstration areas, scene validation sites and small distribution sites;
- Connecting open source publishing halls, standard assessment workshops, data element living halls and enterprise service spaces through the semi-outdoor gallery;
- Set up shades, rain gardens, mobile seats and all-weather work interfaces;
- Only dissensitized public running data are shown at night.

Two rings: public innovation and safety.

** The Public Innovation Ring** targets urban roads, colleges and green spaces, linking public exhibition halls, factor service halls, open-source and shared meetings, results presentations, talent services, and roof gardens. The public and businesses can follow the ring to complete the entire process of "know-know-need-to-access elements-seeing results."

It does not cross public walking lines and reduces the impact on public space and communities through greening, building volume and acoustic barriers.

Three Interfaces

1. ** Urban service interface: **main roads and public transport, entrance squares, factor service halls, business services and presentation space***; 2. ** Campus innovation interface*: open-source communities, joint research offices, standard assessment workshops and credible data sandboxes for universities and scientific institutions; 3. ** Park ecological interface**: low-density public functions, rain gardens, resting platforms and energy science facilities for green areas, avoiding the direct movement of equipment discharges, heat discharges and high noise facilities towards the park.

Six.

Group	Corresponding elements	Core content	Space characteristics
Talent Corps	Talent.	Joint research offices, international talent communities, youth apartments, night collaboration spaces	Adjacent campus interface, mixed residence, R & D, socialization
Enterprise	Enterprise	Incubation, acceleration, unicorn building, investment financing and legal services	It can be divided and reorganized along the street.
Open Source Group	Open Source	Open source publishing hall, public code wall, contributing space	The first floor is open, connecting the courtyard to the city entrance.
Data group	Data	Credible data sandboxes, data element living room, compliance audit	Leveled delegation, key data above security marker
Standard Cluster	Standard	Evaluation laboratory, security governance, red team testing, standard setting	Visitable, reservationable, regulatory governance interface
_Other Organiser	computational power	Endside computational power station, Edge Node, Model Validation Workshop, Near Light Calculator	As a sub-seat enabler, combined with public service, scalable and re-engineered

Six groups form a staggered complex of buildings by sharing courtyards and galleries. Talents, businesses, open-source groups form a public dynamic; data, standards, computational power groups form the controlled core of governance and support — of which **computational power groups serve as the base energy element**, providing the first five groups with a public infrastructure of near-light, marginal caches and certification, which is a source of visual and functional innovation: it does not occupy the dominant mass, but is a key link of emphasis on resilience and energy.

General plane of the concept of the AI origin community (global bottom map + focus area amplified)

General plane of the concept of the AI origin community (global bottom map + focus area amplified)

The master plane uses a two-scale structure of “wide-down map + focus area amplification”: the left map presents the spatial relationship between Zhongzhi Park, the origin community of AI and Dazhongsi in the Jing-Zhang development axis, roads, orbits, water systems and green-field networks; the right map uses AI to support the generation of bird-view concepts for open and innovative neighbourhoods, the development of incubator groups, the community service core, the company paper Node and the Central Green Valley Park.

Space implementation on three levels

Element renewal: redundancy and substitution

- * *Talent*: **establishment of mobility and mutual recognition routes between institutions of higher learning — parks — enterprises, where talent fluctuations in any single institution or enterprise do not constitute ecological breakpoints; establishment of flexible joint study rooms and short-term work units;** - * **Enterprise**: hatching-accelerated-unicorn ladders coexist with redundancies to avoid overdependence on a single lead enterprise or a single track; space can be divided and reorganized to support enterprise growth and exit; - * *Open source*: **code, model and community decentralized distribution without relying on a single hosting platform; public code walls and contributing space are readily re-matched;** - * **Data**: Key data classification, multiple copies, auditable; credible sandbox supporting cross-subject compliance collaboration; - **Standards**: assessment and governance approaches open to participation, layout and avoiding monopolization of standards; - **Commutational power**: as a bottom-type element, redundant supply of the 3rd tier of the

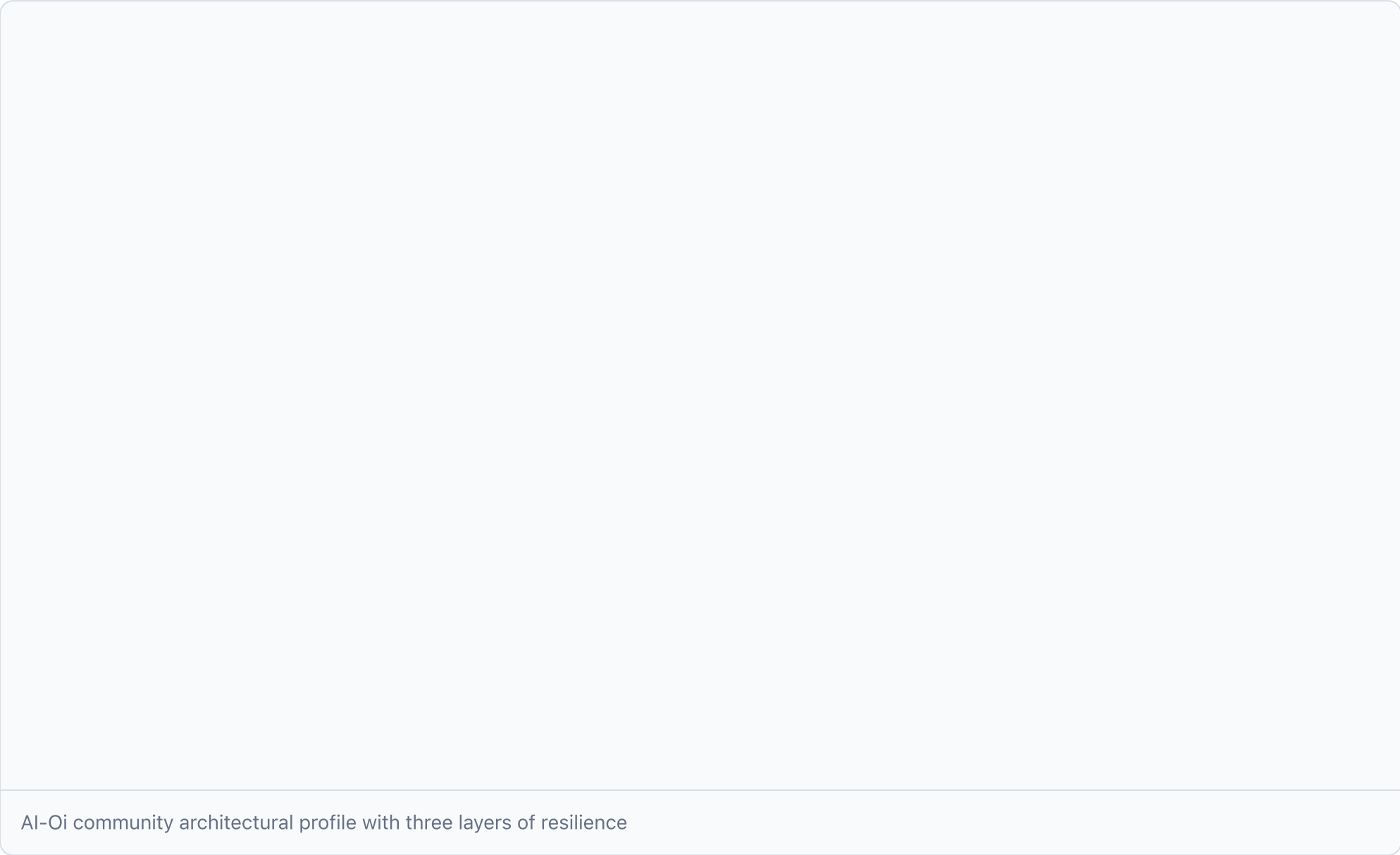
Exterior Training – Zhongzhi Park Whole Inference – Origin close to Light Count – Edge of Dazhongsi' Commutation Power Network; local retention of the source point for near lightweight, edge cache and authentication workshops, with cross-scale movement of tasks in any one-time malfunction, creating an alternative to the

near light + area recalculation + cross-scale movement'.

Spaceresilience: Resilience and Reconstructability

AI communities of origin use a composite profile of “public space below, collaborative space above, key data and standards above, energy systems above”, avoiding the release of key data, standards and governance functions into flood-prone underground spaces.

- Key data sandboxes, standard assessment and governance controls are set up above the surface floors after flood control, structural and fire review, using module, partition and modular design; - Underground space, in principle, only parking and non-critical support rooms, without core data, unified governance controls, critical network nodes and a single power supply control system; - Construction entrances, ramps, vents and pipelines are protected; garage ramps are isolated from the entrances; - The group uses a phased, disassembly structure that allows for partial re-engineering or updating without interruption of core functions.



AI-Oi community architectural profile with three layers of resilience The profile is vertically structured as “public base-cooperative middle-level data and standard safety-energy roofs”: public space connects urban streets, shared courtyards and park eco-interfaces; collaborative space supports joint research, open-source contributions and flexible restructuring of business incubation; key data, standard assessments, red team testing and compliance audits are located above flood safety control lines; underground only parking, rainwater management and non-critical auxiliary functions are retained. Elements resilience, spatial resilience and institutional resilience are integrated throughout the building and are organized separately through public, controlled and transport-wide streams.

Institutional resilience: self-healing and the right to speak

- * *Multi-Intelligence Collaborative Governance: **Urban smart body Auxiliary Identification Activity Mobility Breakpoints, Public Space Thermal Power, Facility Maintenance, Business Service Demand and Activity Security Risks, without substitute for planning clearance, non-export of unauthorized personal images and no claims of official implementation commitments;*** - ***Standards and Governance Output:** Translating assessments, red team tests, security ratings and compliance audits into accessible, bookable, regulated public interfaces, making “how standards are developed, how models are governed” a core competency for perception and output; - * *Open source and ruleover**: Open source community governance, data flow rules and algorithms guidelines open to participation, versionability, auditability, resulting in institutional products that can be reused by other cities.

Design relationships with water systems, parks and urban environments

Water is not a shortcut to cool, but a safe border.

Zhongzhi Park approaches the Qing River and the Moon River, Dazhongsì approaches the river, and the communities of origin in AI still need to verify urban water accumulation and groundwater.

- Key data, standards and governance functions are above the reviewed security marker; - Underground space does not perform the only critical function; - (b) Installation of inverted water for entrances, ramps, vents and pipes; - Management of cooling water, fire water, rainwater and sewage subsystems; - (b) Set up leakage detection and auto-disconnection in key areas; - Rain gardens are responsible for day-to-day savings, but are not a substitute for the own protection of critical facilities; - Based on the multi-smart “four-prevention system” of scene IV, pre-introduce flood ranges and create scenarios (capability endorsement of scene IV does not constitute implementation commitments) before heavy rains and sub-standard floods.

The park is not a back yard, it's a public innovation interface.

The first floor of the building will be partially accessible and accessible, avoiding the use of large-scale walls to cut off public space.

Central Park Belt: From Landscape to Ecoresilience Disk

The central park belt, released from high-speed transport, is the most distinctive public space in the community of origin and the hardware base of the intellectual-carbon platform.

- * Plant configuration as carbon-sink hardware: **priority is given to high carbon sinks, low-conservation, native-climate-adapted soil species, using the Joe-Irrigated-grass-plough structure to improve the efficiency of the per-unit leaf area index and carbon sequestration; through vegetation health monitoring, irrigation and shearing strategies are dynamically adjusted to avoid “greening” high carbon maintenance.** - * Water body design as climate relief hardware: retention and expansion of rain-fed wetlands, ecological ditches and retention pools to perform the triple function of heavy rainfall, water purification and evaporation cooling; wetland plants and microbial communities are simultaneously subject to carbon sinks and methane emissions monitoring and are included in the dual-carbon accounting. - * Water penetrating and sponge facilities as water carbon synergetic hardware: **water penetrating, submersible green areas, eco-pools to reduce surface runoff and heat island effects by increasing precipitation; penetration rates and water content of the matrix are incorporated into online monitoring to provide data support for “light rain does not accumulate, heavy rain does not flood, water has resilience”.** - *active mobility network as carbon-mitigating hardware: campus, business and public transport in the park connected by foot to the cycling path, identified through passenger flow statistics and travel, quantifying the carbon-reducing contribution of “active motor vehicles instead of motor vehicles”.

Through the Wisdom-Carbon platform, these “hardware” have been continuously monitored, harmonized and dynamically optimized, transforming the central park belt from static landscapes to ** an eco-infrastructure for active participation in the operation of urban resilience**.

People flow, logistics, data flow and fire lines

Stream	Entry and Path	Design requirements
Public and business flows	City entrance -- factor service hall -- shared courtyard -- exhibition hall -- park	Open to continuous, barrier-free, but core safety zone
Researcher flow line	Campus entrance - Identification - Joint Research Office/Certification Workshop - Controlled Data Area	Level-level authorization, access by appointment, no overlap with equipment logistics
Equipment and transport logistics	Backfield entrance - platform for unloading - equipment elevator - general power and energy groups	Independent closure for replacement and full-time maintenance of large equipment
Fire and emergency flow lines	Urban roads - ring fire routes - rescue sites - fire subsectors	Full-time access, without conflict with parking, activities and casual displays

Public display data, scientific research data, business data, city-sensitive data and governance control data go into different networks and safe areas, respectively. Public exhibition halls can only call desensitized, delayed or simulated data; data, standards and governance control networks are isolated from public networks.

Building patterns and urban style

AI Place communities use low-middle, organized and staggered building forms:

- Talent, business, open-source groups form continuous, active urban interfaces along the street; - Data, standard groups are located inside the group to cover the sun, piercing components and maintain the technical image of the gallery organization; - The height of the building is controlled by the surrounding campus, community and city interfaces, and the conceptual phase does not pre-empt the height; - (b) The inclusion of roof energy and photovoltaic equipment in the fifth dimension design; - Night visions express only public entrances, shared courtyards and operational status and do not create a large media presence; - Public art combines the industrial memory of Jing-Zhang Railways, the history of computing development and the culture of innovation in Central Guan.

Space modules to reference ratios

The following percentage is a reference to ** conceptual priorities for the urban design phase, which is used to guide the relationship between the six factor spaces rather than the final statutory indicator.

Functional category	Reference priority/scale area	Design intent
Talent, enterprise, open source group	40%—50%	The elements that form the source of original infusion and public vitality
Data, standards, governance cluster	20%—25%	Carrying credible data flow and governance voice output
Computational power, edge and authentication (underground)	15%—20%	For near lightweight, marginal caches, validation of workshops and energy synergies as substrate energy enablers
Transport, logistics and security support	10%—15%	Provision of equipment transportation, fire rescue and routine services

Retention of the main space to talent, enterprise and open sources has led to the “source of environmental innovation” rather than the “engineer room”; and the modification of the general power from the “compressed corner” to the “underground energy element” - its size in the field block is used for the near-light, marginal cache and validation workshops, and for the physical interface of the energy-computational power-city safety interface, rather than being cut to nothing. 30-50 acres of work units are only conceptualized scales, ultimately based on official focus areas and regular detailed conditions.

Design drawings and simulations

The results should result in at least seven designs, not just system frames:

1. * General scene plan: **representation of one session, two rings, three interfaces and six clusters**; 2. * Front floor open space **map**: expression of links between the public, business, research and parks; 3. * Architectural profile: **representation of the public base, collaboration mid-level, key data and relationship to standard safety layers, energy roofs and flood resistance**; 4. * Quadrate **chart**: expression of human flow, logistics, data flow and fire stream; 5. * One-day scenario: **expression of the use of scientific research, business, public and urban governance**; 6. * Impact Switch Simulation *: **Expressing Talent/Enterprise/Computational Power/Provisional Power/Elemental Restructuring and Space Response**; 7. * Central Park has a map of the intellectual-carbon **platform**: expression of sensory nodes, carbon sequesters, water bodies and water penthouse layout, and dual-carbon cockpit interface.

Digital sandpads should allow the switch to “normal-concern-restricted-carry-emergency-restoration” to indicate open areas, closed areas, factor mission locations, power supply routes, water collection risks and changes in personnel evacuation in different states.

AI Innovation Ecology, Talent Image and AI+ scene

The chapter focuses on the AI+ scene in the public service, stock update, smart transport, soft intelligence and rail veins, and forms the complete service unit with the client, space carrier, data boundary, manual review, failed exit and operator.

4.2.7 Three types of AI+ scenario

The inventory update cloud applet is a light quantitative user interface for the above-mentioned closed loops. The following three scenarios translate “Space Asset Identification AI Auxiliary Decision-Making → Precision Space Activation → Smart Operating Schedule → Feedback Optimization” into a perceived user experience.



AI Camp City Platform Resource Square interface



Map interface for space assets for Camp AI platform

Resource maps

* *Planning object**: local governments, national finance platforms, space operators, innovative enterprises, community leaders, citizens.

* *Space position**: liveable assets covering low-use areas along the Jing-Zhang corridor, sub-bridge space, idle buildings, edge green areas, etc.

* *Wisdom experience*: **1. * Resource Square browsing**: First page into Resource Square, screening for "full/hotspot assets/light assets/heavy assets" and rapid positioning by spatial type and operating model; *** Multi-dimensional Labeling: support for retrieving by city, land type, label, sorting, AI recommends potential matching assets based on user history preferences and project**

graphics; * Resource map positioning: switch to resource maps to quickly view asset distribution, tenure boundaries and perimeter interfaces on real floor maps to reduce the cost of footing on the ground; * Transaction case reference**: showcasing historical transaction cases through the “resource column” to provide business and profit models for operators to draw on.

** Data and privacy boundary*:* open space asset information is dissensitized; tenure and financial sensitive information is open only to accredited operators.

** Operating entity*:* national platform + source of official innovation operator + third-party space operator.

Scene 2: Request to operate — asset details and “I’ll run”



Thematic interface on resources for the AI Camp City platform



Resource details

* Planning object*: Intent operators, community leaders, innovative enterprises, creative teams.

Spatial location: any online asset in the Jing-Zhang gallery.

Intelligence experience: 1.* Asset details card: **access to information on the status of space, subject of tenure, area of use, building area, lease offer, current status photographs, terms of business concessions, location and project profile, plot details;**

2. AI Adaptation assessment: the system automatically assesses the feasibility of functionality mix, using the proposal and revenue measurement based on asset characteristics and the project plan submitted by the operator; **3. One key application for operation: the operator clicks on the offer of “I'm going to operate (quick application)” and the platform is automatically transferred to the National Finance Platform/Property Review;** **4. Rapid consultation and interface*:** support for “quick consulting” access points to enable the operation to communicate directly with the client or owner of the platform on policy, property rights, alteration requirements, etc.

** Data and privacy boundaries*:* operator ' s business plans and contacts are available only to authorized reviewers; public information does not contain personal privacy.

** Operator*:* National Platform + Proprietor + source of official innovation.



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AI Camp City platform update request release interface



Public collaboration forum interface for the AI Camp City platform

Forum discussions



Policy advice

* Planning object*: Periphery, innovator, community manager, designer, urban renewal implementer.

Spatial position: Online platform + acupuncture micro-update point + community space.

Wise experience: 1.* Demand release: **users click on the bottom “demand” button to fill out the demand title, description of demand, contact details, type of demand, choice of type, selection of city and upload of live photographs, needs for rent, location, adaptation, cooperation etc.;** 2. Demand square matches: **the system matches the needs of citizens/enterprises with the already online assets of the platform with AI and reverses potential operational opportunities to the property party;** 3. Forum

discussion: **discussion in the forum around specific assets or updates, people, designers, operators can build joint proposals, point reviews, and create a climate of transparent governance;** 4. Policy information query*: a central display of policies related to park-side services, inventory asset pool activity, urban renewal, etc., in the policy information section, to lower the information threshold for operators and citizens.

* Data and privacy boundaries*: sensitive information, such as contact details, can be selected for public domain; Forum content needs to be validated and displayed.

* Operator*: Community House Council + urban renewal implementer + small program operator.

4.3.3 Category V AI+ traffic scene

Five AI+ scenarios bring "AI native transport" to an accessible and operational place: each scene corresponds to a real traffic problem — brain mobility, end distribution, public transport timing, travel services, security of movement — and gives a complete closed circle of "space-service-governance", with each client, spatial location, system architecture, data governance, operator and exit mechanism.

The five AI+ scenarios are not stand-alone devices, but are operated in a single set of *intelligent traffic management platforms* — first with the prototype of the platform and seven major modules, and then with five scenarios in each case.

The Platform brings the movement and governance of the five scenarios to an entry point, following the principle of "*Governance before Technical Devices*": AI only produces recommendations, evacuations, closures, movement orders are confirmed; data are minimised, dissensitized, and traces are traceable. The Platform is not an independent product, but an operational support layer for the urban signature proposal.

4.3.3.1 Prototype of platform



Prototype of Smart Traffic Management Platform

Smart traffic management platform dashboard prototype: top module toggle + base map of park + running monitoring details bullet windows + KPI + five-state switch (protocol to further productization)

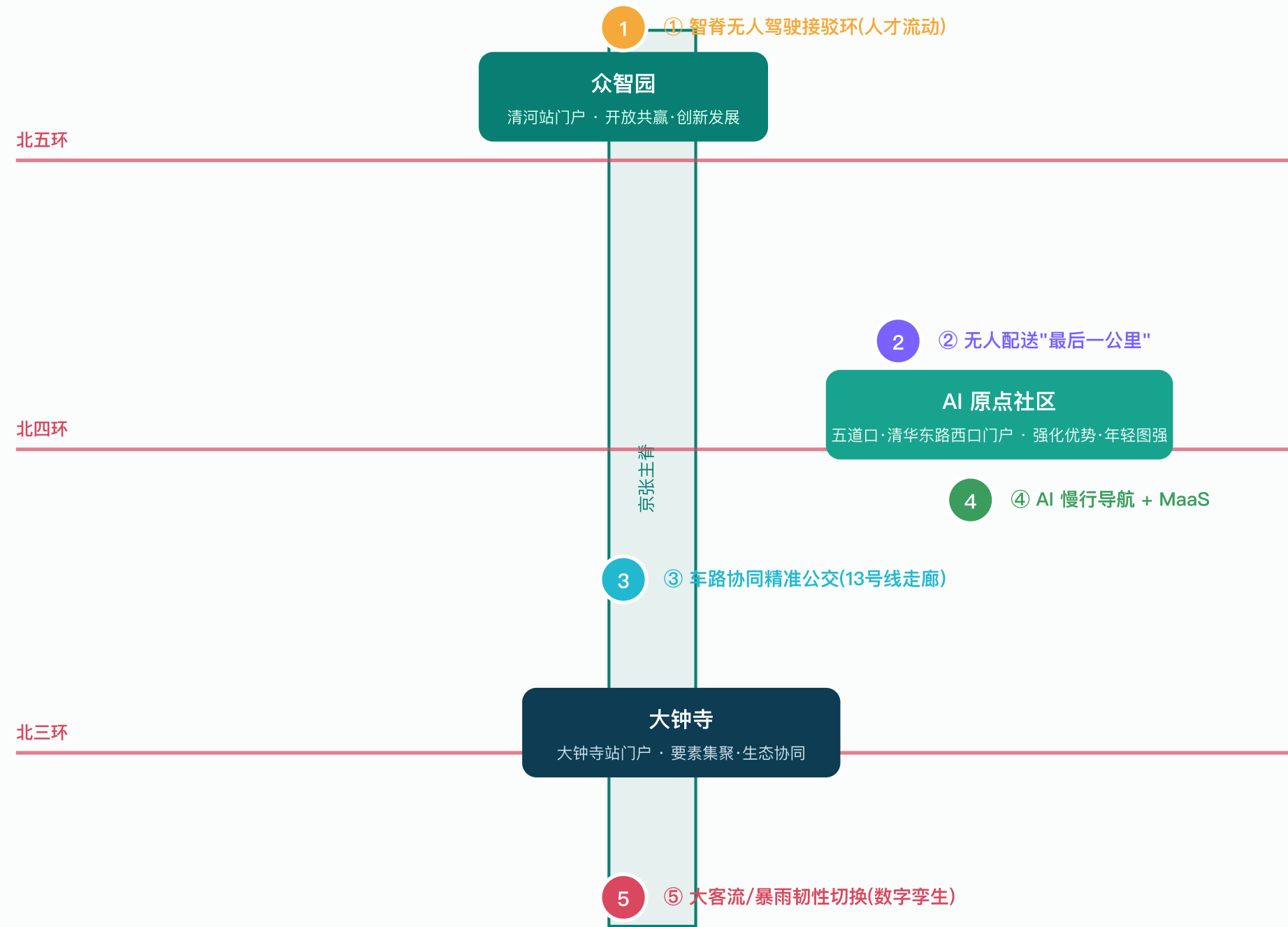
4.3.3.2 Seven functional modules and five scene mapping

Module	Correspond	Main functions
Operation monitoring	Global situation	Flow/road/water level real-time perception, fault indicators and alarms
Access dispatch	scene one.	Unattended appointments, dispatch, terminal prediction
Public transport synergies	Scene three. Precision bus.	Vehicle-infrastructure cooperation time, bus priority
Distribution Management	Scene two, no one to deliver.	Distribution orders, routes and docking points
resilience Switch	scene 5 switch	Normal focus on a limited load emergency recovery 5-state switch
Maas Travel Services	scene IV Maas	Course planning, multiplication advice, carbon credits
Data and governance	Cross scene	Data classification, desensitization, audit marks, manual review

The following scenario draws on the "vehicle-infrastructure traffic / precision public transport / unmanned + Maas / digital twin" capability and the "three-line integration / global parking / cloud twinning / low-altitude logistics" framework of smart traffic solutions.* *Capability is a methodological reference and does not constitute a commitment to the construction or implementation of any subject.**

五类 AI+ 交通场景落位

场景与一带三核的空间关系(概念示意,非精确底图)



五场景对应第 9 节;数据最小化 · 可解释 · 人工复核 · 明确运营主体

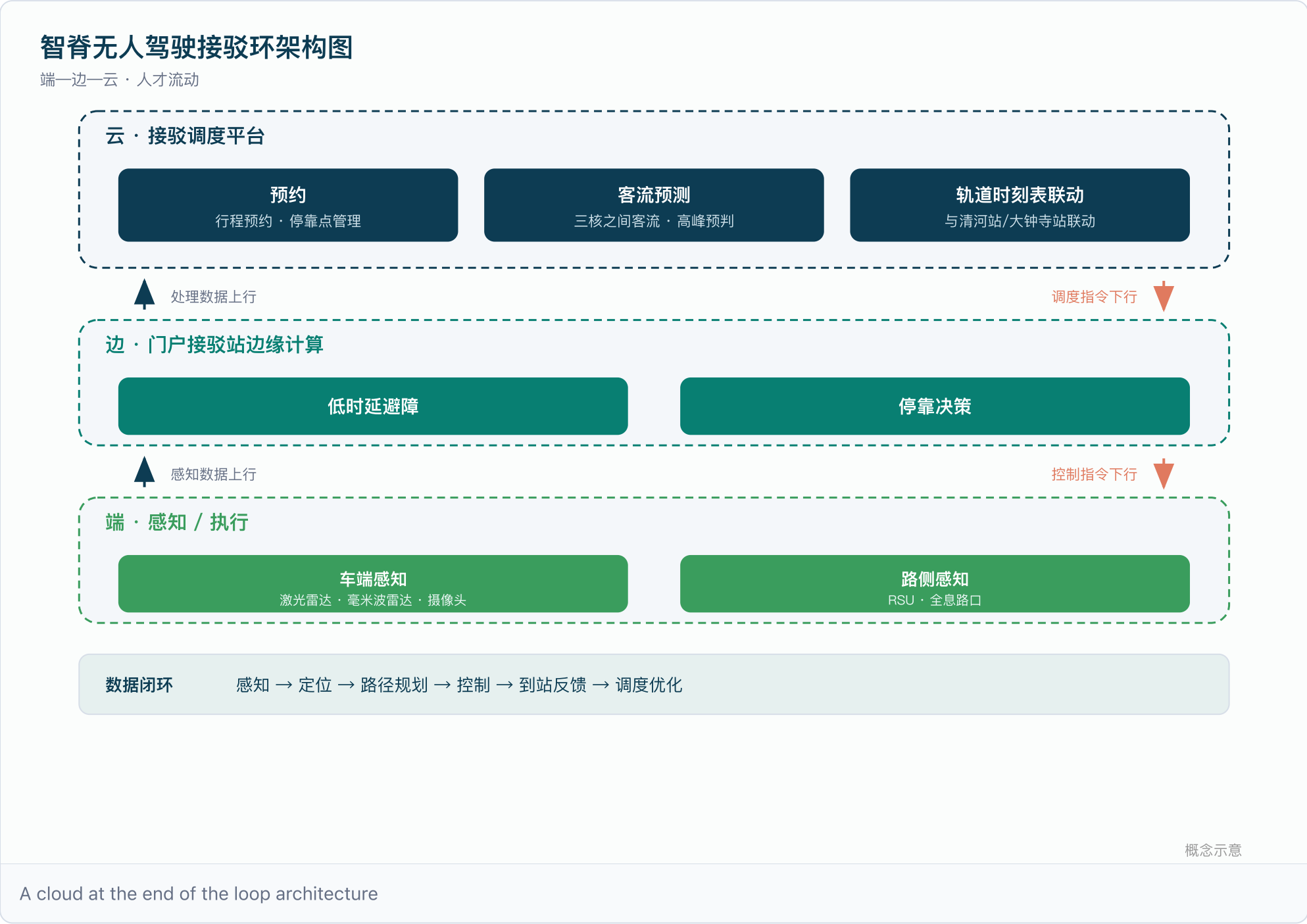
概念示意 · 待官方边界精调

Scene one: Unmanned access ring (brain flow)

Possibility and object: backbone connections between the three cores, service commuters, visitors and high-level train-to-passengers at the Qing River station to reduce the mobility of talent between the three cores.** Space position: along the Qing River station — Dazhongsi station, low-speed unconnected access (hydrogen power/electrically powered minibus, design speeds of 20-40 kilometres per hour), station set at 300-500 metres conceptual distance, combined with the main spinal cord and three nuclear portals.

System architecture (end-side-wind)

- **end:** vehicle-borne laser radar/mm wave radar/cam, roadside RSU and hologram; - * *Edge:* **Computation units on the edges of portals and access points to assume low-temporal avoidance and docking decisions;** - * **Cloud:** access scheduling platform, appointment, passenger flow forecasting, link to orbital clock; - * *Data closed loop*:* Perception positioning → path planning * *Control to station feedback* * Schedule optimization.



** Indicator and validation**: delay in response to access, commuting time between the three cores, rate of arrival; test at closed/semi-open segment before opening.

** Implementation and exit**: in the cost class; piloting of the Qinghu station — Dazhongsi station line; stop if safety is not met and return to routine access.

** Discrepancies (lightpoints)**: running on the Jing-Zhang main ridge, using the orbital portal at the end, servicing the flow of talent rather than viewing — no access is the blood vessels of the movement of talent between the three nuclears, not a tourist tourist tool.

Scene 2: No one delivers "last mile"

** Positioning and Targeting: terminal distribution at park/community/high school, service park enterprises, community residents and college teachers and students, bringing "computational power near + life" to the last kilometre.* Space position*: distribution point and terminal storage between Zhongzhi Park and the community of origin, combined with low-altitude drone platforms.

System architecture (end-side-wind)

- **end**: low-speed ground unmanned vehicles + low-altitude instant delivery drones; - ** border: distribution and movement control of terminal warehousing;* - *** Cloud**: distribution scheduling platform, order tracking and route management; - ** Data closering**: Dialing distributions → Handovers → Receipts and Abnormal Schedule Duplicate.



* *Data and governance* *: orders and deliveries are minimized and recipient information is minimized; no-fly zones, lanes and landing points are confirmed by airspace and security management; logistics flow lines are independent of evacuation routes by time or break.

* *Manual review and operation* *: main operator is source of operational innovation + logistics operator; distribution anomalies are artificially covered; non-digital replacement for traditional delivery + self-depositors.

* *Indicator and validation* *: distribution response time delay, abnormal bottom rate, end-to-end interface accuracy.

* *Implementation and exit* *: low-cost (renewal of existing warehousing and roads); pilot Zhongzhi Park single campus; security or complaints fall short.

* *Discrepancies (lightpoints)* *: reuse orbit/road redundancy, distribution and walk time break, high frequency small pieces for AI talent - drop low distribution to "computational power close to + life".

Scene III: fine public transport

* *Positioning & Object*: **Timeliness of public transport along and across corridors, serving bus passengers, supporting the suture of the east and west.** *Space position* *: Integrated conversion of conventional public transport along the corridor of line 13 by cloud-side-car, hologram point at intersection.

System architecture (end-side-wind)

- **end (roadside)**: RSU, mm wave radar, hologram, signal collector; - * *Edge*: **calculation of the edge of the road, time of dynamic signal matching**; - * **Cloud**: Transport dispatch centre, class density and route optimization; - * *Data closed loop* *: → bus priority back to station ** scheduling optimization when roadside perception dynamically pairing.



Scenario 3 System architecture: end (roadside perception) - side (frontside border control) - cloud (bus dispatch centre) * Service flow*: Roadside perception → dynamic mix ** bus priority to station information return → schedule optimization.

* Data and governance*: collection of vehicle location, speed, passenger flow data only, without personal identification; signal control and priority strategy confirmed by traffic management and retained manual takeover.

* Manual review and operation*: the main operator is the transport operator + source of operational innovation; signal strategy manual confirmation; non-digital replacement for fixed time frame + manual movement.

* Indicator and validation*: ratio of traffic to station, rate of adaptation during signal matching, increase in peak points.

* Implementation and exit*: in the cost class (roadside retrofit); piloting a single public transport corridor in the corridor of Line 13; when performance falls short of the target, backsliding.

* Discrepancies (lightpoints)*: the suture of goods in the corridor of line 13 for precision public transport services — the timing of public transport to cross-council connections instead of a single hub.

Scenario IV: AI active mobility navigation integrated with Maas

* Positioning and Object: **One-Stop Activity Mobility + Track + Access Out to serve residents, teachers, students, entrepreneurs and visitors.** Space is down* with the Jing-Zhang vibrator belt as an interface, low-invasive sense and interpretable guidance to identify Active Mobility break points, crowded nodes and accessibility needs.

System architecture (end-side-wind)

- **end**: low-invasive, interpretable screen; - * Edge node, active mobility breakpoint and congestion recognition; - * Cloud: **Maas platform, itinerary planning, multiplication advice, carbon credits**; - ** **Data closed loop**: process planning → conversion recommendation → action mobile navigation → carbon fraction feedback → breakpoint update.

AI 慢行导航与 MaaS 一体化出行架构图

端一边一云



概念示意

A cloud at the end of the Maas architecture

Scene IV System Structure: End (Human Flow/Guide) - Side (Reversion of Dynamic Belt) - Cloud (MaaS Platform) * Service flow*: process planning (orbit + active migration + access + shared bicycle)

* Data and governance*: only human flow and path data are aggregated without personal portraits or commercial recommendations; content for orientation and recommendation is dissensitized and manually reviewed; accessibility findings are subject to professional review.

* Manual review and operation*: the main operator is source of operational innovation + travel service platform; guide and recommend dissensitisation clearance; non-digital replacement for static marking + manual queries.

* Indicator and validation*: 5/10/15 minutes of active migration availability, active migration breakpoint recognition, multiplier recommended adoption rate.

* Implementation and exit*: low cost; pilot in Jing-Zhang main ridge segment; privacy or misleading subject is recommended below.

* Discrepancies (lightpoints)*: Maas Service Active Mobility Main Ridge and Tri-nucleic connection, core of which is "identify Active Mobility Breakpoints" rather than simply recommended route.

Scene 5: The Great Flow to resilience

* Location and object: **traffic emergency supported by digital twin, service manager (transport, emergency, water, park operator), public de-sensitization display of awareness.** Space down*: global AI event week bus flow and storm floods "Pre-motion-reverse" in conjunction with 4.4 water-saving safety "four-prevention system".

System architecture (end-side-wind)

- **end**: Skyland Waterworks Awareness Network (rain, water level, passenger flow, road conditions); - * *Edge: Zhongzhi Park City Centre for Logic, multi-smart "four-prevention system" closed*; - * **Cloud**: platform for urban operations, transport, emergency response; - * *Data closed loop**: perception → Four-prevention system → Duplicate → manual decision-making ** implementation feedback.



Scenario 5 System architecture: End (Skyland Waterworks Awareness) - Side (City Logic Center) - Four-prevention system - Clouds * Toggle logic*: state of concern (passage/water level deviation) → restrictive mode (downgrading of non-critical activities, access encryption) → emergency state (separation section, cross-nodal takeover, evacuation route activation) ** recovery (inspection and phased restoration).

* Data and governance*: only structured monitoring and desensitization indicators are addressed and no personal tracks are collected; all evacuation, blockage and movement decisions are confirmed, and AI only outputs recommendations.

* Manual review and operation*: the main operator is the Urban Operating Sector + Zhongzhi Park City Centre of Dictionary Twin Technical Support (the conceptual phase is a competency reference); non-digital replacement is a preset + manual command.

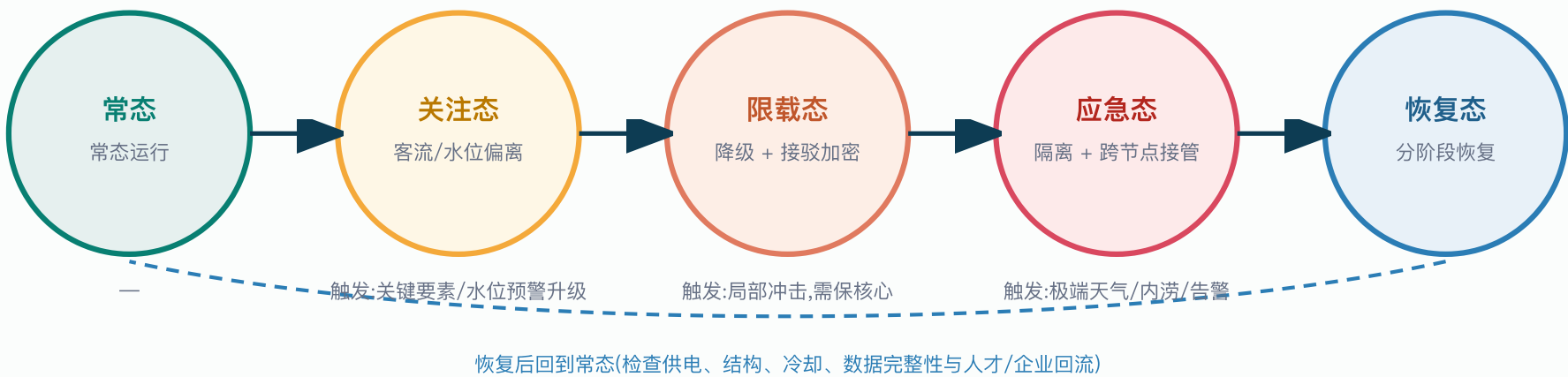
* Indicator and validation*: transition time, pre-proposald advance, cross-node take-over success rate (upplay of marker digital twin).

* Implementation and exit*: in the cost class (re-use of the Four-Provention system capability); pilot, single-stage global AI activities week; misreporting/frustration and adjustment model.

* Discrepancies (lightpoints)*: extension of resilience to heavy rains, mixed with water safety forfour-prevention system in the Jing-Zhang water system (Clean/Mount/Trop).

交通韧性切换:常态 → 关注态 → 限载态 → 应急态 → 恢复态

以可观测指标触发,与 4.5 节水安全"四预"衔接;疏散/封路/调度由人确认



概念示意 · 数字孪生推演,AI 仅输出建议、人做决定

Traffic resilience toggle five-state diagrams

Traffic resilience to quinquennium.

Intelligent traffic and digital twin water case data are endorsements of capability and are formally included in the `sources.json` registration sources prior to submission of the package and confirmed for external reference, and do not constitute a commitment of the proposal to the construction or implementation of any subject.

4.4.5 Three types of AI+ scenario: three types of user trip

The following three scenarios are aimed at three core categories of users — scientific teams, start-ups and citizens — showing how the elements are “one-stop” acquired, how the elements are “diagnosed”, how the source of original innovation is “sensitized”.

Scenario one: "One-stop access to source ecology" by the scientific team

One-stop eco-source digitizing system for scientific teams

One-stop eco-source digitizing system for scientific teams

** Planning object**: Colleges and teams of scientific institutions (teachers and students, post-doctoral, independent researchers), entering from the campus innovation interface.

Spatial Path: School Innovation Interface → Identity Verification Elements Sharing Courtyard (Results Submission Desk) → Local Validation Workshop results returned.

** Process*: **1. * Submission and classification**: Team submission of summary of results, data security phase (public/authorization/restricted) and required elements (computational power, data, evaluation, open source hosting); **2. * Elements: System proposal for a combination of elements based on the “minimum element, near-computational power, compliance priority” principle** *: small-scale validation stays in local certification workshops, large-scale training and centralized reasoning moved to Zhongzhi Park or outlying areas; **3. * Security evaluation**: results returned to local low-time testing and, following a red team test and security rating by the standard team, the decision was made whether to access the enterprise, the open-source community or the city; **4. * Left a mark and rediscussion**** : all-way auditable marks, team can look back at resource consumption and compliance findings.

** Data and privacy boundary**: security phase for processing results metadata only and team-initiated submissions; campus data and scientific results required authorization; no personal research behaviour trajectory collected.

** Manual review**: security classification and feed findings are confirmed by security personnel in conjunction with the team and the system is recommended only.

** Operator**: source of original innovation operator + standard assessment laboratory.

Scene II: "Ecoresilience Clinic"

Start-up Enterprise Ecoresilience Outpatient Digital System

Start-up Enterprise Ecoresilience Outpatient Digital System

- * *Planning object**: Start-up team and growth business, presenting business needs in factor service halls.
- * *Process (seven steps)**: demand diagnostic → safety classification factor recommendation → model fit ** test validation → on line → running double disk.
- * *Core feature - Single point dependency assessment**: Clinics also produce "single point dependence" reports, indicating which elements of an enterprise are over-centralized on a single supply side (single cloud, single data source, single computational power channel) and making redundant recommendations for setting up a "resilience" from park slogans to enforceable business decisions.
- * *Data and privacy boundary**: processing business and compliance information only from the enterprise, using credible sandboxes, Zhongzhi Park or peripherals.
- * *Manual review* *: the final proposal was jointly confirmed by enterprises, technicians and security personnel, keeping the windows below the line closed.
- * *Operator**: source of original innovation operator + corporate service team.

Scene III: Citizens enter the “discoverable reminisce”



Visible resilience Digital Sandboard

* Client*: Peripheral residents and the public.

Spatial path: Public Innovation Ring Pavilion → Sharing courtyard → Park Eco-interface.

2. * Participatory activities*: youth AI course, model security discussion, open-source contribution experience, pilot testing of new products.

* Data and privacy boundary*: public areas use only desensitized, delayed or simulated data and do not enter data, standards and governance safety zones; no personal tracks are collected and no commercial recommendations are made.

* Manual review*: displays were released after sensitization and manual review.

* Operator*: source of official innovation operator + public education partner.

4.4.6 Deepening scenes of urban operations

The following three scenarios raise the source of spatial innovation from “in-garden principles” to “city running”: water safety “four-prevention systems” demonstrates how AI’s original reasoning supports spatial recollection under extreme weather, resilience toggle to demonstrate a differentiated response to a volatile shock, and the Dual-carbon platform demonstrates the measurable operation of the ecological chassis.

Scenario IV: Real-time reasoning for urban running - Water Safety

Water Safety City Operating cockpit

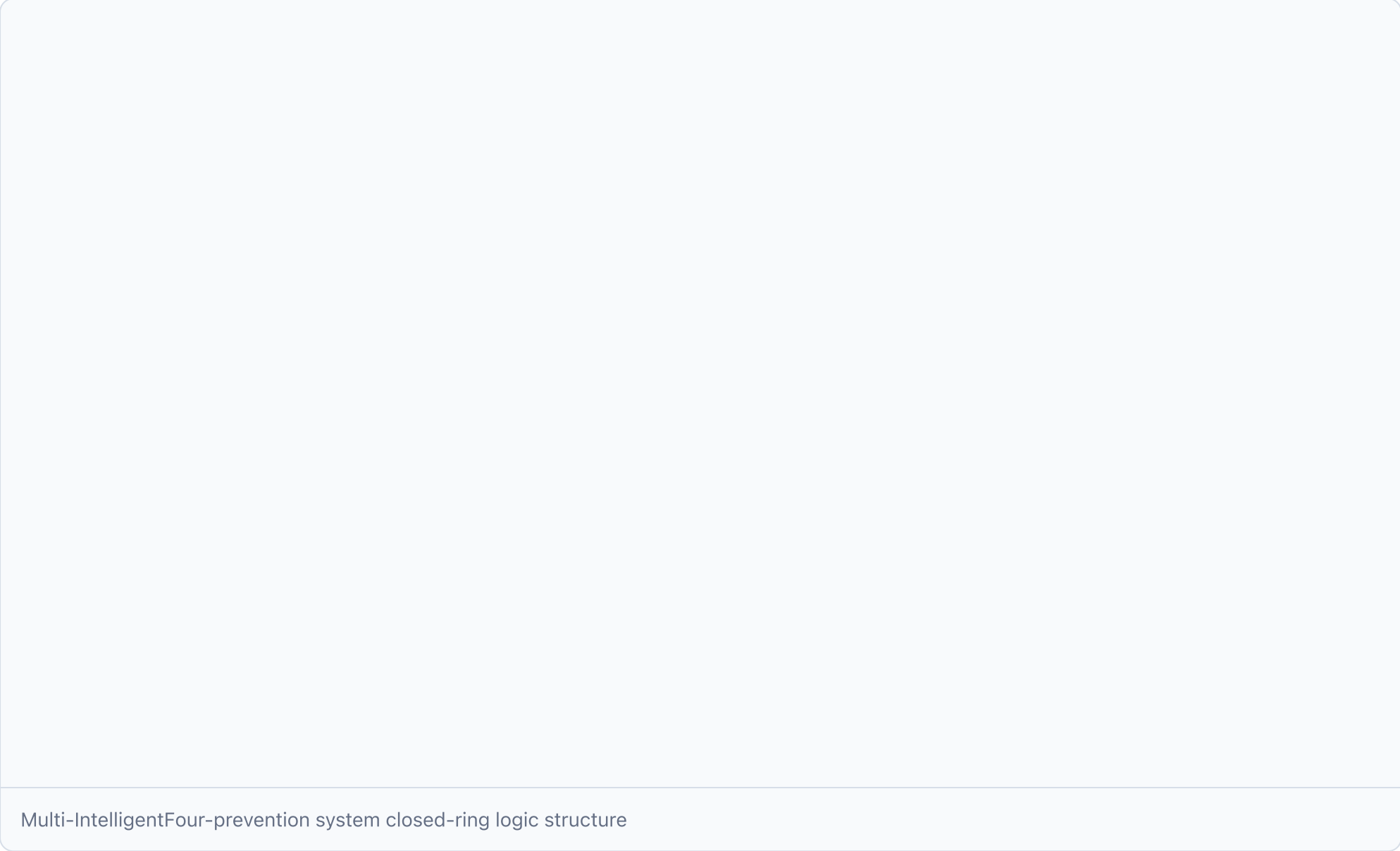
Water Safety City Operating cockpit

* *Position**: This is a key scenario for moving "resilience source of abstraction into urban reality, using water security as an entry point to demonstrate how AI's original reasoning supports extreme weather and spatial resilience under urban flooding. It also relies on the Jing-Zhang Railway to transform urban hydro-capacity into a locally accessible and visible resilience service, based on the innovative Inland Qing River, the Moon River, and the true water system of trans-rivers.

* *Planning object**: Authority managers (city running, water, emergency, park operators); public sensitization displays.

Spatial position: the reasoning central power is concentrated in the urban reasoning centre of Zhongzhi Park, and the origin community of AI and the Qing River/Mount/Turk River system are mobilized through a high-speed network, and do not concentrate the reasoning equipment in the communities of origin. The Qing River, the Moon River, and the trans-river are part of the built-up water system, which is structured along with the problem structure of the Ningbo River "basin basin + urban flooding + underground space + lifeline" and provides for direct migration.

AI Origin structure (multi-intellectual "four-prevention system" closed loop): organized according to the competition "AI Native, multi-intellectual collaboration", while non-traditional water informatization:



Multi-Intelligent Four-prevention system closed-ring logic structure

1. **Agent (Monitoring)**: bringing together the "Skyland Waterworker" sensor network — rainfall, river level, gates, pumping stations, piped networks, videos, satellites, drones and ground patrol robots to upload structured information; 2. **Agent (model)**: Meteorological-hydrography-hydrodynamic coupling, advance prediction of water levels and flooding risks; 3. ** Early warning Agent (thresholds/knowledge): **threshold triggers and expert knowledge analysis, automatically generate risk lists and send them to managers with authority***; 4. ** Preview of Agent (simulation)*: 2-3D combined flood flood flooding model to promote the consequences of "how a certain gate is deployed" "what happens if a section fails"; 5. ** Precedent Agent (mobilization): **proposal to connect drainage control, pumping stations, gates and cross-nodes***; 6. ** Duplicate Agent (over time)*: Post-post comparison of forecasts - measurements, iterative model parameters.

** Manual review and privacy border**: all drainage, gates, evacuation, fire and public safety decisions must be confirmed, AI only issues recommendations; only structured monitoring data and desensitization indicators are processed and no personal tracks are collected and no personal images are exported.

** Operating entity**: Zhongzhi Park City Centre for Dictionary + Urban Operations Management + Digital technology support (concept stage for competency reference, specific service relationships to be formally validated).

Scene five: Treasure versus resilience under impact



resilience to digitize the political screen

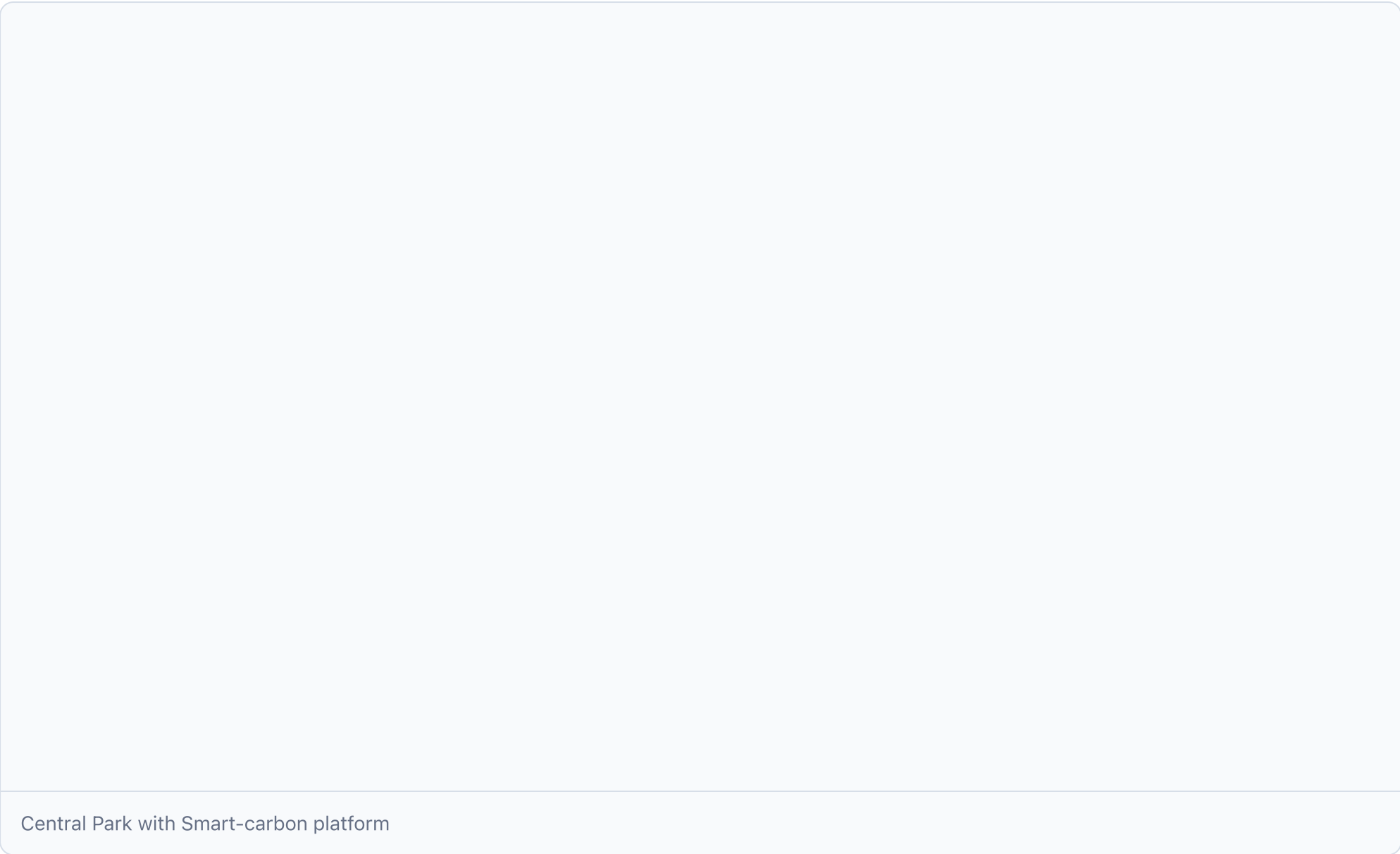
Switches to cover the "Element-Space-Systems" three layers, triggered by observable indicators and linked to the "four-prevention system" of water safety in scenario four:

- **Concerning state** (trigger: key factor indicator deviation from normal or water security early warning upgrade): increase in the frequency of monitoring of key elements — brain drain, enterprise emigration, data risk, supply of general power, water level leakage, power supply and structure;
- **Lock-down mode** (trigger: local shocks, core needs): outward relocation or downgrading of non-critical innovation activities to ensure continuity of core R & D, credible data flows with urban public services;
- ** Emergency state** (trigger: extreme weather/inner flooding/power or structural alarm): isolation of faulty subdivisions (physical or data subdivisions), cross-node taking over critical operations (Zhongzhi Park/outsidecomputal power take over, off-site data copies enabled), critical functionality security shutdowns;
- **Recovery status** (trigger: shock relief): phased recovery after checking power supply, structure, cooling, data integrity and brain/enterprise return.

The water security “four-prevention system” forecast-forecast-forecast-forward-forward is one of the decision-making bases for the “Concerned state-restricted-carrying emergency” transition. The process is presented in a simplified, dissensitized manner at the exhibition hall, making the “source of official innovation” a daily education that can be understood by the public.

Communicational powerforce map: direct effect of the above-mentioned switch on Level 3 computational power network — the normal task is to recalculate by “near lightweight + area” free movement; near lightweight landing, non-critical training/generation moves to Zhongzhi Park and the periphery to ensure continuity of core R & D with urban public services; cross-level taking over of critical operations under emergency response, introduction of off-site data copies, upgrading of the commutation power from “one room” to “one network-wide portable disaster”.

Scenario six: Central Park with intellectual-carbon platform



Central Park with Smart-carbon platform

** Positioning:* **Linear park belts formed at the heart of the community of origin in AI should not be understood only as “green packaging of smart computing facilities” but should be upgraded to * a perceived, measurable and manageable eco-resilience chassis** through the intellectual-carbon platform. It is both the daily resting space of source of environmental innovation and a composite infrastructure for urban carbon sinks, climate regulation and environmental awareness.

** Planning object*:* Peripheral population, commuters, researchers, business visitors, urban ecological management.

Spatial position:

- ** Perception node:* **installation of micrometeorological stations along park belts, anaerobic ion monitors, online monitoring of water quality, soil moisture and carbon flux towers, vegetation health spectrum probes;** - ** Carbon cell block:* "Quantifiable green infrastructure" is established as a model area for Joe-Imbush-grassed carbon comorage, rain-fed wetlands, water-drilling; - ** Presentation interface:* **the introduction of a dual-carbon information screen in the park ' s eco-interface and shared courtyard to show real-time the impact of park carbon sinks, cooling contributions, rainwater recharge, air quality improvement;** - ** Managing the cockpit:* data feed into the operations centre for urban water security "our-prevention system", energy management, computational power movement.

** Platform architecture (third floor)*:*

1. ** Perception layer: credible ecological data collection**

- Energy consumption: sub-energy consumption for buildings, transport, general power, lighting, air conditioning; - Sources of carbon emissions: vehicular traffic, catering, waste, refrigerant leakage, etc.; - Carbon absorption channels: vegetation cover type, leaf area index, soil organic carbon, wetland water quality and methane flux; - Environmental mass: Temperature, PM2.5/PM10, negative ion, noise, heat infrared surface temperature.

2. ** Accounting level: harmonized carbon accounting against 1.5 degrees**

- Development of sector-level carbon emissions-carbon sink dynamic equilibrium models; - Accounting by sub-accounts for construction, transport, green land, water systems, general power; - For the indicator "1.5 Model Lifestyle Zones", output of recommendations for compliance rates, gaps and optimization.

3. Appendix: three user interfaces

- **Public screen:** displaying "how much carbon has been sucked, how much temperature has been cooled and how much air has been cleansed in today's park", turning the abstract dual-carbon goal into a perceptible daily experience; - * *Management of the cockpit: provision of early warning of abnormal energy consumption to park operators, inventory of carbon-sink assets, greening conservation advice*; - * **Planning feedback interface:** linking design parameters such as park vegetation configuration, water area, water perforation rate, etc. to carbon sink effects, and an update of the design of surrounding plots under reverse direction.

** with the ecological integration of:

- (b) An energy consumption and carbon emissions access platform for the Centre of Intelligence and Accounting to explore a "green government power certification" mechanism; - Provide business tenants with carbon footprints and abatement advice as a source of data for ESG reporting; - Create a "low-carbon living laboratory" for the talent community, where behavioural data such as riding, walking, sharing office, etc. are translated into visualized low-carbon achievements.

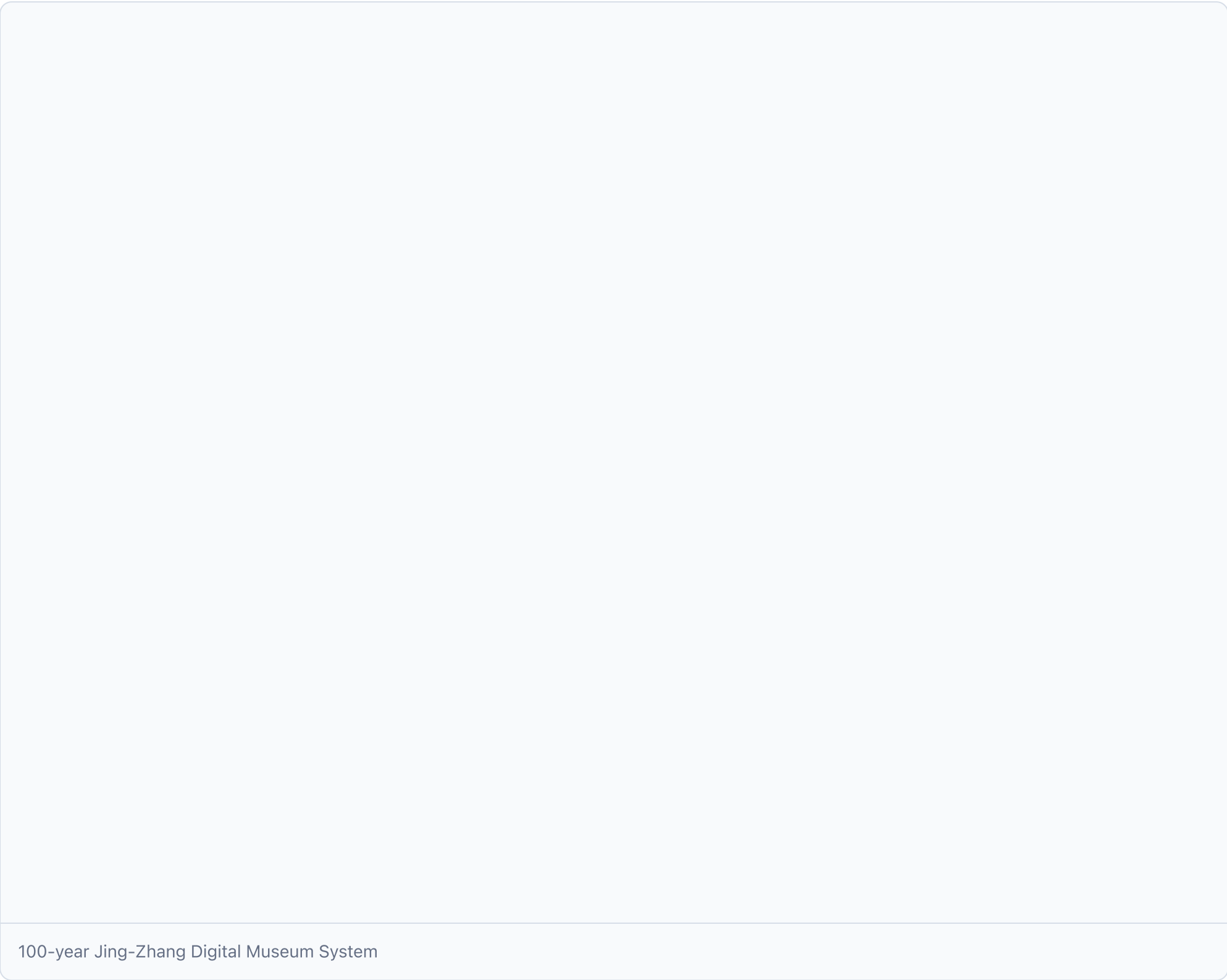
* *Data and privacy boundary**: collection of only regional aggregate data and operation of public facilities without tracking individual behaviour tracks; access to enterprise energy consumption data after authorization.

* *Manual review**: carbon accounting methodologies, target targets and proposals for significant emission reductions are validated by third-party verification and planners.

* *Operator**: source of environmental innovation operator + urban eco-environment sector + technical service provider.

4.5.7 Three types of AI+ scenario

Site one: Visitors to the 100-year Jing-Zhang Digital Museum



100-year Jing-Zhang Digital Museum System * Planning object*: national and foreign tourists, primary and secondary students, university research teams.

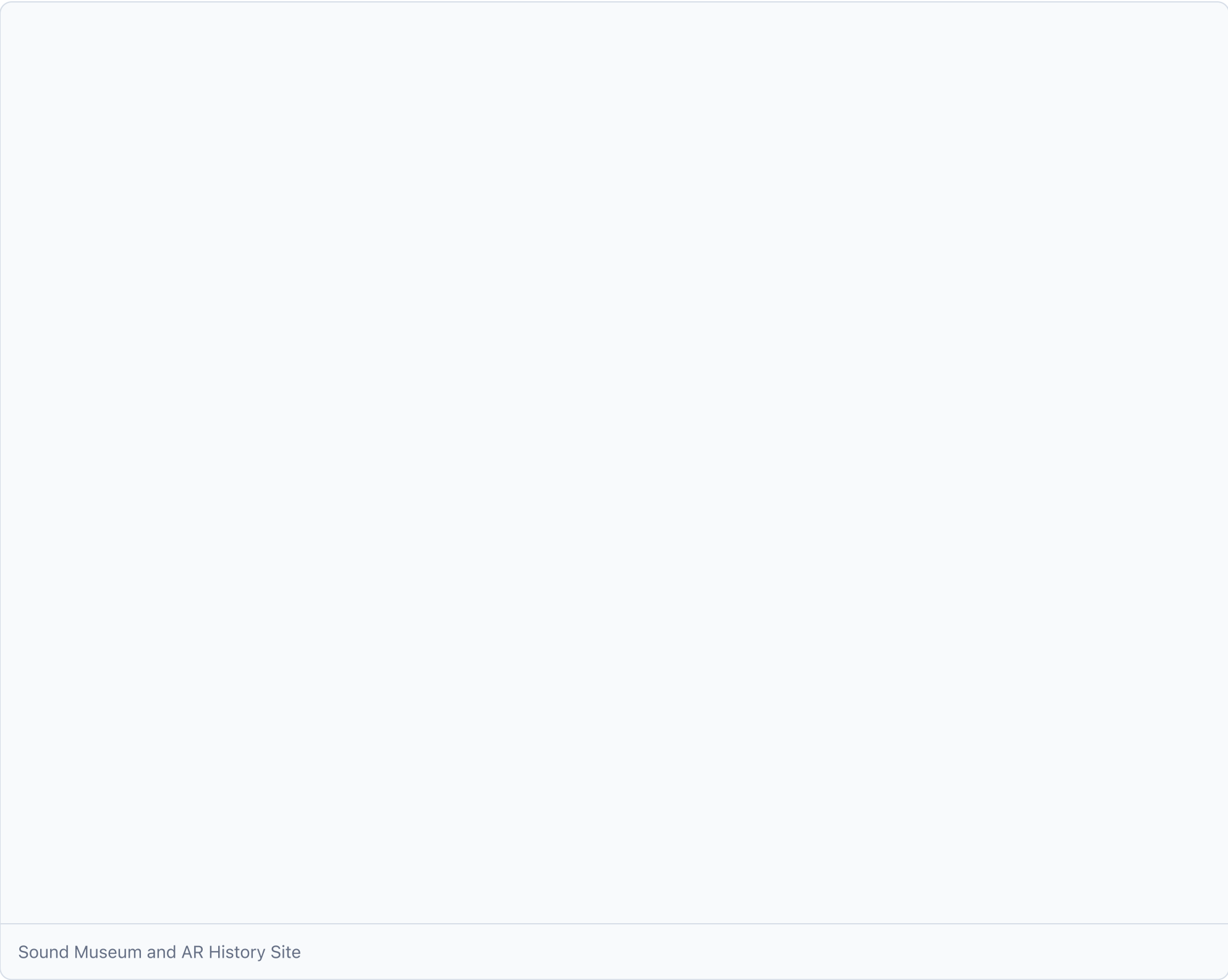
Spatial location: railway museum community at the Qinghua Park station node or the West Straight Gate node.

* Wisdom experience: **1. *Appointment and subject orientation:** Tourists choose engineering history lines, patriotism education lines or AI innovation lines, systems allocate time between traffic and the carrying capacity of sites; **2. digital twin display:** indoor museums display Jing-Zhang rail lines and key technologies through insulated cinemas, three-dimensional models; outdoor scene restoration areas add 1910/1960/2008 station history images; **3. Interactive cards:** VR drivers, car visions, text-making meals, commemorative stamps; **4.* Co-entry***: tourists can upload "Jing-Zhang memory" or creative content and enter digital archives or public art walls after review.

* Data and privacy boundaries*: only images of bookings, research feedback and authorized sharing are processed; no individual biometrics or tracks are collected.

* Operating entity*: museum operator + human security sector + educational partner.

Scene 2: Citizens enter the Sound Museum and AR History Site.



Sound Museum and AR History Site

* Planning object*: Peripheral residents, railway lovers, cultural visitors.

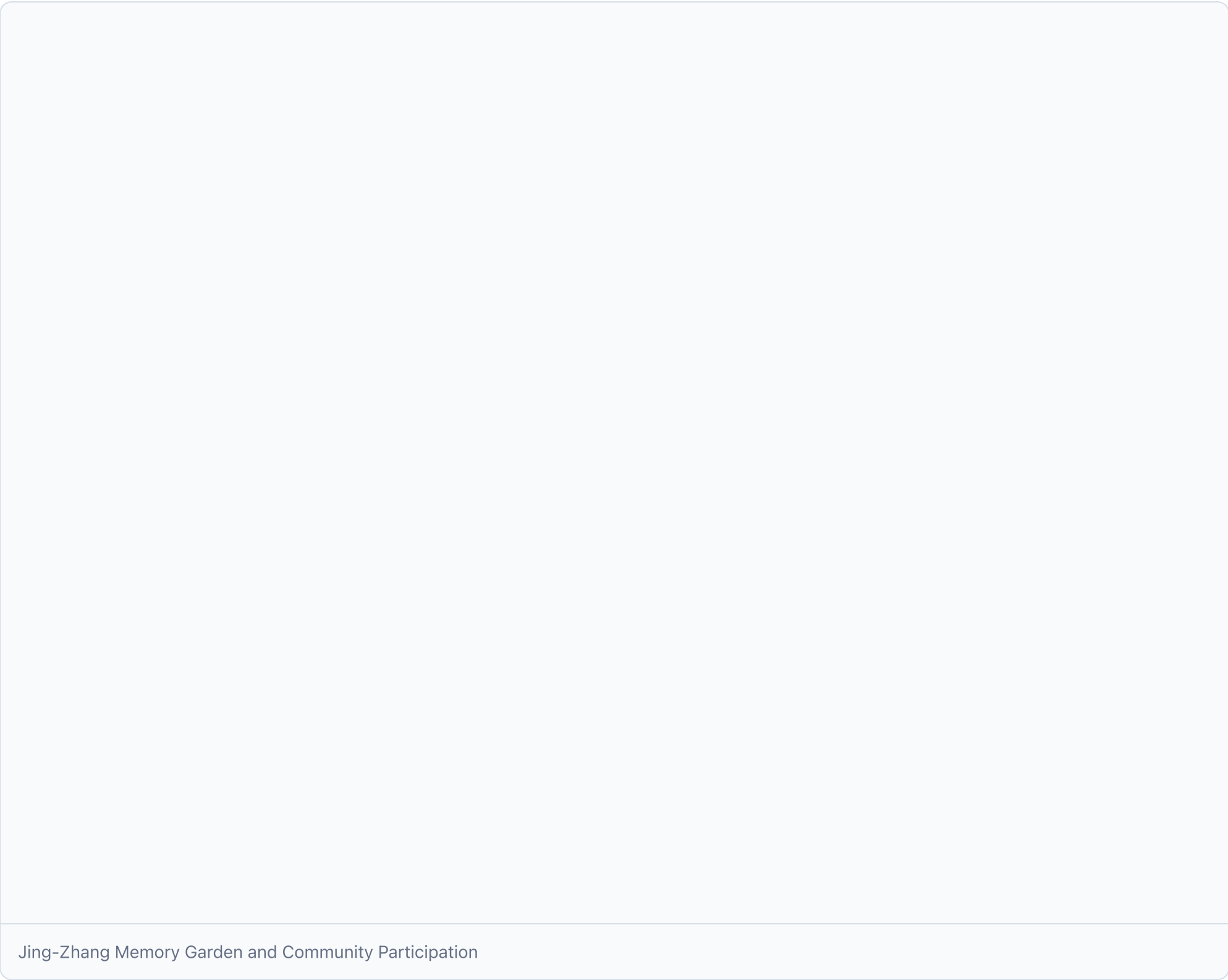
* Space landing*: in 1960, the station in Qinghua Park was transformed into a sound museum with an AR experience point in front of the station.

(b) 2.**AR historical site**: 1909 drive-through ceremony, 1949 "Go to Kyoto" special arrival;

* Data and privacy boundary*: oral history and voice gathering require a written consent; public broadcasts are sensitized and validated.

* Operator*: Museum operator + Community House Council + Oral History Agency.

Scene 3: The community created the Jing-Zhang Memory Garden.



Jing-Zhang Memory Garden and Community Participation

* Planning object*: Peripheral residents, community organizations, senior railway workers, families of new immigrants.

* Space landing*: small pocket parks such as mouths, front stations, community entrances, etc.

* Wisdom experience: **1. * Memory collection kiosks**: small audio/video booths at the entrance to the garden where residents can tell stories about themselves and the Jing-Zhang Railway; **2. * Co-planting: residents are involved in the selection of plants, the naming of small items, the design of layouts, and community memory is placed in space**; **3. * Digital message wall**: old photographs, old stories are reviewed and displayed on public screens or online platforms; **4. * Generational activities**: old railway workers tell their work experiences to young people, who use digital tools to help them make family memory books.

* Data and privacy boundaries*: oral history and old photographs require a written consent; public broadcast is dissensitized and authorized; no personal routines are tracked.

* Operating entity*: Community Houses + Museum Operator + source of original innovation.

Land use, building size and retention proposal

The update follows “evidence before judgment”: the railway heritage gives priority to repair, the storage space available for the structure is reversible, the demolition is subject to retrofitting after tenure, structure, security, security and implementation have been confirmed, and the construction of new ones to fill the service and resilience gap. All size conclusions maintain the conceptual design attributes. [standard: MNR-LAND-USE-CLASIFICATION-GUIDE] [depth: retain_renovate_demolish] [Spatial data](#)

4.2 Fine activation of camp city strategy *AI+ stock asset update cloud (space asset identification * AI supporting decision-making * precision space activation * smart operating schedule * feedback optimization *

4.2 AI+ finely activate inventory assets to update cloud

Operating against inefficient, debris, idle space along the Jing-Zhang Railway innovation zone, transforming the concept of "stocked assets update cloud" into a matching asset operating system; using the "space asset identification, AI support decision-making, precision space activation, smart operation movement control, feedback optimization" as a closed loop, turning space from sleeping assets to innovative cash flow and talent sticky carriers.

4.2.1 Strategic positioning: transforming space stocks into an entry point for innovative operations

Along the Jing-Zhang Railway innovation belt, there are a large number of inefficient spaces that are deposited as a result of railway re-routing, industrial upgrading, complex tenure: grey space under the bridge, idle stations, old industrial complexes, green areas on the edges, vacant office floors, etc. These spaces are not “negative assets” for urban development, but ** “sleeping assets” that have not yet been accurately matched to suitable projects and operators.

This strategy draws on the concept of "stock asset renewal cloud" —* *space for space adaptation projects*,* space for projects,* and *understands fine activation as** a light operating system running through 4.1-4.3: **it is not built on large-scale dismantling, but rather on space asset identification, AI support decision-making, precision space activation, smart operation movement and feedback optimization, making inefficient space a low-cost, flexible interface for innovation and life service entry.** 4.3 In other words,* 4.3, *the answer is “how the space in the innovative belt can feed itself and make blood on a continuous basis” rather than “the new building of as many as 10,000 square metres”*.*



Jing-Zhang Space Asset Inventory Update Cloud Applet home page

Jing-Zhang Space Asset Inventory Update Cloud Applet home page

On the front page of the applet, the four main portals, "Resource Square, Resource Maps, Policy Information, Forums", are the light interface between asset discovery, spatial positioning, rule queries and community management, and the inefficient Jing-Zhang corridor, moving from "sleeping assets" to "dynamic operations".

4.2.2 Space asset identification: from sleeping assets to operational data

Space asset identification is the starting point for inventory assets to update cloud closed loops. Only by digitizing and structuring the “household” of inefficient space can reliable input be provided for subsequent AI decision-making, matching and operation.

Four categories of inefficient space

The active space along the Jing-Zhang corridor can be grouped into four categories, each with different activation potential and constraints:

Space type	Typical object	Core issues	Activate Potential
Ashes below the bridge.	Under rail/road bridge, under shelf	Complex tenure, poor accessibility, low environmental quality	Traffic nodes, community activities, innovative services
Unreserved station/industrial complex	Old station, welding plant, warehouse, industrial yard.	Aging buildings, outdated functions, fragmentation of property rights	Museums, joint offices, creative markets, research
Under-office/coup building	At night, there's an empty office building and a corner dress.	Timeless, single business	24-hour service street, social consumption, community activities
Peripheral Green / Pocket Park	Street corner green, abandoned corner, small park	Inadequate facilities, low usage and lack of maintenance	Community gardens, flexible playgrounds, pet friendly spaces

Space assets database fields

Creation of a database of inefficient space assets on the Jing-Zhang corridor to generate a searchable, comparable and traceable digital identity card for each asset:

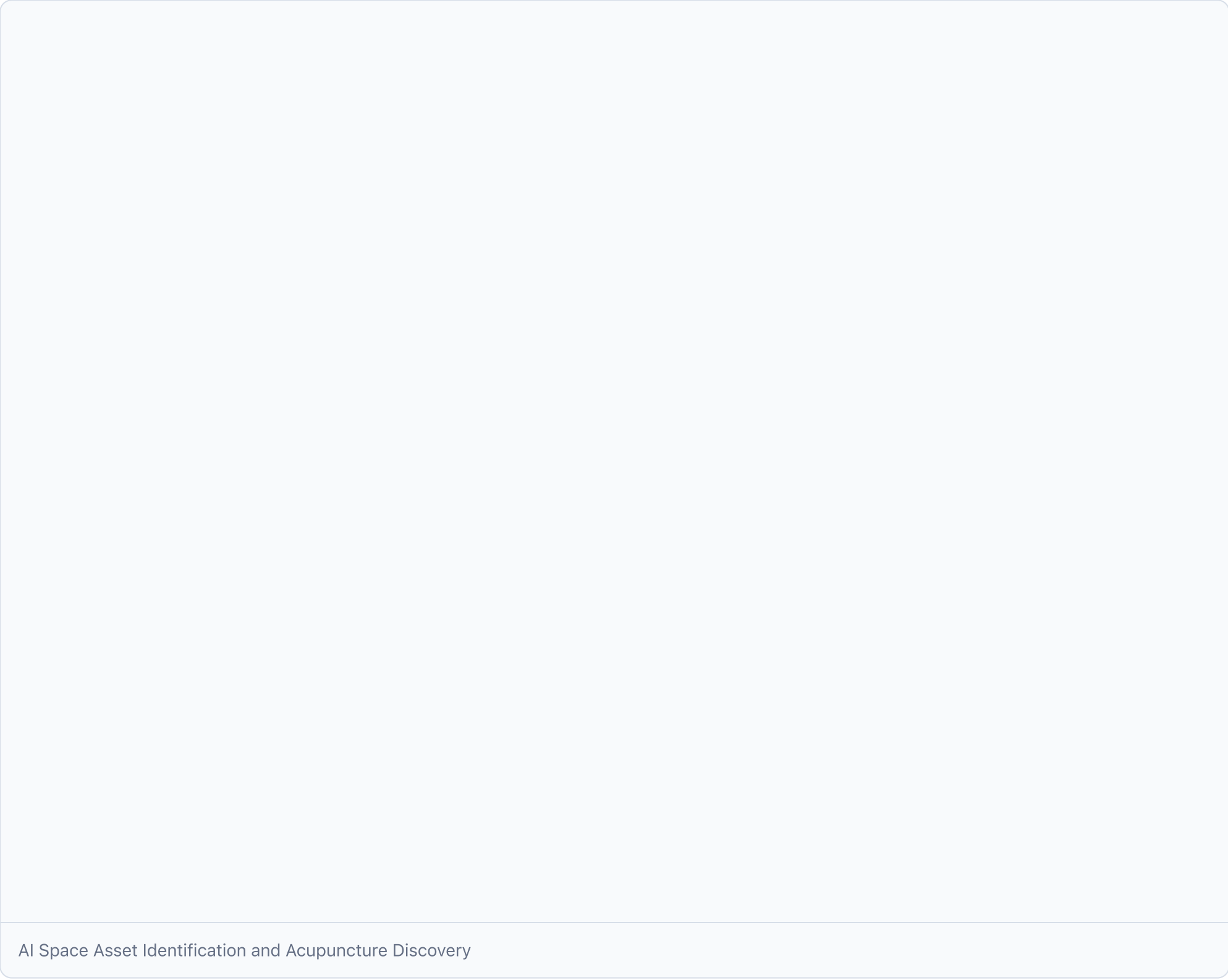
- * *Basic information*: **asset number, name, coordinates, administrative division, street, area of use, building area, volume rate, number of layers**; - * **Title and law**: subject of title, nature of land, status of land title, lease/use status, planning restrictions, fire and structural safety level; - * *Status assessment*: **current functionality, vacancy rate, physical condition, environmental quality, accessibility, peri-urban traffic congestion, municipal capacity surplus**; - * **Accompanying facilities**: distance from orbital transport stations, coverage of public transport stations, parking conditions, public Wi-Fi, charging stakes, accessibility; - * *Economy and operations*: **lease offers, historical rents/revenues, forward estimate of adaptation costs, measurement of operational gains, classification of heavy/light assets**; - * **Labels and portraits**: space type labels, suitable business labels, target population labels, activated difficulty labels, policy adaptation labels; - * *Multimedia and documentation*: **status photographs, aerial videos, cloud-pointing models, mapping drawings, scanned copies of title documents**; - * **Dynamic status**: upline status, matching status, operating status, performance rating, complaints and feedback records.

AI Indicators for spatial vitality assessment

Integrated static endowment and dynamic running data, building spatial dynamic assessment models, providing quantitative basis for screening and prioritization of acupuncture points:

Indicator dimensions	Specific indicators	Data sources
Bit Value	Distance from orbital station, public transport coverage, network accessibility, functional density	OSM, Goth, API, cell phone.
Population needs	Peripheral population density, share of innovative talent, age structure, facility gap index	Census, POI, research questionnaire
Current status dynamics	Day/night human heat, length of stay, social media card density	Cell phone mail, Wi-Fi probe, popular comment/red book
Space quality	Greening coverage, lighting, accessibility, environmental quality	Streetscape Image Identification, IOT Sensor
Operational potential	Property rights clarity, adaptation elasticity, strong policy support, market rental expectations	Tenure databases, policy documents, market comparable cases
Cost of intervention	Volume of renovation work, difficulty of municipal access, approval period, risk of demolition	Site survey, planning conditions, expert assessment

Model output 0-100 spatial vitality index, with three types of "prior activation" "potential observation" "suspension" recommendations.



AI Space Asset Identification and Acupuncture Discovery

Figure 1: AI Space Asset Identification and Acupuncture Point Discovery - Integration of Asset Database, Dynamic Evaluation Model and GIS Spatial Analysis, automatic identification of high potential acupuncture points along the Jing-Zhang corridor and generation of

priority ranking and preliminary business recommendations.

Resource Square and Resource Maps are visual entry points for asset identification results: Resource Square supports screening by spatial type, heavy/light assets, label combinations; resource maps display the distribution of assets, tenure boundaries and perimeters on the real bottom map, and reduce the cost of ground tripping.

4.2.3 AI Auxiliary decision-making: acupuncture point identification, program generation and subject matching

Based on asset digitization, AI assumed the role of supporting decision-making in “diagnostic-match-generated” to convert vague spatial issues into a list of implementable projects.

Acupuncture recognition logic

AI Acupuncture Point Identification follows a five-step process of "Data Integration Candidate Screening Comprehensive Rating Prioritization On Site Validation":

1. ** Data integration: **integration of asset databases, human heat, POI distribution, mobile phone messages, public complaints, planning conditions and social data***; 2. ** Candidate screening*: a hard threshold is set — property rights are relatively clear or potentially harmonized, there are no significant structural security risks and basic accessibility; 3. ** Comprehensive scoring: **calculation of a composite score for each point based on a dynamic evaluation indicator and identification of short panels (e.g., facilities with large gaps but high human flows)***; 4. ** Prioritization*: annual acupuncture bank is generated in conjunction with implementation difficulties, demonstration effects, investment scale and statusholder support; 5. ** On-site proofing**: Rapid Step Survey and Community Interview at High Priority Point to amend the "data blind zone" in AI judgment.

Spatial digital image

Generate a dynamic “space digital image” for each asset as a unified language for matching operational decision-making:

- ** Basic portrait: **location, size, tenure, current status function, constraints***; - ** Dynamic image*: human flow time series, mass composition, type of activity, peak valley; - **Demography needs**: population, industry, infrastructure gaps in the 5/10 minute walk around the perimeter; - **Opportunity image**: fit business, potential operator, expected yield range, superimposed 4.3/4.4/4.5 strategy; - ** Risk image**: structural hazards, property rights disputes, policy constraints, neighbourhood risk avoidance, municipal bottlenecks.

Space-project-operator matching mechanism

Establishment of a tripartite two-way matching model that harmonizes the “space search project” and “project search for space” into an algorithm framework:

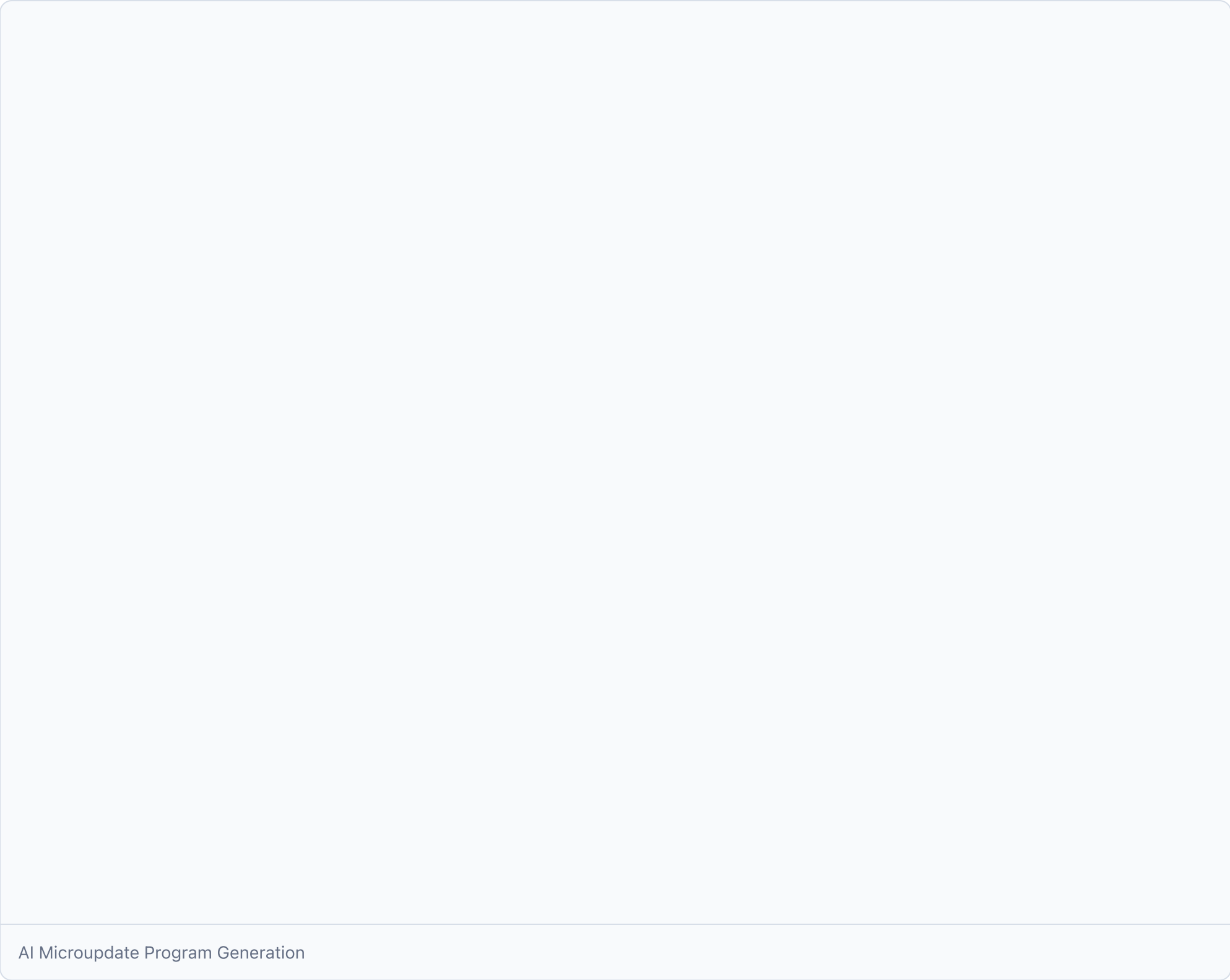
- ** Space side input: **digital portraits, binding conditions, owner intentions, revenue expectations, acceptable mode of operation (rent/assembly/sharing/public goods)***; - ** Project side input*: business plan, target audience, investment scale, business experience, social benefits, time-use demand; - ** Operating entity input: **subject type (State/private/community leaders/high school teams), credit rating, historical performance, competency labels, geographical preferences***; - ** Matching algorithm*: generate recommended ranking based on multi-purpose optimization (space compatibility, expected benefits, social values, risk control, operational sustainability); - **Closed decision**: recommended results enter the resource details-I'm running the application channel, property owners review online, and the platform keeps track of performance and feedback to the model.

Micro-update proposal generation process

For selected acupuncture points, AI supports the generation of "light, fast, quick" micro-updates:

1. ** Diagnosis of problems: **identification point slots (lack of seats, lack of lighting, lack of greening, lack of functionality, lack of accessibility)***; 2. ** User image*: analysis of the spatial and temporal distribution and needs of the main population groups around the perimeter (tripers, residents, students, tourists); 3. ** Toolbox matching: **a suitable mix of modules from standardized toolboxes (painting, furniture, kiosks, greening, lighting, digital facilities)***; 4. **Scene packaging*: superimplementing feature (coffee cafes, community kitchens, shared conference rooms, express delivery stations, riding stations, pet friendly facilities); 5. ** Cost estimate: **based on unit unit costs of modules and project volume output rough butt***; 6. ** Programme output*: production of plan maps, inventory of

materials, time series for implementation, business model proposals and expected results; 7. **Public feedback:** online platform publicity proposals, residential areas, comments, voting, AI compilation of views and supporting optimization.



*AI Microupdate Program Generation * Figure 2: AI Micro-update Program Generation** - Based on point-based diagnosis, user image and standardized toolbox, AI auto-generated multiple microsupplies for community, designer and operator selection, voting and iterative.

The asset details page is the AI Localization Interface to support decision-making: by clicking on the resource card, you can view the status of the space, the owner, the current status picture, the terms of the concession, the location and project profile, the plot details, and the appliance and revenue measurement generated by AI.

4.2.4 Precision space activation: micro-updating toolbox, innovation box and community embedding

AI decisions end up in specific space adaptations and functional embedding. This phase emphasizes “small cuts, quick effects, accumulation”, where low-cost acupuncture interventions activate sleeping assets.

Standardized Micro-update Toolkit

Creates a replicable, low-cost, updated Menu, which is quickly grouped according to the pointboard:

- * *Ground and interface:* **water-penetration renovations, rail pillows, mobile platforms, modular benches, vertical green walls;** - *
- Facilities and services:** mobile furniture, modularized kiosks, public Wi-Fi, charging stakes, smart lights, express containers, direct drinking water; - *
- Ecology and landscape:* **flower tanks, rain gardens, sunshields, pet friendly facilities, riding parking racks;** - *
- Numbers and interactions:** information screens, two-dimensional code guides, NFC labels, environmental monitoring sensors, human flow counters.

Embedded in geochemical scenes

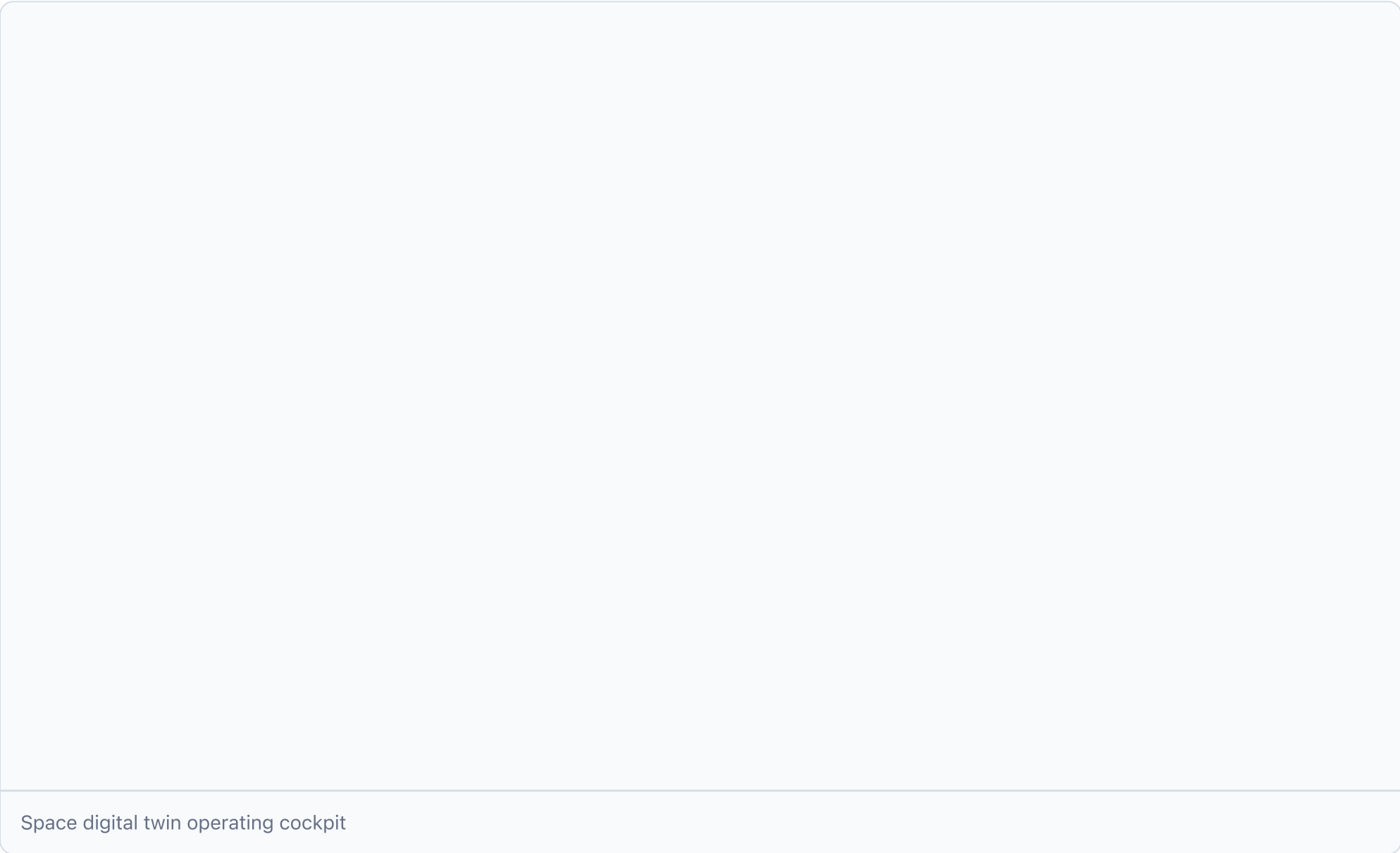
Based on the characteristics of the population in the vicinity of the point, the differential function is embedded in order to avoid the "thousands":

- * *commuting node*: **coffee bar, fast breakfast, temporary work, riding service station**; - * **Resident community**: community kitchens, shared meeting rooms, express delivery stations, waiting spaces for children, pet friendly corners; - * *peri-university*: **Open-source living room, joint discussion room, improvisation corner, 24-hour self-study module**; - * **Node of the Cultural Brigade**: railway theme presentation, Bronze retail, light food, AR card point.

Innovation Box and Community Space Network

Small community spaces such as "Innovation Box" "Joint Discussion Room" "Open Source Lodge" on 5-10 minute walking scale in the innovation zone:

- * *Functional positioning*: **support for occasional communication, improvisation, cross-border cooperation and small activities**; - ***Spatial form**: modularized, mobile, time-sensitive, 20-100 m2 area; - * *Service embedded*: **integration of short-term rental apartments, shared kitchens, laundry and gymnastics, childcare, pet care etc. around community spaces to reduce life frictions of innovative talent**; - * **Business-community-sharing interface**: Encourage enterprises to open conference rooms, exhibition halls, green spaces and parking resources to communities in exchange for public service points or brand names to be exposed and to create mutually beneficial mechanisms.



Space digital twin operating cockpit * *Figure 3: Spacedigital twin operating cockpit** - Consolidation of asset location, human flow heat, facility state, operational performance and community activities into a visual interface to support managers in real-time space postures and rapid decision-making.

4.2.5 Smart operating schedule: time-use and dynamic matching

The same space serves different groups of people over different time periods and is at the core of the inventory update cloud strategy to improve asset efficiency. AI Operations schedule optimizes the four variables: time, space, user, activity.

Four types of spatial time partitioning strategy

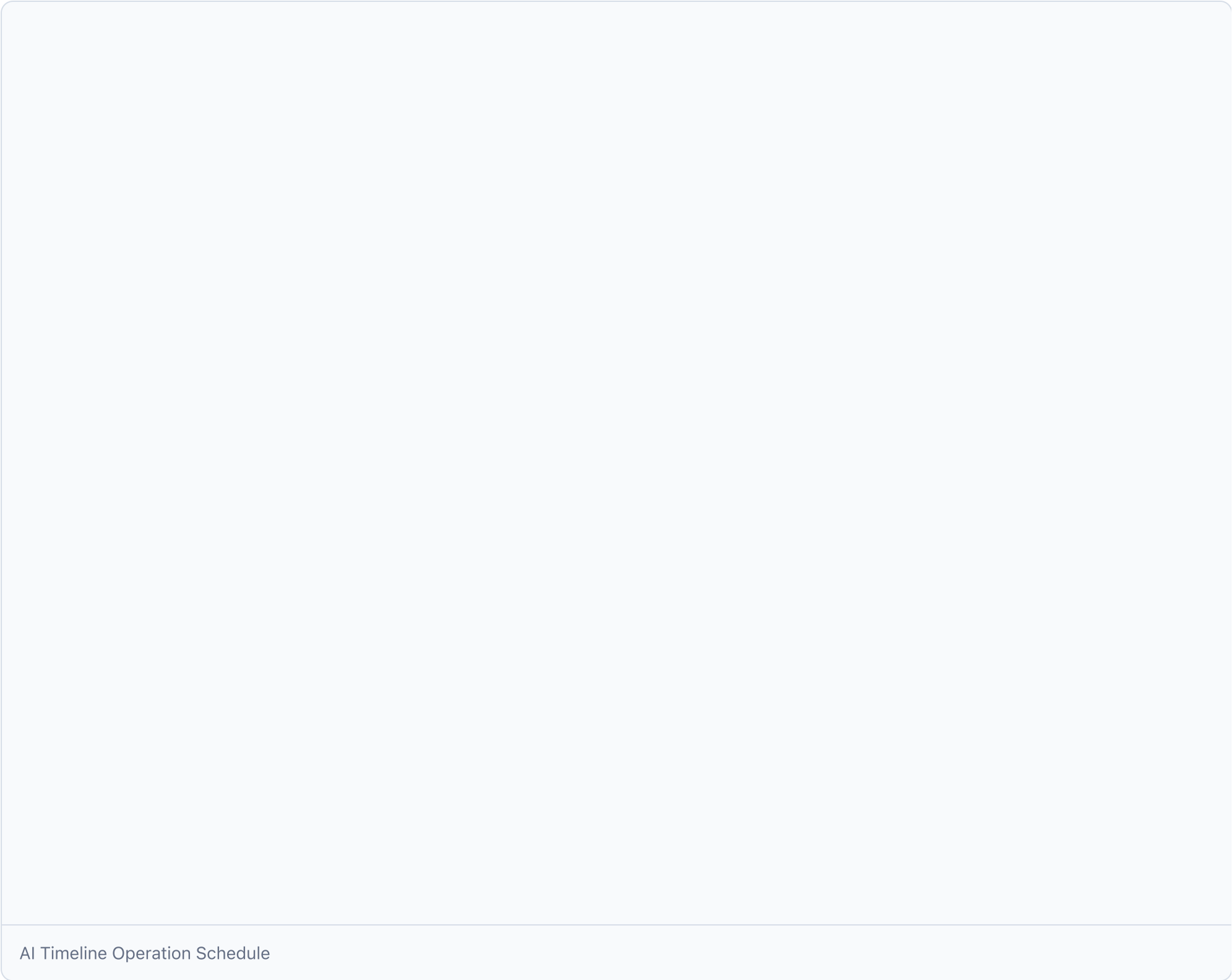
Space type	Mixed/temporal policy	Typical time allocation	Load function
Ashes below the bridge.	+ Night community activities	9 - 18:00 Shared office/show; 18:00 - 22:00 Community fitness/market	A hotel, a theater, a weekend fair.
Unreserved station/industrial complex	Workday industrial service + weekend cultural consumption	Business road shows, joint offices on working days; weekend exhibitions, research, creative fairs	Business living room, study class, flashroom
Under-office/coup building	Day living services + night social consumption	Day coffee, convenience stores, light meals, night taverns, talk shows, community events	24 Hours of Innovation Services Street
Peripheral Green / Pocket Park	Daily leisure + festival	Walking on weekdays, lunch break; weekends/holidays theme fairs, open-air films	Community gardens, flexible playgrounds

AI Timely operating dispatch mechanism

- * *Time period particle size*: **a year-round booking calendar in 1 hour or 30 minutes**; - * **User hierarchy**: different types of residents, business services, community activities, business operations, maintenance of cleaning, different priorities and pricing rules; - * *Dynamic pricing*: **peak-time market prices, peak-time membership prices, public-interest periods are low or free**; - * **Conflict resolution**: when multiple subjects apply for the same period of time, automatic decisions are made on the basis of "security maintenance > long-term contracted lease > pro bono > commercial activity > ad hoc application"; - * *Smart appointment*: **users view available time slots online in real time, one-key appointment, payment, access to doorbars/pass codes**; - * **Handover management**: facility handover, cleaning and security responsibilities are clearly defined at different times, reducing transaction costs for multiple parties.

Functional hybrid access and protocol template

- * *Access list*: **development of a catalogue of incentives, restrictions, prohibitions, allowing for a moderate mix of commercial, cultural and innovative services, subject to ensuring safety and compliance**; - * **Template of agreements**: a standardized time-lapse operating agreement with clear use of subjects, time periods, fees, transfer of facilities, security management, insurance and liability for default; - * *Light asset blending**: introduction of diverse subjects such as professional operators, community leaders, university start-ups, etc., reducing the burden on government and property owners through short rents, alliances, share of proceeds, etc.



AI Timeline Operation Schedule

AI Timeline Operation Schedule * Figure 4: AI Timed Operating Schedule* — Based on appointment data, flow prediction and priority rules, automatic time allocation is generated to achieve efficient time-use security in the same space.

4.2.6 Feedback Optimization: Agent and Shared Platform for Community Operations

The last link in stock asset upgrading is the continuous collection of user feedback, making community self-organization a continuous driving force for space operations.

Community Operations

Deployment of an intelligent community to operate Agent as a “connector” between residents, operators and managers:

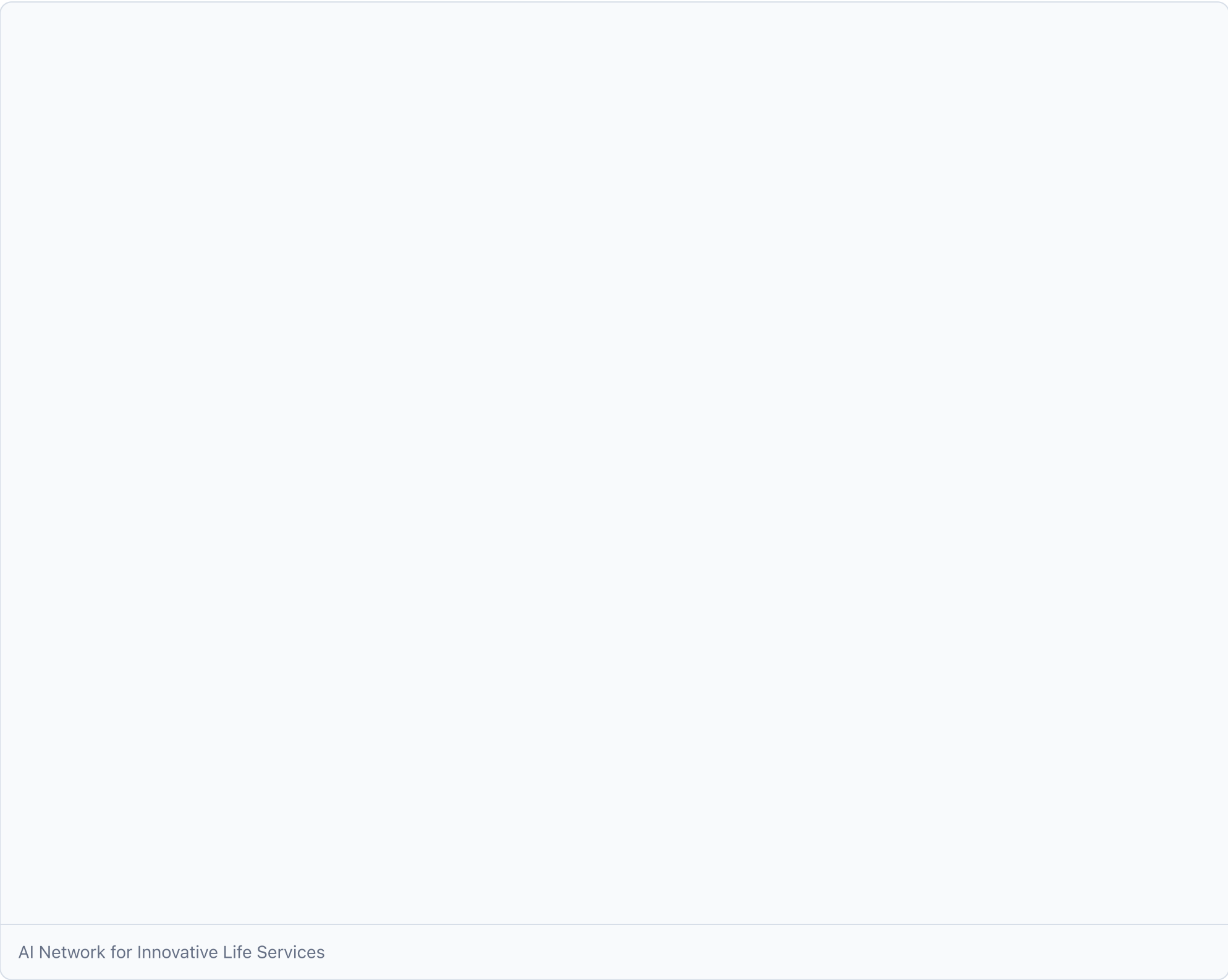
- * Needs awareness: **automatic analysis of forum posts, demand release, voting data and activity feedback, and identification of high-frequency requirements (e.g., “needs night exercise” “want to increase child-care waiting areas”);**
- *Space matching: automatic recommendation of suitable acupuncture points or community spaces, depending on the type of demand, size and timing of the population, and generation of draft proposal of activities;
- * Resource alignment: **matching of activities with potential leaders, volunteers, collaborators and policy support;**
- * Smart reminder: notification of activities to target population and reminder of facility maintenance and handover to operators;
- * Feedback learning: **automatic collection of satisfaction, number of participants, energy consumption and safety events at the end of the activity, and updating of space graphics and operational strategies;**
- * Score Incentives: Recording the contribution of people participating in co-creation, voluntary service and activities, exchange for access to public spaces or creative interests.

Demand release shared with Forum

- * *Requiring publication: users fill in demand titles, descriptions, types, locations and site photographs and publish rent-seeking, address-seeking, adaptation, cooperation etc.*; - ***AI Match**: The system automatically matches the needs of citizens/enterprises with the assets on the platform that are already on line, and reverses potential operational opportunities to property owners; - * *Forum discussion: discussion around asset-specific or renewal issues, co-planning by residents, designers, operators and comments to create a climate of transparency and co-governance*; - * **Policy information**: a central display of policies related to inventory inventory activity, urban renewal, park support services, etc., to lower the information threshold.

Performance feedback and model iterative

- * *Operating Performance Watch: real-time display of asset usage, returns, satisfaction, energy consumption and safety events*; - ***Issue closed loop**: establishment of a single stream for complaints, failures, misuses, etc., to track the outcome; - * *Model iterative**: flow of operational data back to asset databases and matching algorithms to continuously optimize acupuncture point identification, proposal generation and subject matching.



AI Network for Innovative Life Services * *Figure 5: AI Network for Innovative Living Services** — Agent, demand release, forum co-management and points incentive to translate the demand for people and innovative talent into space operations and micro-upgrading proposals in real time to form self-growing service networks.

The Demand Dissemination and Forum is a synergetic portal for feedback optimization: a single key to the population ' s needs, AI automatically matching space with potential operators, and a forum where discussions are built around specific assets to create a climate of transparent governance.

4.2.8 Corresponding to chapter III issues identified

Chapter III core issues	Response to this strategy
AI IRR does not match the facility system	Acupunctures, innovation boxes, life services embedded in the day-to-day interaction and life services
Ineffective reuse along the dual-iron space corridor	Rematch the project with the operator by upgrading the cloud platform of the inventory of assets and increasing the value of assets in inefficient spaces along the railway corridor
Conflict between complex transport systems and poor urban traffic management	Embedding of microspaces around the bridge station, such as waiting times, cycling services, temporary office space, and increasing the experience of transposition and continuity of activity
Contradiction between historical and cultural preservation and spatial renewal	Acupuncture intervention with low disturbance of the heritage's surrounding space, combined functions that allow the historical space to carry contemporary life services

4.2.9 General mapping of intelligent means

urban design	Wisdom.	Output
Asset identification	Survey of space assets + database of tenure and planning conditions + classification of heavy/light assets + potential rating model	Jing-Zhang gallery inefficient space asset map and asset details card
Asset discovery	Resource Square Classification Search, Resource Map Positioning, Hotspots and Recommendations for Completed Cases	List of space assets matching user needs
Project Match	AI Space — Project Matching Engine, Gain Measurement, Business Concession Audit	Adaptation operational proposal and time-use agreements
Operational requirements	"I'll run" fast application, online clearance, quick consultation.	Operating intentions are linked to property rights.
Two-way Match	Project requirements release, forum discussions, policy queries	Dynamic brokering between demand and supply
Microudate Management	Demand crowdsourcing, heat sequencing, proposal publicity, implementation tracking	Annual acupuncture update project bank and implementation list
Space operations	Appointments for shared platforms, human flow monitoring, energy consumption and security alerts	Efficient and safe time-consuming operation of inefficient space
Community governance	Community image, activity matching, record-setting, points incentive	Self-organized and innovative community networks

AI Inventory Asset Updates Cloud Intelligence Closed Ring General Mapping

* *Figure 6: AI Stock Assets Updates the Cloud Intelligence Closed Ring General Map** - From Space Asset Identification, AI Auxiliary Decision-Making, Accurate Space Activation, Smart Operating Movement to Feedback Optimization to create a "data-calculations-space-operate-work-to-take" continuous and iterative closure of urban space assets.

4.2.10 Interface with other responses

- * *With 4.1 continuous spectra policy: **Innovative boxes and community spaces are endpoints of the innovation service spectrum, providing flexible bottom extensions for incubation, acceleration, outcome transformation nodes;*** - * *With the 4.3 menu **dedicated to the smart-line strategy:*** acupuncture micro-upgrading points and sub-bridge space can be used as an active mobility network station, riding service station and waiting space; - * *With 4.4 soft resilience calculation strategy: **the time-use of inefficient space can overlap with the open-source walkway to increase urban redundancy and emergency response capacity;*** - * **Harmonized with the Articular Extension Development Strategy:** Industrial retrofitting spaces can be integrated into inventory assets update cloud platforms, allowing historical spaces to carry cultural narratives and innovative life services simultaneously.

4.2.11 Evaluation indicators

- (b) Inefficient spatial asset survey coverage and asset detail card completeness rates; - Accuracy of the number of assets on line in Resource Square and the classification of heavy/light assets; - Number and implementation rate of acupuncture micronodes; - Number of items used for functional mixing and time-use; - (b) Number of inventory assets updating cloud platform registered operators, matching success rates and conversion rates for "I am running" applications; - (a) The number of project needs releases matched the number of transactions to supply and demand; - Active user and residential/brainer participation at the Forum; - Length and reuse of innovation boxes/community spaces; - (a) Life service coverage and 5 minutes ' availability; - (b) The rate of return on assets or the rate of increase in rent after inefficient space operation; - Public survey scores of satisfaction with surrounding space; - Average cycle of micro-update projects from demand submission to landing.

4.2.12 Design of boundaries

This strategy is a conceptual strategy aimed at the urban design competition, rather than a stand-alone proposal. Specific tenure, planning conditions, fire and municipal capacity, and operating costs and benefits models of inefficient spaces need to be based on

official information and pilot measurements.

Main references

1. The 100-year Jing-Zhang AI Innovation Belt Open Job Book for Smart Bodies is a local summary of clean-up rights. 2. Proceedings of the Internal Seminar on the Jin-Zhang Railway Project, 18 August 2026. 3. 14 Inventory Renewal Clouds: Small Procedures for Urban Space Assets (2024.10): Updates cloud concepts for inventory assets “for space adaptation projects, space for projects”, as well as three-phase operating models for asset exploration, asset packaging, information dissemination. 4. China Institute for Urban Planning and Design: " The concept of a proposal for the integration of the park at the Jing-Zhang Railway Sites " [CCP], March 2020. 5. China Institute of Architectural and Design Research, Ltd.: " A draft of the Convergence Concept Program for Jing-Zhang Railway Site Parks ", 2020. 6. [AI Map/map-three-layers.html](#) ' and AI map/osm-data.js ': temporary scope and OSM road, orbit, water system and green field background data, only for design research and space relationship identification.

Prior to the formal incorporation of the submission package, the specific page number and quoted boundary of external information should be registered at

`sources.json` ' and the unknown conditions of location, title, structural security, town and fire should be registered at

`assumptions.json` '.

Transport, track, municipalities and public services

The Jing-Zhang Site Park forms a North-South active mobility main ridge with orbits bearing the regional backbone and transport, connection and customized services constitute a travel menu; the main computational power hub, Edge Node and applications are designed in conjunction with energy, water safety, fire, communications and public utilities. The engineering line and capacity are subject to professional review. [data: geometry/Roads.geojson#ROAD-001]



It is necessary to map 04 traffic, city, smart and blue green systems.

4.3 Menu-specific smart-line strategy * Response to "complex transport systems and management backwardness" (traffic track active migration, including active migration sutures and sub-bridge space activation, and response to "two irons with low utility")*

Transport trajectories active migration – making ridges mobile.

The Jing-Zhang Railway Site Park has been transformed from an "earth-west rail barrier" to a "North-South public ridge." Based on the spatial structure of "three cores, multiple points, " and "Blue Green Activity Complex", it offers a reviewable road network, track, active mobility, parking and external traffic proposal: an active ridge with blue green, orbital backbone, AI raw mobility to enable mobile system service people to move, and a near-supply and site test.

4.3.1 Transport propositions and strategies

4.3.1.1 Thesis: ridges are first and foremost mobile systems

This proposal takes over the overall positioning of the "100-year Jing-Zhang AI Innovation Belt" —* *world-class* — * world-class — transportation is this mobile system of supply of intellectual innovation:* *the blood vessels of brain flow, the operational chassis of near-*

supply, the scenery experiment. **The Jing-Zhang Site Park is not a "twilight" because it is a green corridor, but because it carries both * longitudinal (North-South consistency) and horizontal (West-West connectivity) mobility functions.**[source]

Transport systems are governed by the principle of "** Governance before Technology Devices*": **AI Primary Transport is not a stack of equipment, flaunting technology, but a system of open contributions, real testing, public safety and long-term operating organization, with each scenario having * data minimization, interpretability, manual review and clear operational subjects**; and Soft resilience as the robustness of a mobile system is reflected in the emergency scene and indicator system.

4.3.1.2 Transport development strategy

Policy	Content	Drop Point
Activate mobile priority	Jing-Zhang corridor and surroundings are dominated by walking and riding behaviour, with mobile traffic giving way to active mobility	With main ridges, complex rings.
Public transport priority	Orbital + conventional bus + the connection constitutes the backbone and secures both positive and preferential access	Orbital portal, public transport corridor
TOD Integration	The perimeter of the site is integrated (concept, no pre-intensity)	Qing Ha station, Dazhongsi station, 5 junctions, Qinghua East Road
Demand management	Guide green travel with parking supply, time sharing, no access	Parking partition, P+R, Maas

There are two types of transport subdivisions: **active mobility priority area** (Jing-Zhang Site Park and Tri-nuclear public space, motor vehicle speed limits, traffic concessions) and ** public transport priority area** (transport priority, access encryption within 500 metres of the orbital portal). ** The boundary of the subdivision is proposed as a concept, pending formal review with transport organization*.*

4.3.2 Traffic skeletons and space structures

4.3.2.1 Status of the Internet

This road is marked by a "ring line + vertical axis" skeleton. Three round roads (northern three/four/five rings) and Kyuchi are responsible for rapid traffic. Central Guan Village Street, College Roads, etc. are responsible for major north-south traffic, and the east road is responsible for east-south communication. Jing-Zhang corridor itself is a railway.

4.3.2.2 Status of orbit

Track	Relationships	Key Nodes
Line 13	Around the old Jing-Zhang line, most vertebrae	West Naumen-Dazhongsi-Know-Chun Road-Five mouths-Upland-Clean River
Line 4	North-South side of the west, service schools and Central Guan Village	Central Kanamura - Great East Gate - People's University - Straight Gate West
Line 10	Ring line, switch from thing to thing.	♪ I know the way ♪
Jing-Zhang high iron	Northern portal hub	Qinghu Station
16/12/15/ Chang Horizon	Complementary network, line pending.	Qinghua East Luxi junction, etc.

Lines and sites are conceptually defined, and precise line levels, access points, transjects and traffic are subject to OSM/Official Data Nucleus.

4.3.2.3 Status of public transport, active mobility, parking and external traffic

- * *Public transport: regular public transport runs along major roads, insufficient connection to orbital stations and lack of access and customized public transport organizations*; - **active modility*: Jing-Zhang main ridges are of strong continuity, but there are few blocks across the street and lots of walls, active modility network "wireless"; - (a) *Parking: the status quo is dominated by land parcel construction, lack of P+R and time-sharing organizations and inadequate static and dynamic traffic links*; - ***External traffic*: the Qing River station is a high iron portal, but the `stop-city' integrated access to regional freight organizations needs to be strengthened.

4.3.2.4 Status diagnosis

1. * *Triple cutting: three horizontal cuts in the northern three/four/five ring, depending on a few nodes for continuity*; 2. * *Site disjointed*: site isolated from industrial space by road, wall, weak access "last kilometre" 3. * *There is insufficient east-west suture: there are few crossing points, many walls and weak links to the campus/community side*; 4. **Ridge free*: active ridge strong, but insufficient network density of feeder, microcycle and active ridge; 5. **Achievability disparity**: 5/10/15 Minutes Availability, active mobility Breakpoint, Public Service Coverage Gaps are extrapolated by Time and Space Analysis and Digital Sandboard "Model B. Innovation Interacting with Activity Mobility", corresponding to `geometry/Roads.geojson` ', `geometry/public_space.geojson` ' school cores.

4.3.2.5 Network systems and faults

4.3.2.5.1 Network hierarchy

Level	Functions	Beneath road (concept)	Principle of treatment
Fast road.	Long-range mobile transport, all closed	North Three-Four-Five Ring, Beijing-Tibet Expressway	Prioritize the intersection with the Jing-Zhang gallery to guarantee continuous cross-border access
Main road	Area traffic	Central Guan Village Street, College Road, Zhi Chun Road	Bus priority, traffic diversion, perfect crossing.
Secondary roads	Zone traffic, diversion main road.	West Turk Road, Cheng Dynasty Road, Dazhongsi East Road, Hu Chong Road	Organisational Microcycle, connecting active mobility with public transport
Secondary roads	Land access, active mobility and microcycle	Parks/intra-community roads (to be completed)	Encrypted feeder network. Breaking the head.

The core action is * *encrypted feeder network, and access to breakout circuit** to increase the density of the feeder network to support active mobility and public transport.

4.3.2.5.2 Road Fragments and Transport Organisation

- * *Urban road break: use of "motorized lanes + non-motorized lanes + sidewalks + green belts" in the main/sub-drive roads, continuous non-motorized lanes, not mixed with motor vehicles (conceptual recommendations)*; - * *Main ridge fractures*: Jing-Zhang corridor uses a composite fracture of "Blue green tops, active mobility center, access side" (see section 4.3.2.8); - * *Crossroad: key intersections such as Dazhongsi, five junctions and Trans-Circumpolar Road, presenting the concept of separation of vehicles, physical mobility across the street, optimization of signal matching, without pre-set canalization and engineering*; - * **Microcycle**: One-way/low-speed microcycles organized with feeder network and intra-campus circuits to reduce cross-border traffic into the core of activity mobile.

4.3.2.6 Public transport system

4.3.2.6.1 Level 3 public transport

Level	Carrier	Functions
Bone layer	Orbital (13/4/10/Jing-Zhang high iron, etc.)	High-capacity rapid communication between the three nuclear systems and beyond
General Layer	Regular bus + precision bus	Main/sub-drive coverage, priority-infrestruction concentration, positive point to station
Bridge	Access to bus, custom bus, unmanned.	Orbital portal last kilometre, trinuclear microcycle.

4.3.2.6.2 Sites and hubs

- **Clean river station:** integrated portal with high iron + track + public transport + connection, to carry out external movements and trans-shipments; - **Dazhongsi station:** orbital transposition + bus hub + quadrant foot connection to carry out southern dispersion; - ** Five junctions, Qinghua East Luxien Junk**: Track + Public Transport + Activity Mobility junction, bearing the burden of University side distribution.

The strength of the integrated development of the site's perimeter-active mobility-industry' is proposed as a concept, with no pre-set capacity and construction height, pending the formal regulatory condition.

4.3.2.7 active mobility system

4.3.2.7.1 Step system classification

- ** City level: **Jing-Zhang Main Ridge (North-South Tunnel, Chong River Station and West Straight Gate)***; - ** Area level:* East-West suture links (cross-ring road, corridor, site quadrant); - ** At the community level**: micro-cycle feeders and public spaces, daily travel of services.

4.3.2.2 Bicycling systems and accessibility

- **Main skeleton:** Jing-Zhang main ridge cycling sidewalk (distant from foot or time); - ** sub-branch: **sutures and composite cycling tracks***; - *** Microcycle:** campus/community cycling lanes, connecting shared bicycle parking; - ** Accessibility**: main ridges, portals, access to the street and public spaces throughout, ramps, blinds, barrier-free access.

4.3.2.7.3 active mobility network indicators (geometrically recalculated)

Core recoverable indicators for the active mobility system include: **active mobility network network density** (active mobility network length/land area), **active mobility coverage** (active mobility buffer area/land area), **active mobility breakpoints** and **5/10/15 minutes active mobility connectivity**. The formula and calculator are set out in section 4.3.5; the exact values are calculated after the official boundary and road data have been completed.

4.3.2.8 A strip of three cores multi-point composite rings

4.3.2.1.1 Area (North-South main ridge)

Jing-Zhang Site Park, as the main ridge, is divided into three sections, each with an orbital portal:

Paragraph	Area	Tie the portal	Lead activities
North	Zhongzhi Park North Four Circles	Clear River Portal	High iron to port, park commuting, river water.
Medium	North Four Rings - Community of Origin - Harun Road	Five mouths, the Qinghua East Road Portal.	Recent school events, open-source interaction, Little Moon River waters
South	Qui Chun Road-Dazhongsi-Si straightgate	Dazhongsi portal	Orbital dispersion, intelligent economy, trans-river water

The main ridge is "Blue Green Up, active mobility Centre, Connected" (walking/riding/Blue Green/connecting), and the width of the ridge is adjusted to the right of the road. The main ridge "Reconstructable Road" is a spatial prerequisite for resilience to switch: in a large event or emergency, the cycling can be temporarily converted to foot evacuation or emergency traffic.

4.3.2.1.2 Three nuclear portals suture with east-west

Each of them became the "orbit-active mobility-industry" portal and took over the official positioning — Zhongzhi Park (open win-win, innovative development), origin communities (enhanced advantage, youth strong), Dazhongsi (elemental convergence, ecological synergy) — traffic portals are the mobile support for these positioning; the stitching of ** street continuity, site integration, and the opening of the wall** have three light-size actions as a priority, without the construction of a bridge:

Suture Node	Type	Main Actions	Dependency conditions
North 5-ring corridor.	Roundabout crossing.	Active mobility up and/or down (conceptual suggestions)	Red line road, traffic organization review
North Quadrilateral.	Roundabout crossing.	Ride and walk continuum, barrier-free	Road red line, subbridge space
Dazhongsi station quadrant	Site Integration	All right, we're going to have a four-dimensional walk.	Orbital stations, intersections, municipal pipelines
Five mouths.	Site + Historical Memory	Jing-Zhang Railway Memory Node + Operation Mobile Suture	Sites, public spaces, security conditions

The specific form of nodes (top-crossing, penetrating, underground connectivity) are conceptual suggestions and do not constitute conclusions on the feasibility of work.

4.3.2.83 Complex rings

The composite ring is a closed form of traffic in the "three nuclear spots along the line": it is based on the water system (Clean River - Light Moon River - Turnover River) and higher schools, gardens, community public spaces, which form a number of active mobility circuits linking the main ridge to the Trinucle.

4.3.2.9 Static traffic and parking

4.3.2.9.1 Parking areas

- ** Core strict control area** (Jing-Zhang Ridge and Tri-nuclear public space): public transport and active mobility are dominated, parking supplies are tight and sharing is encouraged; - ** Peripheral Zones** (Zones and Communities): Mainly structured, shared over time with P+R.

4.3.2.9.2 Parking organization

- **P+R / Parking for Rotation:** Qing River station, Dazhongsi station with P+R (conceptual proposal) to steer the traffic to orbit; - ** Non-motorized vehicle parking: **centralized parking and switching locks between the portal and public spaces, linking shared bicycles***

and not occupying the main sidewalk; - * **Time-sharing:** use of "global smart parking" as a reference to connect static parking with dynamic traffic through data (conceptual suggestions).

The size of parking, berth layout and fees must be based on the official regulatory detailed planning and traffic conditions, and no definitive figures are given in this proposal.

4.3.2.10 External transport and freight logistics

4.3.2.10.1 External transport

The Zing River station is a high-level iron portal in the region.

4.3.2.1.2 Freight and logistics transport

- * *Freight route: cargo takes an outer fast track and a dedicated route to avoid crossing the core area of active mobility (conceptual proposal);* - * **Unmanned distribution logistics:** allocation — ground drones/unmanned delivery — end-point docking, logistics flow line and walk time or break, independent of evacuation route; - * *Low-altitude logistics**: UAV routes and landing points are subject to confirmation of airspace and safety management and no routes are foreseen.

4.3.4 New infrastructure and vehicle-infrastructure cooperation

New infrastructure services "almost light" plus regional recosting "soft supply, without pursuing equipment piles:

- * *Roadside units and holograms: edge computing units, RSU, mm wave radar, smart multipurpose poles, deployed on demand at portals and key intersections;* - ***endside computal power and edge node:** combined with endside computal power station to take on a low-time delay transfer, active mobility breakpoint recognition and distribution coordination; - * *Power exchange and undistributed parking: Portals and campuses for switch/charge, undistributed docking and terminal storage;* - ***digital twin base:** Supporting the flow of people and emergency response with "twins on the clouds" to reuse the 4.4 water-saving "four-prevention system" capability.

Equipment placements, general power configurations and energy conditions are conceptual recommendations that need to be deepened on the basis of electricity, communications, municipal and security conditions.

4.3.5 Indicators Implementation Borders

4.3.5.1 Demand forecasting and carrying (methodological framework)

Transport demand projections are based on a four-stage approach* (*travel generation-distribution-distribution-distribution*), **peak hour demand and road network/orbit/active mobility load assessment, based on official population, employment, mobility and road/orbit information;** *The proposal does not create traffic and load values** and provides a methodological framework only, to be measured by professional teams once the data have been completed.

4.3.5.2 System of indicators (tier 3)

Category	Indicators	Source/ caliber
Geometrically recalculated	Road network density, active mobility network density, active mobility coverage, orbital station 500 m coverage, 5/10/15 minutes availability	Recalculated by coordinates of <code>roads.geojson</code> ', public_space.geojson ' and site
To be official/actually measured	Maximum traffic, network saturation, orbital passenger traffic, parking requirements	Official traffic survey and regulatory detailed conditions
Performance calibration	Accurate rate of access, no-distribution bottom rate, resilience transition time, dissensitisation clearance complete rate	Continuous calibration of operational data

The caliber of geometric recalculation indicators: * *road network density = total length of road / area of use (km/km2);* * **active migration coverage = buffer area of activity / area of use;** * *orbital site coverage = 500 m buffer area / area of use**; * *accessibility* * * * coverage

at 5/10/15 minutes active area / time-scale scales. ** The overall three visual indicators (`site_area_sqm` ', `green_ratio` ', `public_space_ratio` ') are centrally accounted for by the overall indicator layer.

Recent transport implementation projects

Numbering	Item	Type	Mainly dependent
JT-01	Jing-Zhang Main Ridge Active Mobility Breakpoint	active mobility/public space	Road Red Line, Subbridge Space, Transport Organisation
JT-02	Dazhongsi station Quadrant foot connection	Orbital integration/active migration	Orbital stations, intersections, municipal pipelines
JT-03	The Qinghu station is being upgraded.	External traffic/access	Rail, rail, bus stations
JT-04	Secondary network encryption and microcycle	Network/active mobility	Title, regulatory detailed planning, status wall
JT-05	AI access and vehicle-infrestruction pilot	New capital/scenes	Energy, computational power, security, operator

In the short term, reversible pilot and governance agreements (active mobility breakpoints, station connections, secondary microcycles) are being used to upgrade public space and buildings in the medium term, and to expand the AI traffic service with experience over the long term.

Design boundaries

This proposal is a concept for the urban design competition.

The official design scope and priority area boundaries, road redlines, orbital slots and access points, bridge and engineering conditions, traffic and traffic rates, municipal pipelines, fire protection, low-altitude airspace and drone routes, tenure and environmental impacts must be supplemented before the official deepening process is completed.

Main references

1. The 100-year Jing-Zhang AI Innovation Belt Open Job Book for Smart Bodies is a local summary of clean-up rights. 2. Smart traffic solutions: vehicle-infrastructure cooperation, precision bus, unmanned driver + Maas, smart parking, triple integration, global smart parking, cloud twinning, green travel Maas, traffic integration, low-altitude economy, etc., are used only as a reference for capabilities and methods. 3. The Beijing Auto-Driving Automobile Ordinance (2024). 4. Beijing Auto-Driving Road Test and Demonstration Application (Pilot) (2025] 43). 5. The Beijing City Programme for the Legal Promotion of the Innovative Application of Automatic Automobiles (2025). 6. The Beijing "145" top-notch industrial development plan (Intellectual Network Car). 7. National standards: GB/T 51,328 Planning Standards for Integrated Urban Transport Systems, CJJ 37 Design Code for Urban Road Works, GB 50647 Design Code for Urban Road Traffic Facilities. 8. `AI Map/map-three-layers.html` ' and `AI map/osm-data.js` ': temporary scope and OSM road, orbit, water system and green field background data, only for design research and space relationship identification.

Prior to the formal incorporation of the submission package, external information should be registered at `sources.json` ' (including specific sources of global case studies on target and smart traffic solutions) and the unknown conditions of orbital line slots, road red lines, bridge works, low-altitude airspace, town and fire protection should be registered at `assumptions.json` '.

4.4 Soft response strategy *World-class self-adaptation (1 bottom + 5 source)

4.4 Soft resilience extrapolation strategy

As a response to chapter IV, resilience strategy slices, 4.4 defines softness as “the resilience of world-class adaptation”, with the three core strategies of Zhongzhi Park, the community of the AI origin, Dazhongsi, and recasting the six elements of “computational power”+5 source (human, entrepreneurial, open source, data, standards) around “computational power facility elasticity and redundancy design, energy-computational power-city security, white and compatibility for future technological evolution, and with three types of AI+ scenario and urban operation deepening scenes, allowing for the replicating of space-resistant, institutional and three-tiered redevelopment.

4.4.1 Strategy positioning: Softness is a world-class definition of essential innovation

Rather than understanding “resilience” as the resilience of a particular unit, or targeting “many cards and redundant lines”, the strategy provides a more fundamental judgment:** a world-class AI core competitiveness that continues to produce new ideas, new businesses, new standards in the context of the continued volatility of technology paradigms, supply chains, talent and capital cycles.**

This definition combines “soft response” with “world-class approach” to the same thing: instead of being more intensive in the general AI campus, it is ** being stronger in terms of adaptive capacity ** — when model paradigms switch from CNN to Transformer and then switch to the next paradigm, when the supply is subject to external shocks, when capital enters the next cycle, when the brain drain is staged, whether the ecology can replace itself, restructure itself and regenerate itself.

In other words, **4.4 The answer is “how a world-class organization can continue to replace, restructure and regenerate itself in the ongoing upheavals of the technology paradigm, supply chain, talent and capital cycle”, rather than “how many cards, how many redundant lines, how many square metres of machine rooms are to be built.”** It upgrades soft remission from “source of aircraft-level disaster preparedness” to “self-adaptability at the level of aircraft,” using the base of power as a source of talent/business/open source/data/standard, to space and mechanisms.

Accordingly, the AI community of origin is composed of ** elemental structure of the “One Bottom + 5 Source”** and ** three tiers of resilience**:

- ** Six elements (1 bottom + 5 source): *computational power is the enabling element of the bottom *, the first infrastructure for the operation of the five source elements of talent, enterprise, open source, data, standards; the source element is the output subject of the world-class innovation ecology, and the bottom element is their public support; - *Three floors resilience*

1. ** Element resilience (redundancy and supplantability): replacement sources or redundant supplies exist in case of an impact on any element without a single point lapse; 2. *Space resilience (disaster resistance and re-constructability): physical space is flood-resistant, fire-resistant, scalable, re-constructable and critical functions are not placed in vulnerable or isolated locations; 3. * System resilience (self-healing and voice rights)*: standards, rules, open-source governance and multi-intelligence collaboration mechanisms can be iterative, auditable and continuously exported globally.*

Within this framework, **computational power has been established as the “underseat energy element” - rather than as a “chamber park” that captures the visual and functional backbone, but rather as a heavy wall without a scenery:** the foundation does not occupy the main character, but the whole base is in the hands of the main character. Large-scale training and centralization continues to be carried out by training centres in peripheral energy and municipal conditions and by Zhongzhi Park, which is responsible for “AI's autonomous innovation system”; AI, the original community, “outside training - Zhongzhi Park's full-scale reasoning - the original point is near light-Dazhongsi margin” Level 3 commutal power network, which, through high-speed computing networks, has access to regional computal power services on demand, and maintains its own as a “enginehouse park” within a high density area.

** Core scenario: a “source of self-regeneration” - talent flows in open source publishing halls and joint research, businesses grow and reorganize in factor-sharing courtyards, standards are generated in assessment and security governance laboratories and move globally, data flow in credible sandboxes is properly distributed, community power is called on demand as a public service and citizens perceive all of this as a health transition in resilience.**

Jing-Zhang Language: Rail corridor as resilience skeleton

The Jing-Zhang Railway Innovation Belt provides an irreplaceable ** linear spatial skeleton** with ** historical narrative anchor** for this strategy. It is not only a traffic corridor linking the northern part of Haidian, the central part of the city with the outer part, but can be re-read as a **serial element of innovation, the carrying of urban resilience, the continuation of industrial memory**.

- ** Space skeleton: Zhongzhi Park (North Five Rings/Clean River), AI Source Community (North Four Rings North High School-Track-Intension Area), Dazhongsi (North Three Rings/Transverse River) distributed along the Jing-Zhang corridor from north to south, forming the “ Extradition Training - Zhongzhi Park Total Logic and Validation - Source of Origin Elements and Standard Output - Zone Three Margins Response ” complete resilience chain.* - ***Element Channel:** Jing-Zhang rail corridor, together with subway and peri-urban railways, constitutes a multi-modular channel of talent, materials, data and emergency response. In extreme events, the corridor can serve as a trans-regional flow and backup for innovative elements of urban lifeline security. - ** Water safety boundary: water systems such as the Qing River, the Moon River and the trans-river are interwoven with rail corridors, constituting both a space security boundary and a real urban-level application for water security “four-prevention system”* - *** Activation of the historical memory:** Industrial heritage of the Jing-Zhang Railway (orbit, bridge, station memory) has been transformed into an open-source trajectories “resilience commons” and a public art interface, allowing the historical space to carry future innovations and avoid fractures in the renewed art.

Global evidence of the portability of this judgment is evidenced by the fact that the Silicon Valley Stanford-Dune Road system reflects “brain density x venture capital x turnover” and that Kendall Square is an intergenerational update of “university anchor x industrial alternation” and that Israel’s Tel Aviv “Silica Creek” reflects the self-healing of civil-military technology conversion and brain circulation under geo-stress, and that Shenzhen reflects a fast-track paradigm shift from “replacement to innovation” and that one-north from Paris to Singapore reflects the supply of eco-syncs and systems created by the Government-large enterprise, with the village itself a “high school”-head enterprise-unicorn-incubation” model. The above-mentioned reference to the notional orientation of the specific case and application of the border is subject to registration in ‘sources.json’, and does not constitute any factual commitment to this proposal.

4.4.2 Correspond to previous problem identification sections

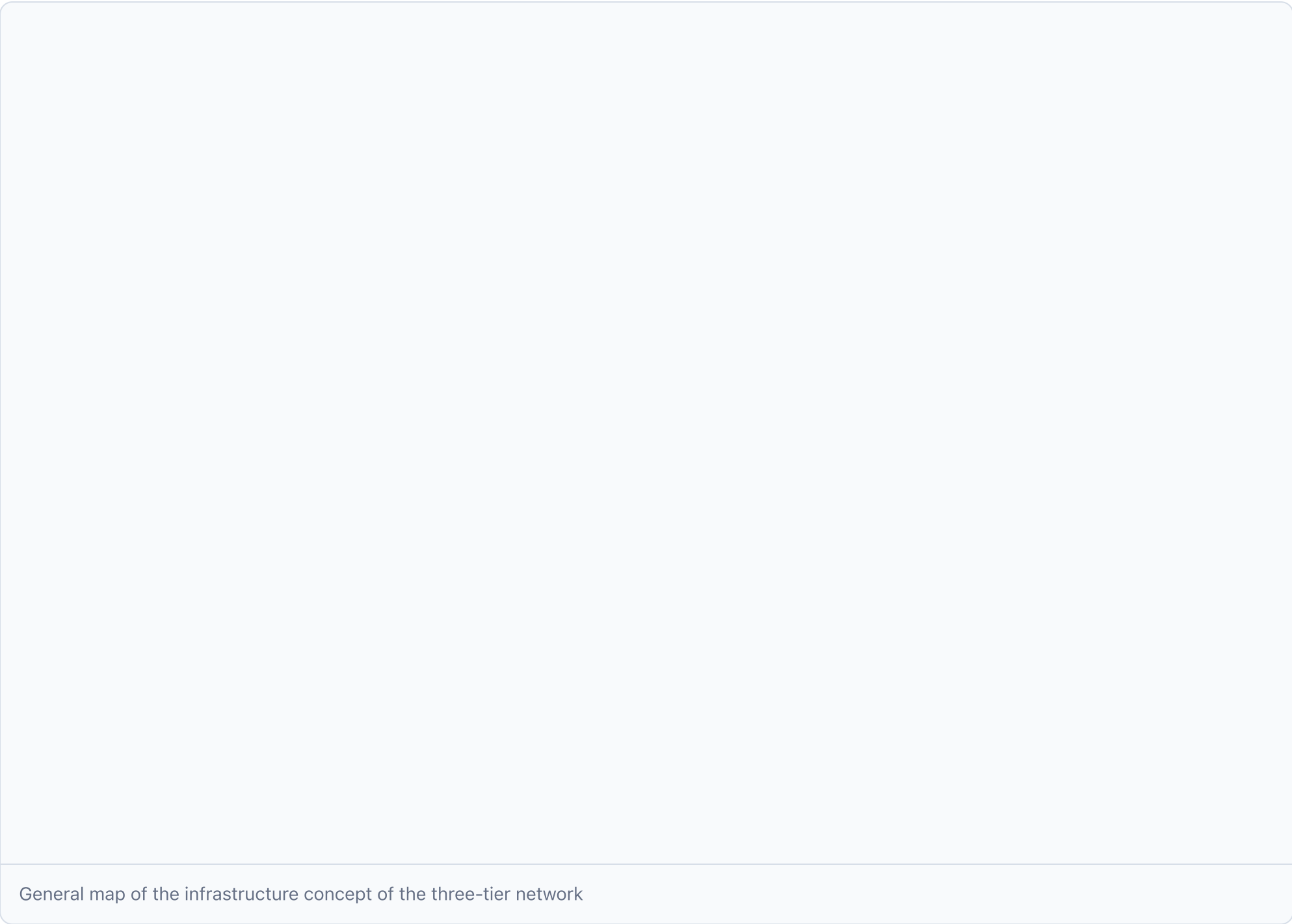
This strategy is not a stand-alone discussion, but rather a clear response to the previous chapter on “Identity identification”, making the “Soft resilience strategy” a verifiable and landable node of response for the entire proposal. Among them, the three-tier network power layout, which is the bottom-type element, responds directly to the diagnosis of chapter 3.4 “Space layout and its security response” (computational power demand density, energy security, disaster risk exposure, network resilience).

Chapter III core issues	Response to this strategy	Key space/wise carriers
The AI Resource Pool does not match the facility system	A courtyard where talent, enterprise, open-source groups and elements are shared, creating a supporting ecology of six elements; reducing element acquisition time through a “one-stop access source ecology” platform	Talent/enterprise/open source group, factor service hall, joint research office
Ineffective reuse along the dual-iron space corridor	Converting the railway corridor and adjacent inefficient spaces into “resilience commons-open-source tracks-innovation service belts” to transform the corridor from a space cleavage to a factor link	Jing-Zhang corridor, sub-bridge space, pre-station space, public innovation ring
Conflict between complex transport systems and poor urban traffic management	Public innovation loops and secure transport entanglements, intelligent travel and multi-modal switching to support human mobility across nodes; data stream segregation secures urban operations	Two circulation lines, active mobility system, data safety area
The space layout of the Centre and its security issue	Establishes the computational power as the "underground element" with the "outside training - Zhongzhi Park full inference - origin is near light at the edge - Dazhongsi" 3rd stage flexible layout of the network power network; the community of origin retains near light and validation workshops to achieve alternative supply by "close light on + area recalculation + cross-scale movement"	Level-III computational power network, end-end computational power station, Edge Node, model validation workshop
Contradiction between historical and cultural preservation and spatial renewal	Jing-Zhang Railway industrial memory integrated into public art, space interface and “open-source walk”, and the historical space turned into a public interface for future innovation	Public art, resilience, open path

4.4.3 Three core strategies

This strategy places the “World-class adaptive capacity” of world-class innovation” on three core strategies that can be implemented, corresponding to the flexible layout of the base seat of the company, the energy-computation power-city security system synergy, and the white compatibility of future technological evolution, which constitutes the “factor reresilience-spaceresilience-system remission.”

Core policy one: flexible layout and redundant design of the base of the computational power (level 3 network)



General map of the infrastructure concept of the third level of the network

As a base element, the organization is based on * "Exterior Training - Zhongzhi Park Full Logic - Origin is close to Light Count - Dazhongsi Edge" Level 3 Commutational Power Network*, which enables the commutational power infrastructure to be neither excluded from high-density built-in zones, nor truly assume its enabling base:

Level	Nodes	The role of general power	Flexible and redundant design
L0 External Training Centre	Energy and municipally appropriate peripherals	Large model pre-training, long-time energy-intensive tasks	High-energy-consumption missions outside, not to compete with high-density areas, water, electricity.
L1 Zhongzhi Park	North Five Ring/Cine River	Total reasoning, chip testing, modelling, security governance (computational power centre)	Recalculate and govern and backstop the source
L2 AI Community of Origin	North Four Circles High School - Interlocking Area	Almost lightweight, edge cache, validation workshop.	Low-time delay scene validation, task up L1/L0
L3 Dazhongsi	North Triangular/Transverse	Lightweight edge node, low-time response in the south, disaster preparedness	It's a time-sensitive Southern scene. It's a remote backup.

The three tiers are interconnected through a high-speed computing network,* any one-level operation in the event of failure, internal flooding, limited power supply or surge in demand can be trans-scaled*: the mission at origin can be moved upwards to Zhongzhi Park or outlying areas, and Zhongzhi Park can be hit by a low-time delay to verify its sinking to Edge Node in Dazhongsi. This “near light +

regional recalculation + cross-scale movement” organization makes the computational power infrastructure itself redundant rather than a single point, constituting the flexible base of source of original innovation.

Core Policy II: Energy - Computal Power - Coordinated Planning for Urban Safety

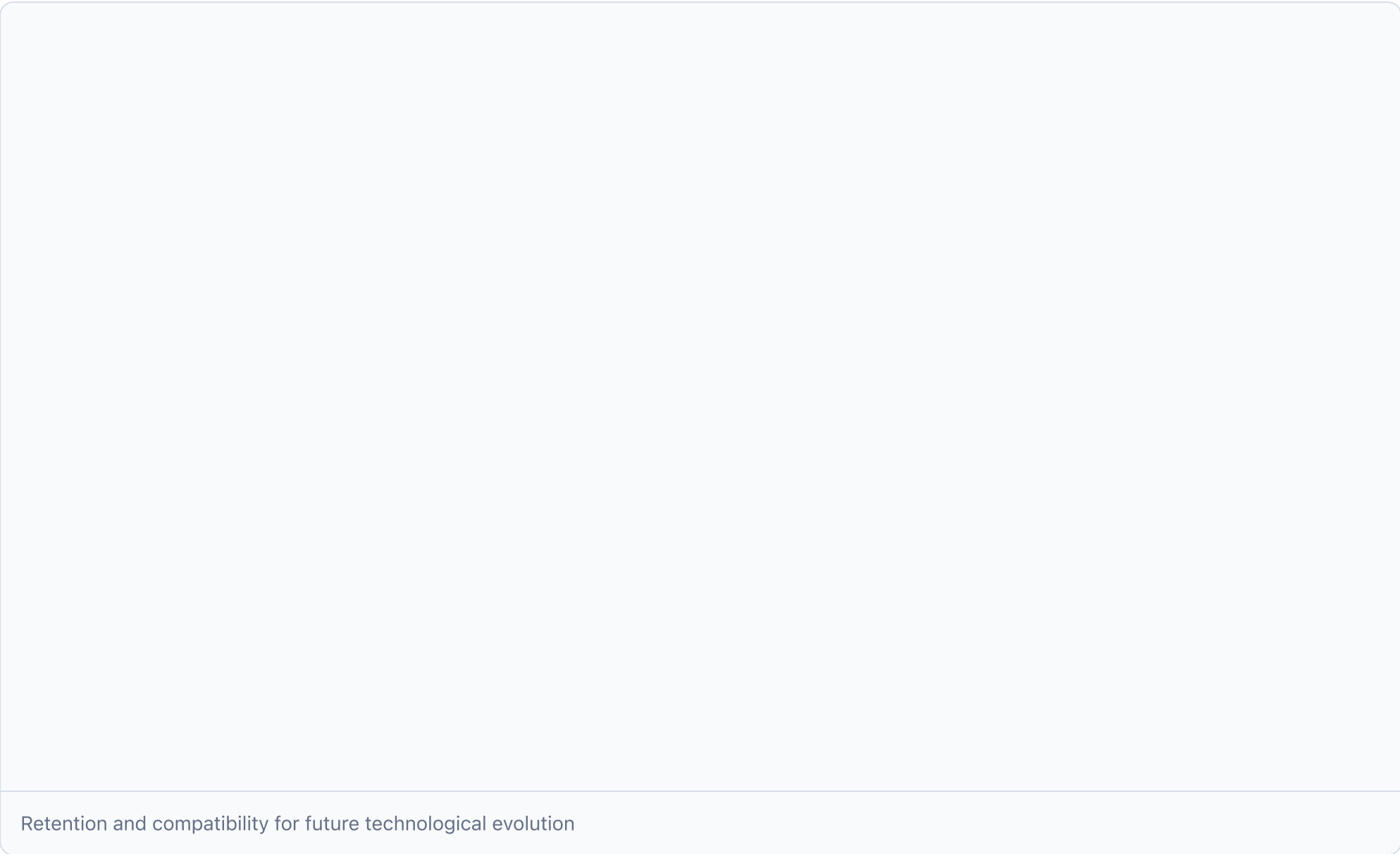
Energy-computal power-cooperative planning for urban safety

Energy-computal power-cooperative planning for urban safety

As a base element, energy supply, energy efficiency, and residual heat use are not “minor issues”, but are the first constraints of access to energy, energy power and urban safety.

- * Supply : **power force group uses “green power as main + storage + two-way power + emergency diesel generators” for multi-source power supply, with key reasoning and governance controls not dependent on a single power source; power reserves to meet future power density increases;** - * **Energy efficiency:** introduction of liquid cooling/high-density cooling for near-light computing centres against $PUE \leq 1.25$; preferential use of free cooling and natural cooling sources to reduce energy consumption and urban heat islands; - * **Residual heat recovery (official co-design): computational power excess heat as a controlled source of heat, giving priority to building life heat and winter heating, and incorporating energy consumption-carbon-dipping accounting for the Smart-carbon platform, in collaboration with stable and hot users (community, public buildings) around them, to shift from passive emissions to measurable low-carbon resources;** - ***computational power-grid synergy:**computational power load participating in peaking valleys and response to demand for active delivery as a stopable load at peaks of the grid or extreme weather in exchange for electricity security for urban public services; - * **Urban Safety Co-operative*:** Commutational Power Group linked to scene Four Water Safety “four-prevention systems” to pre-position key tasks in flood alerts to flood-free Zhongzhi Park/ peripheries, close to light-calculated access to restricted/emergency conditions, and to avoid the possibility that the civil power infrastructure will become a vulnerable point for internal flooding, fire or electricity accidents.

As a conceptual option, specific power capacity, PUE, residual heat size and access are subject to specific demonstration following the availability of official electricity and municipal information; shared halls can demonstrate the direction of recovered heat and energy consumption, but do not write firm commitments for residual heat use that has not yet been implemented.



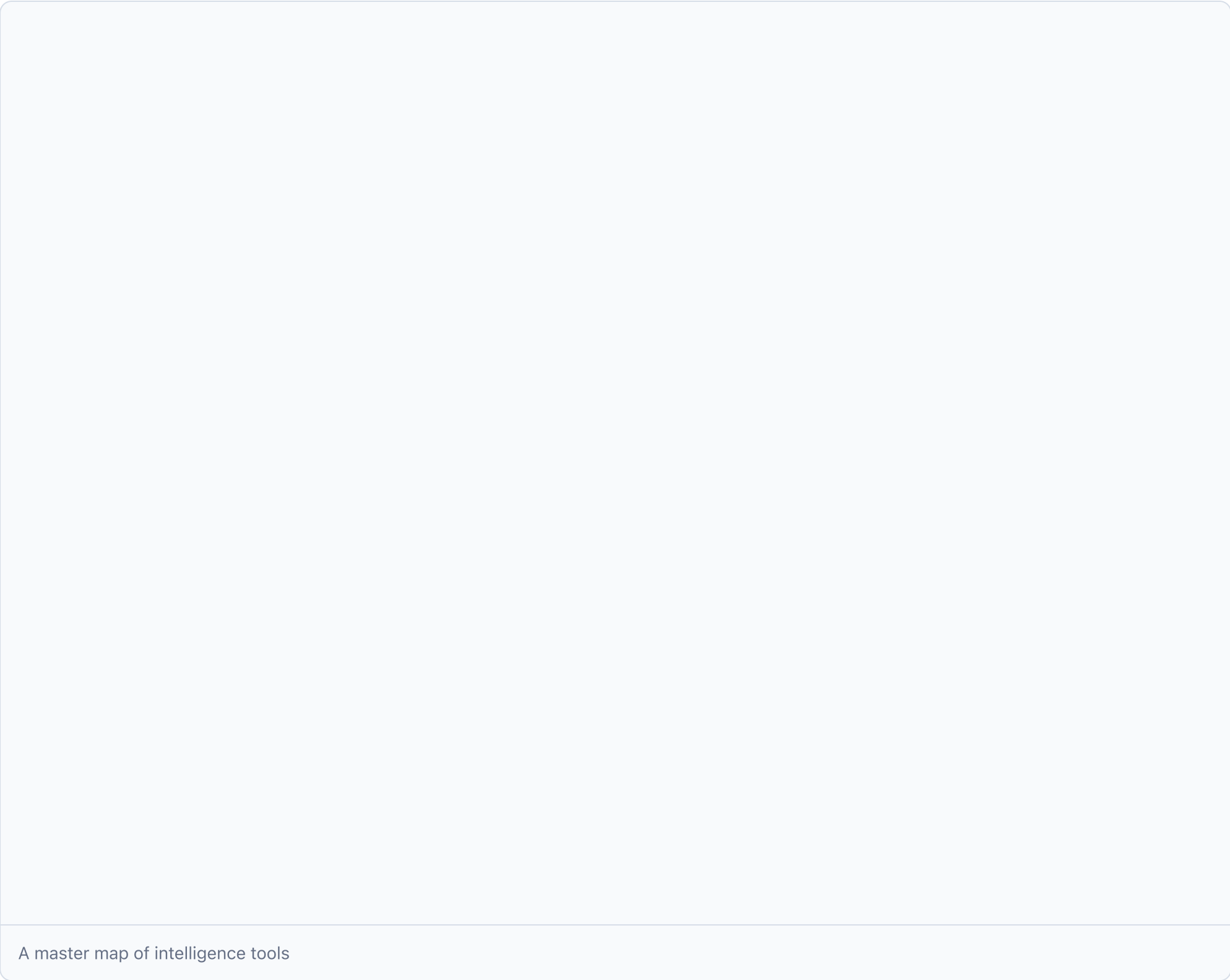
Retention and compatibility for future technological evolution

This strategy provides for an “upgradable, fungible, unlocked” physical and institutional space:

- * *Isomer compatibility*: **GPU / Nationally produced ASIC / FPGA / future architecture interfaces for power distribution, machine cabinets and networks to avoid a single chip route**;
- * *Refrigeration compatibility*: provision for wind cooling-liquid cold switching for near light computing centres, with cooling systems designed to be extended;
- * *Electricity and network retention*: **reserved power capacity and cable pipes at future power density, leaving physical space in the direction of light interconnection, storage, etc.**;
- * **Modularized prefabricated**: modular power group with modularized prefabricated buildings capable of phased expansion and full replacement;
- **System retention**:computational power movement rules, energy consumption quotas and version of governance mechanisms, avoiding premature fixation and locking in technological evolution.

4.4.7 General mapping of smart instruments: from "technology superseding" to "space feedback"

The term “intellectual” in this strategy does not superimpose an information system after the completion of the urban design, but rather a common production tool* *embedded in the urban design** throughout the process of problem diagnosis, proposal generation and operational governance. It serves a core objective: ** to enable urban spaces to perceive their operational status and continuously calibrate on the basis of data feedback.**



A master map of intelligence tools

A map of the smart tools and the urban design layer

urban design	Wisdom.	Issues addressed	Counterpart space/system
Element Configuration	AI factor matching engine, single point dependence assessment	Resource mismatch, factor supply single point failure	Element service hall, ecoresilience clinic
Space Layout	Digital twin + multi-situation simulation + flood/fire stress test	Resilient and re-engineered.	Six groups, shared courtyards, security rings.
Institutional governance	Multi-Intelligence Collaborative Governance, Open Source Overlay, Algorithms Recording Guidance	Standard voice, institutional healing	Standard groups, assessment laboratories, credible data sandboxes
Urban running	Water security "four-prevention system" multi-intelligence closed loops (sension-predict-advance-predation-reverse)	Extreme weather/internal flooding/lifeline failure response	Zhongzhi Park Urban Centre for Logic, Qinghe/Mount/Trans-river Network
Public participation	resilience Digital Sandboard, Ecological Health Information Tape	Perceptible, educational, participatory	Sharing courtyards, park eco-interfaces, exhibition halls

Intelligence system feedback loops on space design

Not only does the smart system serve operations, but its operational data also reverses space design:

- ** Thermal mobility of talent** - ** Enterprise growth data** determine incubation space expansion/restriction, lease strategy updates and service package upgrades; - ** Open source activity** - ** Data compliance traffic log** Optimizing the zoning strategy, audit process and capacity planning for credible sandboxes; - **Communicational power utilization rate versus mission response** Dynamic adjustment of local edge node to external movement ratio; - ** Water safety preview results** - ** Public space thermal and active mobility breakpoint** Optimizing public innovation loop paths, seat shades and night interfaces.

People working together.

AI is not a substitute for judgment, but combines human professional experience with the machine's large-scale computing capability, creating a ****AI generation -- people make choices -- system implementation -- data feedback -- optimization*:

- ** Planning phase: **AI Multi-Scenario Program for Multi-source Data Generation*** - ** Operating phase:* continuous monitoring of intelligence systems for key indicators such as talent, enterprise, data, computational power and water safety - **Governance phase:** AI supporting the completion of model assessments, red team tests, security classification review of security personnel standards and rules, transparent release of data over time

** Principle: All final decisions relating to public safety, personal privacy, planning clearance and resource allocation must be confirmed; AI only exports recommendations, evolutions and options and does not claim official implementation commitments.**

Map with Planning Operating System

The smart instrument of this strategy is the example of an outline 1.3 “Planning operating system” at the resilience dimension:

- ** Public value orientation: **in the direction of “World-class adaptive capacity”, rather than the size of the chamber or the size of the computational power;*** - ** Floor and spatial model base:* AI Map data base, third-tier computational power network and three-layer resilience profile make up a computing space model base; - ** Joint HR judgment: **security classification, red team testing, resilience switch “AI recommendation + confirmation”, no substitute for planning approval and public safety decisions;*** - ** Rules and project implementation vehicle:* standard assessment, algorithm filing guidelines, open source governance rules as an iterative implementation vehicle; - ** Urban run feedback loop*:* water security “four-prevention system”, dual-carbon accounting, computational power dispatch data correction space design to form closed loops.

4.4.8 Interface with other countermeasures: resilience nodes in the continuum

This strategy is not an isolated “computational powerline” island, but a key node in chapter IV, “Discovery of the response”, that bites each other.

- ** With 4.1 Continuous spectra policy: **AI Origin Community is the resource pole on the Jing-Zhang Railway innovation belt “Innovation Service Spectra”, creating a continuum of bands through the rail corridor and along the incubation nodes, enterprise acceleration nodes, outcome transformation nodes to avoid over-one-point convergence.*** - ** With 4.2 fine activation of camp city strategy:* “Ecoreresilience Clinic” and “Team of Talents” in the communities of origin provide a diagnostic tool, talent support and scene validation platform for low-impact minor upgrading of the surrounding and old buildings. - ** With 4.3 menus dedicated to the smart-line strategy: **double-barrier lanes and smart travel systems to support talent to cross-node between Zhongzhi Park, origin communities and Dazhongsi; public innovation ring and security entanglements to achieve “quick arrival, slow communication”.*** - **Initiative development strategy with 4.5.:* the industrial heritage of the Jing-Zhang Railway is transformed into an “open-source trajectories” “resilience communal” and a public art vehicle whose linear features of historical space overlap with the networked features of future innovation.

Through the above interface, the soft-resilience calculation strategy expands the term “intellectual means” from a single technology landscape to a holistic approach that cuts across the urban landscape, operational governance and public participation, echoing the general idea of chapter IV, “Measuring urban problems with wisdom in smart cities”.

Four interface exits: from scene to application

The strategy ended with four applications in chapter 5:

- * *Planning approval: **resilience profile, flood-resistant high-mark, third-stage computational power network layout as a reviewable and comparable option element***; - * **Investment decision**: redundancy, energy security and single-point dependency assessment of the base of the project, supporting the updating of project prioritization and benefit-risk judgment; - * *Urban operations: **water security “four-prevention systems”, dual-carbon platform, communal power - energy coordinated movement into urban operations monitoring and emergency response***; - * **Public life**: resilience digital sandpads, eco-health information tapes, dual-carbon public screens, transforming the source of environmental innovation into perceptible daily experience.

4.4.9 Evaluation indicators

- (b) Talent mobility and return rates; - (b) The cycle of enterprise birth, growth and safe exit, and single point dependence; - Open-source community activity and hosting redundancy of codes/models; - Number and audit completeness of credible data compliance; - Number of releases and external adoptions of standards, assessments and governance methods; - Time for scientific teams and enterprises to acquire factor services; - Local lightweight validation mission response times; - (b) The length of the first-level open interface and public space use; - (a) Element redundancy as an alternative to success and cross-node takeover; - Key data and flood risk exposure for governance functions; - Early warning of internal flooding for water safety "four-prevention system" (12 hours ahead of plum typhoon); - Water level/inundation prediction error (for Yao Yumu 0.03 m scale); - (a) The risk list automatically generates bar numbers and time limits (article 539); - Inundation of preview coverage and frequency of updates; - Early fire detection, breakdown isolation and emergency transition times; - Continuous availability of public services in case of failure; - Public display of data desensitization and manual clearance complete rate; - Annual carbon remittances per unit area of central park (tons of CO₂/ha per year); - (a) The contribution of the park belt to the mitigation of the hot island in the surrounding area (the decrease in surface temperature); - (b) Rainwater wetlands and annual rainfall recharge and run-off factors; - Real-time compliance rates for environmental indicators such as negative ion, PM_{2.5} and noise; - (a) Activity mobile travel sharing and the resulting behaviour to reduce carbon; - On-line monitoring coverage of energy consumption by the Centre and the campus; - Rate of error between carbon accounting results and third-party verification of the dual-carbon platform; - 1.5 Rate of achievement of indicators for model lifestyle areas.

** With chapter III 3.4: The evaluation indicator for this strategy is not a separate stove, but a direct response to the diagnosis of chapter III 3.4: "The space layout of the extrapolation centre and its safety" -

- **computational power demand density**: "Elemental redundancy replacement success rate" for "local light-calculated validation mission response time lags ", to be carried over by a layered supply of the Level 3 network; - * *Energy security: **energy-computational power synergetic indicators for “two-way coverage, long power storage, availability of emergency firewood, PUE compliance rate”*** - * **DRR exposure**: Corresponding to "Key data and flood risk exposure for governance functions", implemented by raising of critical layers and inundation of subdistricts; - * *Network response**: Corresponding to the "Current takeover success rate" for "Current move success rate" for missions with a failure of any node, implemented through high-speed network interconnections and triple-state switching.

4.4.10 Design of boundaries

This strategy is a concept program for the urban design competition. AI's site community-specific plots, 30-50 acres of work units, six-group spatial structures and proportions are all design conditions, not approved site selection, use, size or engineering conclusions.

The data should be completed by reproducing the total surface, building size, energy consumption, drawings and simulation indicators.

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1. The 100-year Jing-Zhang AI Innovation Belt Open Job Book for Smart Bodies is a local summary of clean-up rights. 2. Proceedings of the Internal Seminar on the Jin-Zhang Railway Project, 18 August 2026. 3. Beijing City Bureau for Economics and Informatization, Beijing City Communications Authority: Implementation Programme for Construction of Infrastructure in Beijing City Commutational Power (2024-2027). [https://jxj.beijing.gov.cn/zwgk/2024zcwj/202406/t20240621_372303.html] (https://jxj.beijing.gov.cn/zwgk/2024zcwj/2024400621_372303.html) 4. Departments such as the National Development Reform Commission: Specialized Action Plan for Green Low Carbon Development in Data Centres.

[https://www.ndrc.gov.cn/xwdt/tzgg/202407/t2020723_1391895_ext.html]

(https://www.ndrc.gov.cn/xwdt/tzgg/2024077/t200723_13918995_ext.html) 5. [AI Map/map-three-layers.html](#) ' and AI map/osm-data.js ': temporary scope and OSM road, orbit, water system and green field background data, only for design research and space relationship identification.

Blue green space, public space and urban style

The old Jing-Zhang Railway, Ching River, Light Moon River and the open spaces along its routes form a patrilogy-ecological-public life network that sustains railway memory through historical remembrance, innovative sharing, and park ecology. Greenfield-public space ratios come from submitted design models and are subject to uniform recalculation once official borders are released. [depth: blue_green_public_space] [Spatial data](#)

4.5 Articular continuity development strategy *Jing-Zhang Railway Industrial Heritage activation (cultural roots — cultural evolution — cultural regeneration; linear parks and cultural troupe — retrofitting — digital reproduction)*

4.5 AI+ Sequence of Renaissance

Using the industrial heritage of the Jing-Zhang Railway as the spiritual chassis, the redefinition of the artery as the world-class Alsorce of cultural innovation, the three-tiered strategy of “Cultural Roots—Cultural Evolution—Cultural Renewal” is designed to build a sensory, participatory and iterative historical narrative system; and to transform railway memory from spatial fragmentation into innovative cohesion.

4.5 Articular Continuing Development Strategy

4.5.1 Strategy positioning: the artery is a soft operating system with an innovative belt

The Jing-Zhang Railway, the first State-owned rail line designed and built by the Chinese nation on its own initiative, shares the spirit of “self-reliance, boldness, excellence” with the original innovative demands of the AI era. The nine-kilometre linear corridor of the Jing-Zhang Railway innovation belt is both a physical archive of industrial heritage and a space witness of the “railway age” to the “smart age”.

Instead of creating a separate “cultural city scene”, the strategy understands the art of art as* *a soft operating system that runs through 4.1-4.4: it does not compete for a dominant space for itself, but rather enhances the sense of belonging, recognizableness and global diffusion of innovation by providing historical narratives, site identity and public participation mechanisms. In other words,* *4.5 The answer is “What makes this innovation remember and want to stay” rather than “how big a cultural park is to be built”.

4.5.2 Space structure: one axis, three belts, five curtains, multiple nodes

Using the old Jing-Zhang Railway line as a skeletal skeletal frame, forming the north-south * *cultural narrative axis** and spreading three functional bands, five thematic curtains and several accessible nodes along the axis, the 9 km corridor became a continuous system of cultural experience.

Axis I: Jing-Zhang Railway Cultural Axis

The maintenance of cultural axis continuity through the retention/rehabilitation of the old rail lines, ground marking, gravel belts, steel tracks and walker roads, which are linked to key nodes such as the West Straight Gate, the five mouths, the Qinghua Garden, the Zing Chun Road and the Qing River.

Three.

- * *Historical monuments: museum communities, station sites, transmitters, mouth-to-mouth memories, memorial squares with emphasis on authenticity, readability and education;* - * *Innovative sharing belts:* open-source tracks, orbital theatres, creative gardens, research classes, business outlets for higher education, businesses and communities to make cultural spaces the generator of innovative activities; - * *Park Ecoregion**: for green and water systems, with rain gardens, wetlands, eco-forest belts, to match cultural narratives with sponge cities, biodiversity.

Five.

Scene	Theme	Space down.	Core intent
Act one: Westgate leaves.	Jing-Zhang Beginnings Old City Portal	West Naumen station, welding station.	Industrial Residues + Old City Update
Act II: Five words of memory	mouth culture - international interaction	Five straight-track memory, 13-line.	Dowmouth Square + 3D Garden
Act 3: Qinghua Park Dialogue	Centenary Station Science Dialogue	Qinghua Park (1910/1960)	Museum community + scene restoration
Act IV: The transformation of the known spring road	urban renewal	In 1960, rewired monuments, old-line gardens.	Old Line Garden + Innovation Block
Act Five: The New Qinghu.	The Iron Age. The Future Ecology.	Qinghu station, Jing-Zhang high-terrain/entry section	Wetland + Smart ecology

Multinodes

The sound museum, the orbital theatre, the written garden, the research class, the railway theme parent park, the AR historical experience point, etc. are embedded between the five curtains, creating a spatial rhythm of “great narrative + small surprise” by linking the main axis to an active mobility network.

4.5.3 Spatial implementation of trilayers

Translating the arteries from abstract concepts to operational three-tiered structures of “protective-reflective-regeneration”, corresponding to heritage background, narrative experience and contemporary functions.

- ** Cultural roots: Identification and static protection** Classification of elements of heritage such as stationary, orbital, bridge, entrance, signal, industrial structures: first-level protected object site maintained, reinforced; second-level protected object adapted but reusable; third-level protected object displayed through ground marking, local restoration, digital archives, etc.
- **Cultural performance: narrative and scene re-emergence** By time stacking (e.g. Qinghua Park's “three stops on the screen”), sound narrative (sound museum), symbol translation (human font lines, Suzhou code, pillow tracks in abstract to lay/furnit/mark), AR/VR historical re-emergence and night-vision devices, historical information is understood and felt in contemporary languages.
- **Cultural regeneration: rejuvenation and participatory renewal** allowed historical space to carry contemporary functions such as open-source dialogue, business road shows, community activities, research education, culture incubation. Key vectors are: a museum community centred on the Qinghua Park station, open-source tracks linking higher education and communities, orbital theatres with industrial shelves, study classes for primary and secondary school students, and community memory gardens designed with the participation of the population.

4.5.4 Core strategy I: Railway heritage linear parks and cultural paths

Using the old Jing-Zhang Railway line as a skeletal skeletal skeletal, a north-south, hierarchical, and continuous narrative linear cultural park and active mobility walkway system. The walkway is not only a vehicle for heritage protection, but also a vehicle for daily commuting, innovative interaction and community activity.

Three modes of expression for the skeleton.

In accordance with the existing conditions of the old railway line and the functions of the surrounding cities, a differentiated spatial expression strategy is used to avoid “one-size-fits-all” replication:

- ** Retention of original line: maintenance of orbits, wood/concrete pillow tracks and ramps at the original site of key sectors such as the Qinghua Park station and the five junctions, to form stepable and tactile rail memory bands; installation of green and resting benches on both sides of the orbit to control human flow density at peak times; - (b) * Symbolic hint:* the visual continuity

of cultural axes is maintained through “indicated old lines” through ground-based copper signage, steel-tracked garden roads, gravel belts, pillow-track muscle preparation, etc., at high-terrestrial entry, cross-road or building-pressed areas; - * *Vertical Extension* *: In areas that cannot be expressed on the ground, such as under the 13th line high-mounted bridge and Jing-Zhang high-mounted section, the axis memory is extended to air and night, using bridge-side rail symbol spray, acoustic devices, photo projection and vertical greening.

Cultural troupe hierarchy

Step Level	Width	Functional positioning	Typical Battery
The main walk.	4—6 m	North-South access to practice mobile corridors to support walking, running, barrier-free access and small activities	Centre 2 m water-penetrating + side 1 m gravel/column belt + walk tree
Steppath	2—3 m	Connect the main walk with surrounding communities, colleges, innovation nodes	Brick through water + rail theme ground marking + low-level lighting
Explore Path	1.2—1.8 m	Access to specific sites, sub-bridge space or ecological nodes	Wood plume/crush path + information column + touch guide

Narrative Information Carrier System

A multi-scale, multi-sensor narrative vehicle along the walkway makes walking itself a historical reading process:

- (a) **Information column and explanation card**: set up a waiting steel information column at key historical event points (e.g., 1909 drive-through ceremony, 1912 Sun Zhongshan inspection, 1949 “Arrival in Kyoto” column), carrying text, 2D code, old photographs and engineering drawings; - * *Ground-based marking system*: **abstracting human font lines, Suzhou code, station name bodies, pillow tracks, textures, taking into account cultural expression and accessibility**; - (c) * **Sound device**: development of a direct sound in the front square of the junction memory point and the front of the station, broadcasting of historical scenes such as old motorbikes, telegraphs and radio stations; - * *Public art intervention**: artists are invited to create devices with steel tracks, pillows, transients, signal lights as materials to form a “big narrative + little surprise” spatial rhythm.

Smart Step Narrative System

Installation of NFC/Bluetooth beacons and 2D codes in information columns, nodes and entrances to sites, and construction of an iterative digital narrative network:

- * *Triggered speech*: **visitors can trigger multilingual speech, old photo comparison, AR ground markings and 3D components decomposed**; - ***Themes guide**: the provision of thematic routes such as the “Professional history line” “Patriotic education line” “AI innovation line” with the system recommending time frames and paths based on real-time passenger flows and site carrying capacity; - * *Heat-driven content optimization*: **Backstage continues to optimize narrative nodes density, length of talk and activity, avoiding peak congestion and dynamic storylines based on anonymous walk-thermal data**; - * **Accessibility services**: voice navigation and touch marks for visually impaired users and sign language video and text guides for hearing impaired users.

Activity planning and day-to-day operations

- * *Day-to-day functions*: **commuter mobility, physical exercise, parent-child walk, lunch break**; - * **weekend function**: railway theme fairs, research classes, business flash announcements, community culture festivals; - **Feleration function**: 100-year Jing-Zhang light show, railway music festival, cultural heritage day thematic event.

This strategy responds directly to chapter III, “The Contradiction between the Protection of Historical Cultures and the Use of Space for Renewal” — the transformation of the railway corridor from the “City Clan” to the “Cultural Cohesion Belt”.

4.5.5 Core strategy II: Industrial memory transformed into a cultural space for innovation

The principle of minimal intervention, reversible, functional activation of industrial heritage and railway structures along the routes has become the generator of innovative activities and a catalyst for community life. The goal of the transformation is not “museumization” but

rather to allow historical space to carry contemporary functions such as open-source dialogue, business road shows, community activities, research education, etc., as the interface for “old industry-new intellectuals.”

Principles for adaptation and space strategy

- * *Minimal intervention: **priority is given to maintaining existing structures, surfaces, roofs and industrial muscles, and additional facilities are implanted in a temporary, disassembly and reversible manner;*** - ***Function complex:** a single memory space no longer binds a single function, but leaves open the possibility of multiple uses such as exhibitions, performances, fairs, conferences, education, etc.; - * *Old and new dialogues: **new material (glass, light steel, perforated aluminium plate) and old material (red bricks, concrete, steel racks) to create identifiable contrasts in colour, quality and structure;*** - * **Security priority:** structural safety, fire protection, municipal capacity and accessibility assessments completed prior to modification and dynamic monitoring files established.

Surplus type and reuse direction

Residual type	Space characteristics	Reuse direction	Load function
Station sites (Old Station, Qinghua Park, 1960, etc.)	Brick structures, slope roofs, large open spaces	Post-renovation implants, voice experience, community meetings	Railway Museum, Sound Museum, Tea Club Space
Industrial emboss/long door hanging (weld tracks, etc.)	Large-wide steel structure, high-fisting volume	Retain structure, with light installations and temporary installations	Orbital theatres, show shows, weekend fairs, fashion announcements
Orbit and mouth facilities	Linear facilities, strong memory	Partial rehabilitation, combining public art with children ' s activities	Railroad theme parent garden, Dojo memory garden, signal machine witness point.
Old line ground and subbridge space	Long, gray space, complicated tenure.	Landscape and Time Use	Old Line Gardens, Under Bridge Parks, Temporary Exhibitions and Flash Shows

Functional implants and scene planning

- * *Innovative launch scene: **launch of new business products, open-source project road shows, science conversion encounters at the stage of the show, orbital theatres, where industry remains as a “hard technology” showcase;*** - * **Community living scenes:** community kitchens, shared meeting rooms, centres for elderly activities, youth-engineered classrooms to bring heritage space back to everyday life; - **Cultural consumption scene:** installation of railway bookstores, creative retail sales, fine coffee, fast-flash halls, creating low-density, high-resolution cultural consumption belts; - * *Educational scene of research and development*:* joint primary and secondary schools, colleges and museums, development of courses on subjects such as railway engineering, industrial heritage, AI innovation, etc.

Wisdom Operations and Time Sharing

The deployment of light-weight human flow monitoring, environmental sensing and energy consumption measures for open storage spaces, and the re-use of exhibitions, fairs, road shows and community meetings in conjunction with appointments to shared platforms:

- * *Time allocation: **working days are dominated by industrial services, research and education, and evenings and weekends by cultural consumption, community activities;*** - ***Appointment and dynamic pricing:** differentiated appointment rules based on space size, facility configuration and time heat, free or low-cost pro bono hours, commercial hours charged at market prices; - * *Security early warning*:* real-time monitoring of human flow density, structural vibrations, humidity and fire-fighting equipment, automatic warning of ultra-threshold values and engagement of field managers.

Toggleable digital display interface

In-house modular digital screens, projection interfaces and removable display walls, etc., in the station rooms, in the booths, so that space content can be quickly switched with the subject of the event:

- * *Historical archive model: **display of old photographs, engineering drawings, oral history***; - * **Business results model:** presentation of cutting-edge technologies and products for the integrated university teams; - * *Community Story Model: **Images of memory, creative content and images of events presented by residents***; - * **Art stand-by model:** support for artists, curators, fast-discovery and evacuation.

4.5.6 Core strategy III: Digitalization of historical scenarios and public participation

Intelligently lowers the threshold of historical understanding and transforms the public from a “watcher” to a “co-founder.” A sustainable and growing Jing-Zhang culture knowledge graph is built through a platform of digital twin, AR/VR scenarios, oral history collection and public participation.

Jing-Zhang Cultural digital twin platform

Three-dimensional laser scanning and tilt photography of heritage-building (constructive) structures and the creation of a BIM/CIM-class digital archive into a searchable, comparable and updated Jing-Zhang cultural digital twin:

- * *Data layer: **integration of multi-source data such as historical maps, old photographs, engineering drawings, structural monitoring data, oral history video, creation product information***; - * **Model layer:** construction of a three-dimensional model of the heritage body, a historical landscape restoration model, a perimeter environment model, support for multi-temporal alternations such as 1909/1960/2008 status quo; - * *knowledge graph layer: **construction of knowledge graph on the Jing-Zhang Railway to support semantic search and association recommendations by node of person, event, location, technology, object***; - * **Open interface layer:** open standards for museums, educational platforms, author developers, urban governance systems API, promoting secondary development and transboundary applications.

AR/VR Historical scene

Set up an AR/VR experience point at key nodes such as Qinghua Park Station, five mouths, West Naumen and Harun Road to make historical events immersed:

- * *Core scene list: - **In 1909, the Jing-Zhang Railway ceremony and the “human-shaped” line went instantaneously**; - **1912 Visit to Jing-Zhang Railway in Sun Zhongshan**; - **In 1949, the entrance exam to the Qinghua Park was specially scheduled**; - **In 1960, the railway line was changed and the Qinghua Park station was relocated**; - **In 2008, Jing-Zhang high-line landing and Qing River station was inaugurated***; - * **Trigger method:** mobile phone scanning station floors, site base sites or specific public art, allowing the superseding of historical images and 3D restoration; entry of immersed cars and sites through VR head-on in museums; - * *Content production pipeline: **AI historical narrative engine produces scripts, spectroscopy and multilingual interpretation based on archival information, combined with manual revision to ensure historical accuracy***; - * **Intergenerational interaction:** Adolescents can use AR tools to add digital notes to historical scenes told by older persons to form family memory books.

Sound and oral history files

The collection of old car sounds, telegraphs, station broadcasts, bells, and oral history of the population forms an audio archive and community memory bank:

- * *Collection process: **establishment of fixed collection points in communities, museums, schools, regular “sound collection days” and “old objects story fairs”***; - * **Soundscape design:** the sound collected is edited as a directional sound view by spatial and historical time slots to be played in sound museums and walk nodes; - * *Privacy and authorization*: oral history collection requires the signing of a written consent, specifying the scope of use and public authority; public broadcast content is dissensitized and censored.

Public participation and co-creation mechanisms

Opens access to old photographs, old stories, creative content, so that people are deeply involved in the planning of memory gardens, public art and events:

- * *Memory pick-up kiosks: **small audio/video kiosks in the garden, in front of the station, where residents can tell their stories about themselves and the Jing-Zhang Railway***; - * **Co-planting and co-design:** participation of the population in plant selection of memory gardens, minor naming, layout design, and community memory into space; - * *Programme announcements and voting: **site***

renovations, public art, campaign planning proposals in small proposals/websites, popular praises, comments, voting; - * **Digital message wall**: old photographs, old stories, reviewed and displayed on public screens or online platforms to support time, location, theme search; - * *Intergenerational activities**: Old railway workers tell young people about their work experience, and young people use digital tools to help older people make family memory books that form a closed link to oral history.

Jing-Zhang IP & SKK

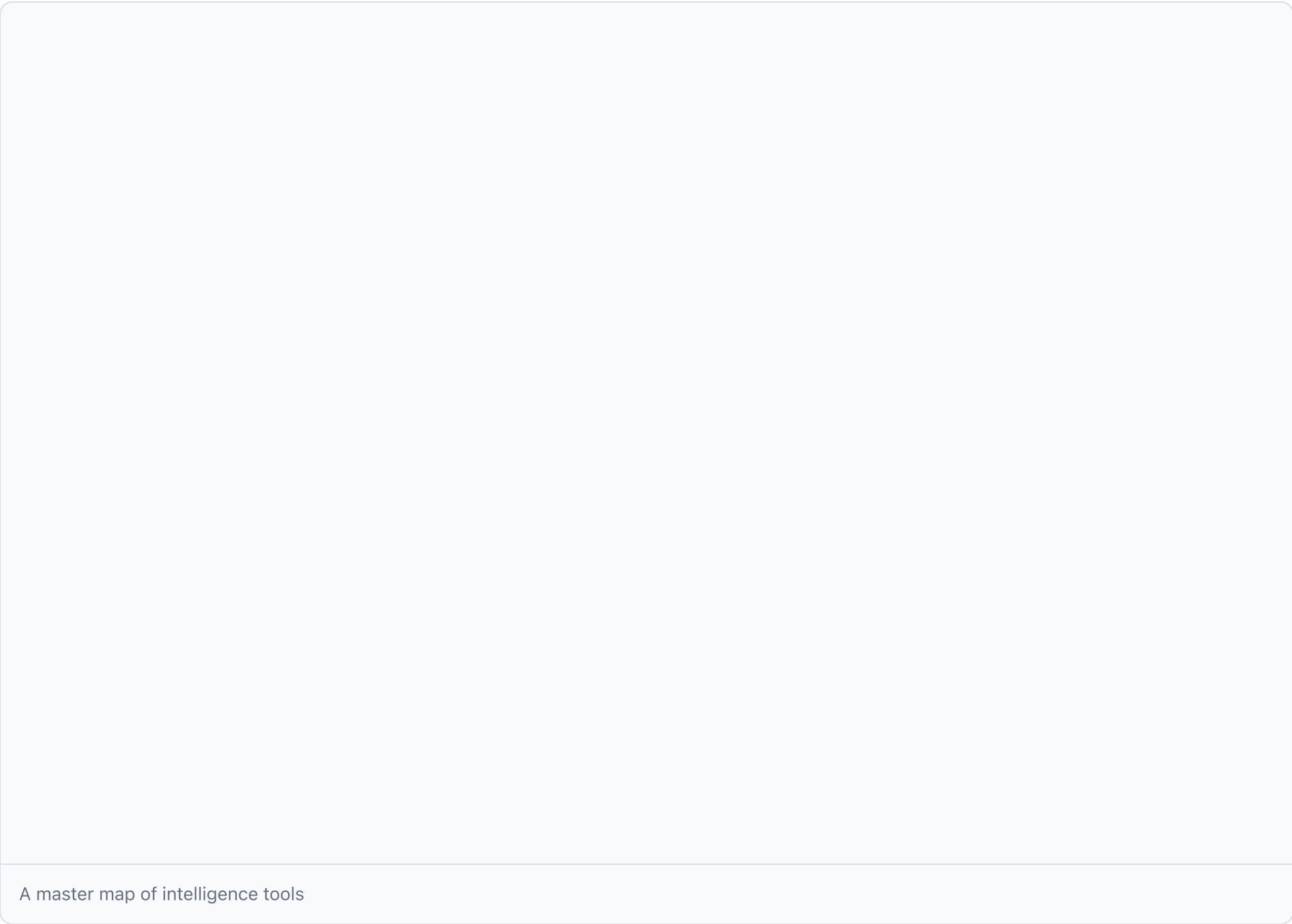
The development of creative products and digital collections based on historical symbols (human font lines, Suzhou code, station signs, pillow tracks) and the establishment of sustainable operating mechanisms:

- **IP system**: development of the main and sub-brands of the “Jing-Zhang Railway” (Jing-Zhang memory, Jing-Zhang intellectual, Jing-Zhang youth, etc.) and harmonization of visual recognition and authorization norms; - * *Product development*: **stationery, clothing, blind boxes, digital collections, AR postcards, railway theme meals, etc.** - ***Incentives against nursing**: Creative income against community renewal, heritage maintenance and public education activities in proportion, creating a virtuous circle of “cultural values - economic gains - protective inputs”.

4.5.8 Corresponding to chapter III issues

Chapter III core issues	Response to this strategy
Contradiction between historical and cultural preservation and spatial renewal	Replace a single museum model with a "linear park + memory retrofit + digital reproduction" to make protected objects a place for both active mobility, innovation and community activity
Ineffective reuse along the dual-iron space corridor	Conversion of old railway lines and inefficient spaces along them into cultural troupes and public activity belts to enhance the spatial value of both sides of the corridor
AI IRR does not match the facility system	Cultural troupes and industrial relics as a combination of low-cost, high-identity innovations that enhance the sense of talent belonging
Conflict between complex transport systems and poor urban traffic management	The old rail line is converted to an active mobility corridor, complementary to the intelligent travel system, with short-range travel and leisure.

4.5.9 General mapping of intelligent means



A master map of intelligence tools

urban design	Wisdom.	Output
Heritage recognition	3D laser scan + BIM/CIM digital archive + structural health monitoring	Queryable, contrastable Jing-Zhang culture digital twin
Narrative Design	AI historical narrative engine, multilingual generation, AR/VR scene reconstruction	Node historical scene experience, multilingual orientation
Space operations	Flow monitoring, reservation system, activity matching	Efficient and safe operation of museums and theatres
Public participation	Community Memory Collection Platform, Oral History Package, Digital Message Wall	Community memory bank, community creation.
Cultural governance	Jing-Zhang IP management and authorisation chain	Sustainably operated creative products and revenue-resilient feeding mechanisms

Technical

Technology map: 4.5 The intellectual artery system can be summarized as a four-tier closed ring of “Sensor-platform-Application-Governance.” The bottom three-dimensional laser scanning, IoT sensory and structural health monitoring completes the digitization of the heritage base and forms the data foundation of Jing-Zhang cultural digital twin; the platform layer relies on BIM/CIM, AI narrative engine and multilingual production to translate static archives into interactive, evolvable knowledge graph; the application level produces dynamic feedback from tourists, citizens, residents, three categories of subjects, on digital museums, AR/VR historical sites, sound museums and community memory gardens; and the governance level ensures sustainable operations and public participation in cultural heritage through human flow monitoring, booking, IP authorization and revenue-back mechanisms.

4.5.10 Interface with other responses

- * With 4.1 continuous spectra policy: **cultural walkways and linear parks are public interfaces for innovative service spectra, providing identifiable cultural anchorages for incubation, acceleration and conversion nodes along the routes;** - * With 4.2 fine-activation camp strategy: industrial retrofitting and community memory gardens provide low-cost, high-emotional space catch; - * With the 4.3 menu-specific smart-line strategy*: an active mobility gallery for the conversion of the old rail line complements the intelligent travel system, with short distance commuting and leisure functions; - ** With the “resilience public” “open-source walk” in 4.4, the cultural narrative of 4.5 can be superseded to enable resilience space to carry both historical memory and innovation.

4.5.11 Evaluation indicators

- The old rail line is active migration corridor pass rate; - Number of industrial retrofit utilization and activation projects; - Coverage of the digital archive of heritage elements; - Number of visits and study teams during the year; - AR/VR Sites and Number of Sound Files; - Number and participation of community memory collection entries; - Number of Jing-Zhang creative products compared with the rate of income for nursing communities; - Public perception of the “Jing-Zhang art” and identification survey scores; - Use of international visitors ' multilingual guides.

4.5.12 Design of boundaries

This strategy is a conceptual strategy aimed at the urban design competition, rather than a stand-alone proposal. The specific line of the old railway line, the level of residual protection, the manner of adaptation, and the digital content must be based on official security assessments, tenure verification, community participation, and regulatory detailed conditions.

Main references

1. The 100-year Jing-Zhang AI Innovation Belt Open Job Book for Smart Bodies is a local summary of clean-up rights. 2. Proceedings of the Internal Seminar on the Jin-Zhang Railway Project, 18 August 2026. 3. China Institute for Urban Planning and Design: " The concept of a proposal for the integration of the park at the Jing-Zhang Railway Sites " [CCP], March 2020. 4. China Institute of Architectural and Design Research, Ltd.: " A draft of the Convergence Concept Program for Jing-Zhang Railway Site Parks ", 2020. 5. AI Map/map-three-layers.html ' and AI map/osm-data.js ': temporary scope and OSM road, orbit, water system and green field background data, only for design research and space relationship identification.

Before being formally included in the submission package, specific page numbers and quoted boundaries for external information should be registered at

`sources.json '`, and unknown conditions such as location, coverage, structural security, town and fire safety should be registered at

`assumptions.json '`.

Update project lists, implementation policies and phased plans

The implementation level is valued as a value export for planning and approval, investment decision-making, urban operations and public life. The immediate priority is reversible pilot, based on an assessment and public feedback expansion in the medium term, with a long-term deepening after regulatory detailed planning, tenure, facility capacity and investment mechanisms have been clarified. The concept does not represent government commitment, approval or investment decisions. [Depth](#) [Depth](#) [Spatial data](#)

Chapter V - Landing of applications - How does planning produce value?

5 Application landing: from outcome to scene

Translating planning results into planning approvals, investment decisions, city operations, four types of applicable and operational interfaces and mechanisms for public life. The four modules are not isolated software, but are closed to the planning operating system of the "Approval of issuance rules - Decision Transformation Project - Operational Security Operations - Public Flow Experiences - Data Return Bottoms" and are anchored in Beijing's "Multi-Role One" platform, Beijing's International Large Data Exchange, HaidianCity Brain, and Jing-Zhang Railway Park.

5.0 General: Planning for the export of four values at the executive level of the operating system

The floor answers the last question of the six-layer incremental logic - ** how planning produces value.** The planned results are not the end of delivery, but the four applicable, operational and iterative interfaces:

| Module | Interface | Breaking the core contradiction. | Main clients | Real anchorage | | 5.1 Planning and approval | AI Complementary planning and compliance governance interface | How dynamic, fungible results are stabilized by statutory processes | By sector, design unit, evaluation expert | Beijing's Multi-System Coordination Platform implements a monitoring information system | | 5.2 Investment decisions | urban renewal business decision-making interface | Inadequate power and poor accounting for low-impact renewal | Implementation, fiscal and social capital | Beijing International Big Data Exchange, Haidian Integrated Data Elements Service | | 5.3 Urban operations | Monitoring of urban running and movement interface | Complex traffic, energy power, operating safety in extreme events | City transport centres, operators, street communities | HaidianCity Brain IOCC, 1+1+2+N | | 5.4 Public Life | Resident tourist city experience interface | Heritage activation, public participation, planning awareness | Residents, tourists, developers, students | Jing-Zhang Railway Site Park (first phase opened) |

Planning approval 5.1 Issuance of space rules 5.2 Conversion of investment decisions into project time series 5.4 Reverting demands and perceptions 5.3

**Key mechanism:* *5.4 Social awareness data on the return of public life, 5.3 Dynamic running data on the return of urban operations, not endpoints, but re-entry into the data foundation, becoming the new input for the next round of problem identification and planning, which constitutes the planning operating system of the "city-run feedback closed" (1.3).

Principles: All final decisions on planning approval, investment decisions, public safety and privacy of individuals must be confirmed; AI only exports recommendations, evolutions and options, without claiming official implementation commitments. The four modules are positioned as "conceptual recommendations/reference options/for further study by professional teams".

5.1 Planning approval - AI Supplementary Planning Compliance Governance



5.1 Planning and approval of prototype screenshot of the compliance pre-trial desk interface

5.1.1 Site theme: dynamic planning results, how to be captured by mandated processes

The core contradiction between traditional planning approvals is the mismatch between the pattern of results and the capacity of the review**: the proposal moves from static drawings to dynamic twinning, multi-scenario evolution, while the review still relies on manual cross-checking of the text, which is slow and difficult to cover the entire space rule, and the public is less able to understand “why the proposal was approved and where the card was issued”.

This module provides a more fundamental judgment:** planning approvals should not be a terminal for planning results, but rather an institutionalized interface for dynamic planning capacity into the statutory system.** It does not replace statutory reviews, but transforms the multi-scenario scenario generated by "human decision making" (5.3.3) into ** review decision support that is reviewable, comparable, traceable, traceable** — allowing planners to shift their energies from "turning the text" to "values".**

5.1.2 Jing-Zhang context: why innovation belts need more compliance governance

- ** Complex tenure:** Site park phase I, covering approximately 13 hectares of railway tenure, was used free of charge for public space construction following an authorization agreement between the Beijing Bureau of China Railways and the Haidian district in 2021 — a multi-subject mandate to update compliance processes that are transparent and leave no trace of natural needs; - ** Multi-rule superimposed:** Specific constraints such as municipal, district-level super-trajectories, Wenbao (railway heritage such as Qinghua station), water services (Clean River, Light Moon River, Turn River), transport, etc., and manual checks of easily missing cross-cutting rules; - ** High level of public interest:* * Railway heritage protection and urban renewal are high levels of tension and public demand for transparency is strong (opposed 5.4); - ** Contest context:** The proposal uses provivional boundaries extensively, and compliance reviews naturally assume the role of gatekeeper in “designing official restraints from provisional presumptions and preventing temporary geometry from being misappropriated as a legal red line”.

5.1.3 Place of scene: the Beijing legal system

- **Data layer:** Implementation of the monitoring information system for the "one map" (online collaboration of three types of planned results, three levels of city-wide integration, quality control, inventory integration); - (b) * *Platform level:* a synergetic platform for "Multi-Corporate One" (2018,381) was fully implemented in 2018, implementing the "Multi-Corporate One, Multi-Certificate One, Multi-System Integration, Multi-System One" and the "Plan for Integrated Implementation" system established by Articles 30 and 31 of the Beijing Urban and Rural Planning Regulations; - **Additional level:** superseding AI compliance pre-screening, program selection, dynamic monitoring early warning capabilities, resulting in a division of labour "statutory platform succession rules, AI capacity-assisted judgment" without altering the tenure and legal validity of any approval.

5.1.4 Corresponding to chapter III issues identified

| Chapter III core issues | Response to this module | Key carrier | | Low-impact redevelopment along the dual rail corridor | Provision of a 'rules against + list of witnesses to be refilled' to update the proposal 's compliance pre-trial | Pre-trial engines for compliance, deviation maps | | Complex traffic conflicts with urban management | Inclusion of orbital security zones, active mobility break points, station city integration requirements in compliance items | Transport - Land Interaction Mapping | | Contradiction between historical and cultural preservation and renewal | Set the protected boundary as a rigid pre-screening item, and touch automatically indicates the basis for protection. | Historical protection - update trade-offs | | AI Creative Resource Collection and Matching | Inclusion of innovative facilities, human housing requirements in the "Plan for integrated implementation" element | Science resources - matching of facilities |

5.1.5 Core functional design

1 AI Pre-compliance (rule versus engine)

The package (geometry + Metrics + Compliance Matrix) uploads and exports the three types of "compliance/violation/reparation" conclusion. Deviation is indicated on the map, giving * *source of the deviation + path of the proposed supplement* *. Key design: * *Deviation* * Rejection ** - The system does not draw conclusions, giving only " which rule is violated, where and how" and avoiding AI black box vetoes.

2. Rapid assessment of proposal selection and impact

Multi-programs run parallel impact assessments on digital twin and output comparable indicator cards and five-dimensional radars (economic/social/ecological/cultural, corresponding to 5.2 multiple goals).

3. Visualization and transparency of the approval process

The list of elements, progress of processing, leave marks on opinions, version diff, full-time process can be traced; the reporting party and the public can look at "where the proposal is, what elements are contained, what changes have been made to successive amendments" (publicity end and 5.4 common entry point).

4. Dynamic monitoring and assessment of early warning

The "Dynamic Monitoring and Assessment Early Warning Mechanism" of Beijing City was used to plan for post-implementation monitoring, to provide timely warning of risks of breaking rigid control requirements, and to change approval from "one-time gate" to "care throughout the life cycle".

5.1.6 Wisdom tools and technical routes

| Wisdom. | Issues addressed | Technical elements | | Rule digitization | Non-calculation of provisions | Policy provisions Structure rules Spatial algorithms (overlapping, adjacent, accessible) | | AI Pre-trial compliance | Manual checks are slow and easily missed | Rules engine + Space Popup + Neutron semantic matching | | Option evaluation | Multiple cases are difficult to compare fairly. | Multi-targeted optimized decomposition + proxy model + 5D radar | | Process transparency | Check black box perception. | Version management, chaining, public readable card | | Dynamic monitoring early warning | Ageing of risks during implementation | Threshold for binding indicators + time and space warning |

5.1.7 Site cards: three typical user-hours

Scenario one: "Self-check for compliance" before the design unit is submitted

* *Planning object:* * Prepared unit / Design team.* *Space path:* **digital twin platform** * **Program package uploads → pre-compliance pre-screening → Self-inspection report returns.** *Process:** 1) Upload proposal package (geometry + metrics + compliance matrix); 2) Three colour labels (red / yellow deviations / green compliance) on a "chart" bottom map; 3) Export deviation source of original innovation + list of material to be completed + recommendations for amendment; 4) Revision of post-checks to meet the bid for formal review.* *Data and privacy boundary:** only process structured proposal data; closed planning data go through both approval and confidential processes.* *Manual review:** pre-screens as supporting conclusions and formal review conclusions are issued by the approving officers.* *Business entity:** rules from the Department + Planning Institute technical team.*

Scenes two: "Multi Case Selection" at the jury.

* *Planning object:** review experts, review bodies.* *Process:** 1) 3 candidate parallels; 2) 5-dimensional radar + impact indicator card; 3) expert voting on evidence-based markers.* *Data and privacy boundary:** proposal data are submitted by the author unit and displayed after dissensitization.* *Manual review:** supported evaluation by selection findings, resulting in recommendations from the review committee.* *Operating entity:** Evaluation organizers.**

Scene III: "Intellectible, Revealable" during the public announcement period

* *Planning object:* * Residents and the public along the line of* (activation 5.4.* *process:* **1) Programme Publicity; 2) Three-minute readable Publicity Card (what to build, what to change, how to affect you); 3) One-click "Support/opposed/additional recommendations"; 4) Opinion structured into the approval stream.** *Data and privacy boundary:** The public sees only de-sensitization content; feedback makes a clear statement of purpose.* *Manual review:** Opinions are grouped and adopted by the approving officers.* *Operating entity:*** Regulation from sector + streets.

5.1.8 Evaluation indicators

Compliance pre-trial automated coverage • Deviation from detection/misreporting • Programs ahead of selection cycle compression • Opinion closed cycle approval transparency index • Dynamic monitoring early warning.

5.1.9 Design of boundaries

Without replacing statutory review and administrative decisions, AI conclusions must be interpretable, reversible, with marks throughout the process; confidential planning data must be managed according to a confidential process; control indicators involving regulatory detailed planning, volume, construction height, etc., should be based on official regulatory detailed planning or an annex to the mission document, and the provincial boundary should not be disguised as a statutory red line.

5.1.10 Basis of reference

Beijing Urban and Rural Planning Ordinance (articles 30, 31) City Planning Territory (2018) Circular (2018) 381 on the Full Implementation of the Mechanism for Working on the Multi-Range Platform (2020) Kyeongfa (2020) 7 Opinions on the Establishment and Monitoring of the Implementation of the System of Multi-Rural Strategic Planning DB11/T 2389-2025 Technical Guidelines for the Transfer of Data on the Results of the Inter-State Strategic Planning. Jing-Zhang Railway Sites

5.2 Investment decision-making - urban renewal business decision-making



5.2 Prototype operational map interface for investment decision-making

5.2.1 The scenario proposition: the source of the renewed drive is "the cleanness of accounts"

Question 2: Diagnosis of the causal chain that was inefficiently redeveloped:** complex property rights + security controls + insufficient power for renewal Space fragmentation and value sank. **The source of the "renewing power deficit" is often "uncountable" - investment efficiency, social benefits, risk exposures cannot be combined in one interface. This module suggests that *urban renewal's competitiveness is shifting from a "land price dividend" to a double payment of "business gain + data assets."***

5.2.2 Jing-Zhang Language: Updates of authentic "books" along railway lines

- ** Low-impact stocks are real: * There are a large number of inefficient industries, storage, unused land and multiple property rights along the bi-rail corridor; - * A successful precedent could lead to an authorized free-of-charge use of 13 hectares of railway tenure in public space for the logo: * Site Park, * Best practice for the selection of urban renewal in 2023; - *The business model can be replicated: *The "city partner" model includes the subjects along the routes in the business sharing; - * Data asset growth pole: ** Haidian has run through a nationally replicable data asset entry path.*

5.2.3 Site location: update project library + three ledgers assessment

The Jing-Zhang Innovation Belt Renewal Project Library is the central carrier covering two wings of the three sectors: the Repository (the "project image" for each of the low-impact projects) The Assessment Ledgers scorecard) The Sorting (which explains the integrated priority) Monitoring (the full life-cycle status tracking).

5.2.4 Corresponding to the identification of Chapter III issues

| Chapter III core issues | Response to this module | Key carrier | | AI IRR does not match the facility | Translating innovative facilities, talent housing gaps into social accounts | Creative Resources - Matching Mapping | | Low-impact redevelopment along the dual rail corridor | Update the potential index assessment on a case-by-case basis and integrate it in order to break "deep" | Update project library, priority engine | | Projection centre layout and security | Inclusion of security and energy redundancy in the "release account" | Energy - computational power map | | Historical and cultural preservation and renewal | Make heritage conservation, and the cost of continuity visible into the Social Bill. | Historical protection - update trade-offs |

5.2.5 Core functional design

Update project priority smart sequencing

Enter the updated potential index x investment benefits x social impact x policy window x science matching, output queuing list + an explanation for each sorting based on the black box sorting. Prioritize the release of the most disequilibrium in the vicinity of higher education and orbital stations and call for the complete "one slice".

2 three ledgers integrated assessment

Economic accounts (value-added land, rent, operating income + increase in the table of data assets) Social accounts (post, public space, job balance, cultural continuity) resilience account (computational power security, exposure to disasters, energy redundancy). Output comprehensive scorecard + sensitivity analysis (conclusion flipping mark).

3 Project life-cycle monitoring

• Implementation of the five-phase operation key node status, funding, monitoring of progress deviations; and automatic generation of unusual warning and assignment of responsibility.

4. Assetization of data

The updated project de-sensitization data will be organized as a registrable data asset, the integrated service centre for data elements of the Northern Digital Institute and Haidian will be interfaced, the registration and entry tables will be completed and the "fourth book" added to the update project.

5.2.6 Wisdom means and technical routes (with authentic and quoted anchor points)

| Wisdom. | Issues addressed | Technical elements | | Update the assessment of potential | Low-impact recognition of experience | Land-use efficiency, accessibility, tenure complexity, updated potential index | | Sort Priority | Get in line and slap your head. | Multi-guide decision-making + interpretable sorting + sensitivity analysis | | Three ledgers. | The economic-social-resilience cannot be matched | Financial model + social impact assessment + risk pricing | | Full-cycle monitoring | Progress reports | Key Node Watchboard + Abnormal Early Warning + Allocation of Responsibility | | Assetization of data | Data value sleeping | Data registration, compliance, entry tables, circulation (to the north) |

** Real anchor:* * Beijing International Big Data Exchange (one of the six major poles in Beijing City's construction of a global digital economy pole) Haidian's first data asset entry table (New Sea Area, April 2024, Wisdom Park Energy / Parking Data Set, Table 101 million, Valuation 228.8 million, Beijing's first green data asset listing) + Haidian's Integrated Data Element Service Centre + Data Element Site Laboratory (2025, January), providing rights, compliance, circulation, trading, assetization full-chain services.

5.2.7 Site cards: three typical user-hours

Scenario one: "Renew the project line" on the national platform

** Planning object:* * * Zone updates implementation entity/national platform.* Space path:* Updates project database * Three ledgers assess integrated ranking of annual implementation plan.* Process:* 1) Low-effectiveness in cataloguing and generation of project graphics; 2) System Export Economics / Social / Renewal leedgers rating card; 3) Integrated priority list with each ranking basis; 4) Development of a draft annual update implementation plan for decision-making.* Data and privacy boundaries:* tenure, investment, fiscal sensitivity data are managed on a confidential basis and are not publicly displayed.* Manual review:* ranking is decision support and the implementation plan is confirmed by the decision-making body.* Operating entity:* District update of implementation body + financial, regulatory and sectoral.

Scenario II: Social capital "A Visible Communication Tool"

* *Planning object:* * Industrial operators, social capital, business clients.* *Process:** Selection of negotiables** Generate a dissensitized version of the "project investment cards" (the leedgers summary + policy conditions + time-to-place)

Scenario III: Progress light for the full project cycle

* *Planning object:** implementer management, responsible department.* *Process:***1) Real-time viewer for node status; 2) delayed/funding gap/automatic early warning of compliance risk; 3) assignment of responsibility to subject matter overhaul; 4) closed loops.** *Data and privacy boundaries:** internal management data hierarchy delegation.* *Manual review:** early warning disposal is manually confirmed.* *Operator:** project implementer.**

5.2.8 Evaluation indicators

Update project time-based • Low-effectiveness release (ha) • Integrated return on investment • Improved job balance • Differing ratio of funded • Number and valuation of data asset input tables (de-sensitization).

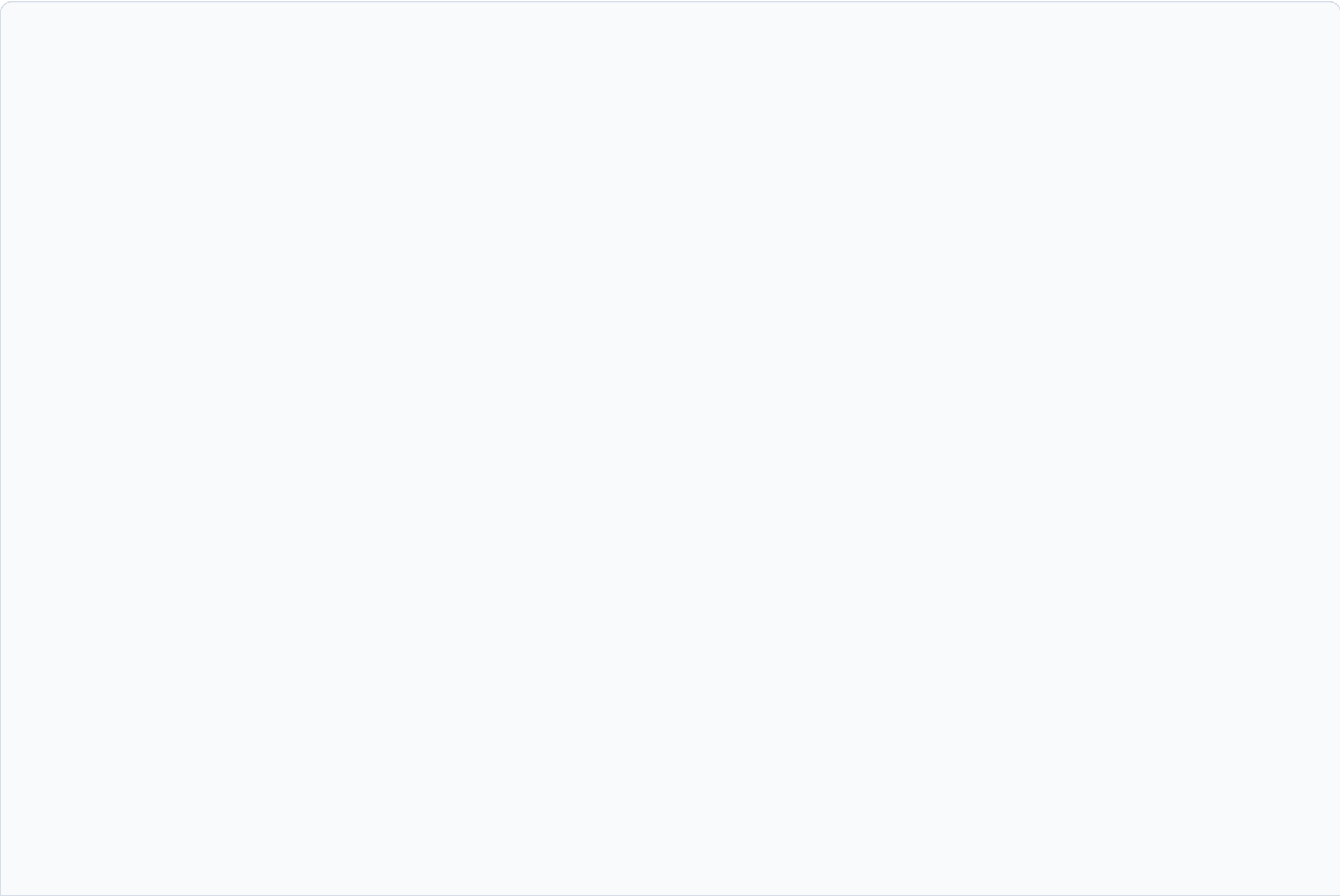
5.2.9 Design boundaries

Sequencing and evaluation are decision-making support and are not a substitute for the statutory bid approval and solicitation process; land, finance and investment-sensitive information is managed on a confidential basis; measurement must be based on assumptions and sources; and data asset entry forms must meet accounting and compliance requirements, supported by professional bodies.

5.2.10 Basis of reference

China Economic Network, Beijing First Green Data Asset List (2024-05-10)

5.3 Urban operations - monitoring of urban operations and movement



5.3 Prototype cut-off of the urban operating resilience running dashboard interface

5.3.1 Site theme: The competitiveness of urban operations is "capability"

The module proposes that* *city operations should not stop with "sensitization-disposal" but should be upgraded with "preventing-prevent-self-adaptation"** – when extreme weather events, large-flowing, commercial power shocks occur, the city can pre-tempo, quickly re-organize, and reset itself afterwards, determining whether the Jing-Zhang innovation belt will fulfil its promise of "resilience".

5.3.2 Jing-Zhang Language: Double Iron, Computational Power and Water System

- * *Multi-modal transport superheavy*:* subways, suburban railways, high irons, urban roads with multiple overlaps along the Jing-Zhang corridor, Chong Ha station - Beijing North Station is a typical multi-modal hub; - **Computational power extension: *Zhongzhi Park carries reasoning and training computational power, peripheral layout training centre, energy-computation power-security synergies as new topics;** - * **The true boundary of water security:** The clean river, the moon river, the trans-river and the rail corridor are intertwined, and the internal flood risk and the "four-prevention system" digital twin have been fully designed in 4.4.7 scenario IV; - * *Has the bottom seat matured:* * HaidianCity Brain has completed the IOCC, 1+1+2+N structure and over 200,000 sensor perception networks.

5.3.3 Site location: no duplicate construction based on HaidianCity Brain IOCC

City Brain Sensory Network, ICA, IOCC command and control capability and "Steps of Time and Space"; superseding 5.3 Dynamic Promotion Models, more than 4.4 Smart "four-prevention system" closed loops, and computational power-traffic scenes. This module is an interface for "Innovative Area Operation" without replacing HaidianCity Brain as a whole.

5.3.4 Corresponding to chapter III issues

| Chapter III core issues | Response to this module | Key carrier | | Complex traffic systems and traffic management are lagging behind | Multi-model real-time monitoring + swap efficiency analysis + congestion warning + induction scheduling | Transport dashboard, multi-smart traffic drive | | Projection centre layout and security | Energy -computal power - Safety co-monitoring, transport and landing of loads under extreme events | Computational power - Energy synergy, resilience switching (link 4.4) | | Low-impact redevelopment along the dual rail corridor | Underbridge, prestation space integrated into operational monitoring | Space monitoring layer | | Historical and cultural preservation and renewal | Safety and experience operational monitoring of heritage sites, crowd centres | Public safety run layer |

5.3.5 Core functional design

1. Real-time transport rollover and movement control

Multi-modal monitoring, multi-plication efficiency analysis, congestion warning, multi-modular induction, and a 6.3 "menu-selection for smart lines" scenario: Qing Ha station - Beijing North Station Top Exchange, sudden large-flow response.

2 Energy-computational power synergetic monitoring

IQC energy consumption, load, PUE, redundancy, real-time monitoring of disaster exposure; energy-computational power-city safety coordinated movement (6.4)

3. Early warning of urban events and emergency response

Extreme weather, heavy passenger flow, integrated warning of safety events + hierarchical response; re-use 5.3.2 extreme event stress test model pre-action.

4 Water Safety "four-prevention system" interface

Access to the main operator interface in closed loops of the four-scenario-based smart body "four-prevention system" to serve as the basis for decision-making in extreme weather; public desensitization presentation.

5. Dynamic resource allocation for urban services

Shared facilities, public transport, emergency resources, community services are dynamically distributed on demand, docking 6.2 "lightly activated camp town".

5.3.6 Wisdom means and technical routes (with real and quoted anchor points)

| Wisdom. | Issues addressed | Technical elements | | Multi-modal traffic promotion | Multiple Synergy | Transport Simulation + Real Time OD + Multi-Intelligence Schedule | | Energy -computal power | computational power and grid security | Load prediction + migration policy + reversion switcher | | Multismart "four-prevention system" | Extreme event response slow. | Perceptions/forecasts/preprints/drives Category 6 | | Dynamic resource movement | Resource mismatch | Supply-and-demand matching + hierarchical authorization + mark-up at all times | | Run feedback design | Data does not return flow space | Operating data → Space design feedback closed loop (link 4.4.6) |

** Real anchor:* * Haidian City Brain (a memorandum signed in 2018 between the Government of the Haidian and the hundredth time; structure "1+1+2+N"; computational power 2028 trillion times/second; region-wide more than 200,000 sensors; access to 10,000 road cameras and 13 boards, over 40 operational systems; convergence of perception data of nearly 5 billion and an additional 60 million daily; construction of a "time and space information map"). Effective cases: scavenger governance violation rate of 26%.4% and identification accuracy of 95% +; snow-sweeping ice advance 30-40 minutes; intelligent traffic lights in the western part of the city are adjusted by traffic and the efficiency of fire trucks increased by about 20%.

5.3.7 Site cards: three typical user-hours

Scene one: "Multi-modular conversion" at peak times

** Planning object:* * Dispatcher, Transport Operator, ** Space Path:** Qing River Station / Beijing North Station hub → Traffic Operator dashboard → Larger Flow Alert → Level Response.** Process:** 1) Multi-modal Real-time Flow Convergence; 2) Switch efficiency and congestion analysis; 3) Super-threshold warning and recommendations for inducement (addition, peak tips, temporary waiting area); 4) Movement Control personnel to implement and leave marks after confirmation.** Data and privacy borders:** Use of polymers, de-sensitized passenger flows and OD data, without collecting personal tracks.** Manual review:** All traffic movement orders are confirmed manually, AI only recommends.** Operating entity:** City Transport Centre + Transport Operator.

Scenario two: "computational power-energy switching" in extreme weather

** Planning object:* **Computational power centre operator, power provider, emergency command.** *Process:** 1) Weather and grid early warning access; 2) Assessment of load migration proposal by 4.4 status (concerning limited load emergency recovery); 3) Automatic generation of alternative "displacement/download/emergency power source" scenarios; (4) Authority to implement cross nodal takeovers after confirmation.** Data and privacy boundaries:** Processing only infrastructure operating data and not user operating data.** Manual review:** Movement control involving public safety must be manually confirmed and traceable over time.** Operating entity:** Zhongzhi Park City Logic Centre + Power and City Operating Sector

Scene 3: Integrated Response to Emergencies

** Planning object:** Emergency Command, Street Community.** Process:** 1) Incident Awareness (Video / Sensor / Public Reporting); 2) Intellectuals Initial Study and Recommendations for Response Levels and Resource Portfolios; 3) Commander Confirmation; 4) Dispositary Disposals; 5) Ex post-repeated and iterative models.** Data and Privacy Borders:** Individual Location Data Anonymized to be Used for Modelling; Sensitive Events Accessed within Mandate.** Manual Review:** Command Decision-Making confirmed, AI Output Options.** Operators:** Emergency Command + Street Community.*

5.3.8 Evaluation indicators

The duration of the response to an event • efficiency swap • energy efficiency • PUE • early warning accuracy/fault rate • resource movement satisfaction • continuous availability of public service failure • early warning of internal flooding / water level prediction error (re-use 4.4 indicator).

5.3.9 Design of boundaries

Real-time movement involving public safety must be manually validated, ordered to be duplicated; public security data should be graduated, the data minimized; personal location data should be anonymously used; this module is an accessor to the "Innovated Area" and does not replace the entire HaidianCity Brain operation; and third-party data are quoted from the source of registration.

5.3.10 Basis of reference

Beijing City People's Government Portal, HaidianCity Brain Ten Sites and Technology Needs Release (2023-01) People.cn Beijing, HaidianCity Brain 2.0 Top Ten Applications Need Release (2023-06-13) Haidian Government, Citizen Dialogue First: City Brain provides smart, fine services for urban management (2023-01) Bermuda Smart Cloud Client Case, HaidianCity Brain (2025-04) Programme 4.4 Site IV (water safety "four-prevention system").

5.4 Public Life - Urban Experience of Residents and Visitors



5.4 Prototype cut-off of the Jing-Zhang experience interface for public life

5.4.1 Site theme: the value of planning, ultimately realized by "human perception "

One link in the causal chain of question 5 is ** insufficient public participation. **This module proposes that** the final decision of the plan is a daily user.** A plan can be perceived, involved, used, determining whether it is a "result" or a "life."* *The public life module is an accessible, participatory, operational urban interface for 9 km of green lanes and their heritage, scenes, services,** and makes this an entry point for social cognitive data (2.1.4) to the bottom.

5.4.2 Jing-Zhang: The park that has grown is the best interface

- * *Part of the space*: * Open to the public in June, 2023, north to east, south to south, and south to Sichuan, approximately 2.4 kilometres long, approximately 16.8 hectares, into the surface linear space released by the Jing-Zhang high iron; - (b) The maintenance of the Qinghua Station (1960), the restoration of the old Jing-Zhang Railway line and the integration of more than 20 railway elements, such as the “human slope”, the memorial dunes, train rollers, etc., and the transformation of the north four-ring railway bridge into the “Sunset Lodge” called the “Jing-Zhang Railway Outdoor Museum”; - **Active Mobility system: * *New greening approximately 120 000 m2, creation of "three green" activity space, 5 new roads to connect to and connect to the breakout***; - * **Places of activity: *20 Other places (Standard football, five-person football, basketball, smart gym, etc.)**; - * **Operating model**: The "city partner" model, which incorporates key units along the routes into business sharing; - * *Long-term view*: * Overall plan for the Beijing North Station in the north to the north, with a total length of about 9 km and 14 km2 of radiation, serving 9 street towns and nearly 20 tertiary institutions; - * *Honours*: * Beijing urban renewal Best Practices in 2023.

5.4.3 Places of scenes: public interface of the organization of "multi-point" sex parks along the routes

| Region | Public experience focuses on | Main carrier | | Five mouths - Qinghua Park Station | Heritage experience + school innovation + community leisure | Qinghua Park Station, nearness school, cultural path | | Zhongzhi Park | Innovation presentation + computational power Cop + sports leisure | Smart Sports Square, industry presentation interface | | Dazhongsi. | + One Station City + International | Front of station space, node of the civil brigade |

5.4.4 Corresponding to chapter III issues identification

| Chapter III core issues | Response to this module | Key carrier | | Contradiction between historical and cultural preservation and renewal | The digital re-emergence of heritage + participatory public activation to move from passive evasive to active experience | AR Orientation, Participatory Activities, Cultural Path | | AI Creative Resource Collection and Matching | Embedding the scene of innovation in the park public interface so that innovation can be seen | Developer walkways, results gallery | | Complex traffic conflicts with urban management | Multi-module Travel One Key Planner + Park Active Mobility Connection to Reduce Last kilometre | Smart Travel Module | | Double-barrel tracks are being redeveloped in an inefficient manner. | Reactivation of the pre-station, sub-bridge spaces into public nodes and renewal of power along the anti-nursing route | Under the bridge, the space is active, the front square. |

5.4.5 Core functional design

1 Platform for participatory planning for the population

Programme announcements, opinion collections, feedback loops (shared with 5.1 portals); community micro-update voting, site acupuncture proposals (6.2).

2. The digital experience of cultural heritage

The Jing-Zhang Railway heritage digital recovery, historical scene AR re-emergence (Cinghua Park Station, Jian Tianwu Story, Centennial Motorcycle); cultural walkway guide; historical photographs, railway technology evolution, forward rail technology demonstration (responding to "Jing-Zhang Heritage Educational School").

3 Smart travel and life service

Multi-modal trip one-key planning: subway + peri-urban railway + heavy iron + active migration + shared connection (6.3); AI Living Services Reach: Medical, Education, Law, Community Services; direct interface with Site Park's "three green" active mobility system.

4 Developer and innovation community interface

Open scenes, data sandboxes, AI application testing and demonstration (responding to Deep Control); open source release, results presentation, developers ' activities landed at the park public interface.

5.4.6 Wisdom tools and technology routes

| Wisdom. | Issues addressed | Technical elements | | Heritage AR Wizard | History is invisible, insensitive. | Site Recognition + History
Image superimpose + Narrative Audio | | Multi-modal travel planning | Last mile. | Real-time multi-modal path + active mobility connection
+ accessibility | | Participatory feedback | Opinions are difficult to structure | Issue card + Group of Views + Closed loop tracking | | Data
Sandbox | Data is not available, it is difficult to use | Desensitive dataset + scene laboratory + Developer API (compliance authorization) | |
Social awareness flow | Perception data not reflow planning | Perceptions / Comment / Behaviour (Desensitization) 2.1.4 Next round of
diagnosis |

5.4.7 Site cards: three typical user-hours

Scene one: Residents are involved in "my zone update."

** Planning object:* * Residents along ** Space path:* * Community participation platform ** Votes on topics* ** Votes on issues* ** Publication of results*
** Process:* ** 1) Update Programme Publicity to produce "3 Minute Readable Cards"* ** Process:* 2) Resident Vote + Additional Recommendations; 3) Structured Opinions; 4) Results feedback (adoption/ interpretation / supplementation) to create a "present-process-receipt" closed loop
** Data and privacy boundaries:* * Collection of only feedback on issues, clear use statements, and no collection of irrelevant personal data. ** Manual review:* * Opinions are reviewed by street and approving officers. ** Operating entity:* * Street + Rules for the Sector (link 5.1).

Scene Two: "The 100-year Jing-Zhang AR Guide for Visitors"

** Planning object:** visitors, citizens, students.* *Space path:** cultural walk entrance ** Site AR Identification of historical scenes and narratives.*
** Process:** 1) Scan station / Track / Monument logo; 2) Superseded historical images (e.g., "1919 Qinghua Park"), technical stories and person narratives; 3) Consistent slopes, memorials, train drives, sundown guesthouses, etc.; 4) Created "my Jing-Zhang footprints".
** Data and privacy boundaries:** location data are used only to guide location, anonymity and closure; historical images and portraits are subject to clearance.* *Manual review:** presented after history, copyright and sensitivity clearance.* *Operator:** Cultural Operating Company + Street+ Educational Cooperation.*

Scenario three: Developer access to Open scene and data sandbox

** Planning object:** Developer, AI Creative Team, University Science Team.* *Process:** 1) Request scene authorization; 2) Use desensitized data sets in data sandboxes (energy consumption, parking, human flow, use of scenes); 3) Development of AI applications; (4) Testing displays in public interfaces; 5) Integration of excellent applications into the results gallery.* *Data and privacy boundary:** Sandbox data dissensitized, authorized, auditable; no personal identifier data provided.* *Manual review:** Application online and data access reviewed.* *Operator:** source of official information + Haidian Integrated Service Centre for Data Elements (United Nations Action 5.2).

5.4.8 Evaluation indicators

Number of public participants/adoptions Number of cultural experiences Number of travel service satisfaction Developer landscape adaptations and sandbox activity Community issue response cycles Length of use of park public spaces connected to active mobility

5.4.9 Design boundaries

Personal information is minimized, anonymous, and explicitly authorized; historical images, trademarks, portraits are subject to clearance; the exhibition ends meet the requirements of the competition "HTML is open offline and does not rely on external services"; public areas use only desensitized data; park operations continue the "city partners" model, which is not a substitute for enhancement.

5.4.10 Basis of reference

The first phase of the construction of the Jing-Zhang Railway Site Park by the Haidian Government is free of charge to the public (2023-06-30). The People.cn Beijing Jing-Zhang Railway Site Park is officially opened (2023-06-29). The Beijing News Jing-Zhang Railway Site Park is officially open to the public (2023-06-29). The Beijing Daily News client, 2023 Beijing urban Renewal Best Practices: the Jing-Zhang Railway Site Park (first phase) project (2023-10) This proposal 4.4 Site III (resilience public)

5.5 Four Module Closed Rings and Data Reflow

| Link | Input | Output | Rewind to | | Planning approval | Programme package + multi-scenario results | Approved space rules, compliance deviation list | Data mapping (policy-project-space map) | | Investment decisions | Approved rules + update potential | Project time series, Bree ledgers calibration, monitoring early warning | System of indicators, progress in implementation, data assets | | Urban operations | Landing Project + Run Data | Operation status, incident warning, dispatch instructions | Data foundation (dynamic running data 2.1.2) | | Public life | Space rules + operating services | Human experience, claims, sensory data | Social awareness data (2.1.4) Next round of diagnosis |

**Key mechanism:* *Reflow of data back into data baseation, allowing the "Planning Operating System" (1.3) to continuously calibrate itself in each rotational generation — this is what it means to "close the feedback from the city running."

5.6 Phased implementation path

| Phase | Time window | Area carrier | Focus. | | Recent Pilots | 0-1 years | Five mouths - Qinghua Park (Slice One) + Site Park One | Public life (AR orientation + travel + participation), planning approval (compliance pre-trial pilot) | | Medium-term expansion | 1-3 years | Three focus areas | Investment decision-making (updated project bank + data asset tables), urban operations (transport monitoring and emergency response) | | Forward governance | 3-5 years | Jing-Zhang Innovation Area (after 9km Green Corridor) | Four-module closed loops + energy power synergy + open scene and capacity trading |

System of indicators, area recovery and compliance matrix

Indicators are classified as geometrically recalculated design model indicators, statutory or engineering indicators relying on official conditions, performance indicators for continuous calibration over the operating period. Full mission and standard coverage is stored in the matrix of machines, and this chapter explains how the indicator supports the design judgment. [Depth: metrics_rea_sqm]

The proportion of public space is calculated by the corresponding design layer and total boundary area. [data: geometry/public_space.geojson#PUBLC-001]

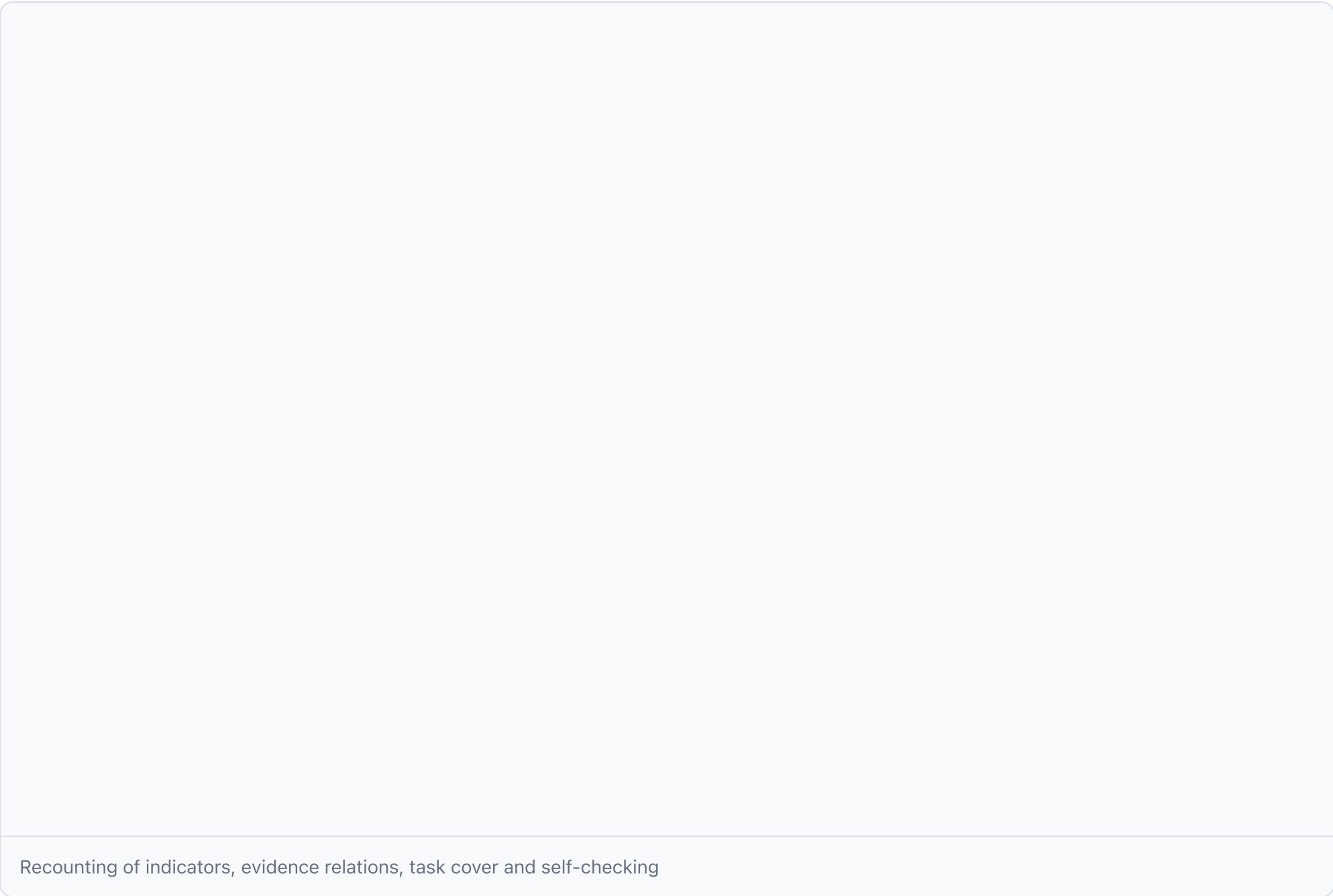


Figure 05 indicators, evidence, mission coverage and self-censorship are required.

5.7 Interface with competition tasks and existing modules

| Competition Task / Existing Modules | Four module acceptances | | Global case | Shenzhen Nanshan "Strategic Downfall" Investment Decision + Urban Operations + One-north Government of Singapore | | Agent 3 AI+ scene | Transport / Services / Consumption / Medical / Education Public Life + Urban Operating Load | | object.4 | "Cultural Experiences" of public life for developers walking walks, showrooms for results, and a wall of honor. | | delegate.6 Long-term operations | Operators of the four modules and a phased mechanism, i.e. long-term operating closed loop | | Five functions: "Intelligent AI Urban Governance." | 5.1 Planning approval + 5.3 Direct urban business counterpart | | Smart governance track | DeepAgent (data check-up / solicitation / synergetic) is available for attachment 5.2 / 5.3 | | 4.4 Soft resilience wisdom | 5.3 Urban operations are the extension of their operating layers (water safety Four-prevention system, resilience switching) | | Chart "Deep Series" | Deep Chat 5.4 Life Services; Deep GIS 5.1 Approval Analysis; Deep Age ## 5.2/5.3 Governance; Deep Control 5.2 Data Assetization + Developer Interface |

5.8 Landing statement (take note of the package submission)

1. The four modules are identified as "conceptual recommendations/reference proposals/for further study by professional teams" and are not written as implementation arrangements have been established by the Government; 2. Indicators for each module enter

metrics.json '; data sources (including public reports from City Brain, North China House, Site Park, etc.) enter

sources.json '; corresponding to the requirements of the bulletin / mission document enter `compliance_matrix.json '; 3. The prototype of the digital twin platform (8.3-1), the AI Planning Assistant interface (8.3-2), and a slice full-processed video (8.3-3) as a common technical base and demonstration vehicle for the four modules;

Description of risk, copyright and compliance

The results package completes the original version of the team, but does not package the conceptual proposal as statutory planning or definitive implementation. The main risks include an error of location and area caused by the temporary boundary, inadequate information on transport, municipalities, electricity, fire protection and the environment, limited information on traffic, traffical power facilities and flood, vibration, noise, heat discharge and sensitive functional conflicts, lack of transparency, privacy disclosure or automatic deviations in algorithms, confusing historical recovery of generation and unclear authorization of external cases, fonts, images and media.

The results of the team are registered separately from third-party data.

References

The reference system includes an announcement of the competition with Agent ' s mission, site packages and planning standards, complete team drafts, and open policies and cases used within the permitted and applicable borders.

The original complete team manuscript is archived by word in `report/narrative.md '. Machine audits are used for retroactive and recalculation and are not a substitute for the original team argument. The additional information must be entered into the body, indicator or spatial conclusion by registering the source, licence, time frame and use limits. [source: SITE-PACKAGE] [source: SOURCE-REGISTRY]